SMITHIAN FOLD THEORY - MATHEMATICS BRANCH PAPER 001

From Fold to Mathematics

An Exact, Parameter-Free and Machine-Closed Derivation of Mathematical Foundations from Smithian Fold Theory

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Open Source Science Platform and Knowledge Tree

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Third clean-room reconstruction - complete V1/V2 Mathematics reconciliation
Twenty-seven registered records - 21,504 generated candidate structures
Complete exact scientific calculator - desktop, browser and local-network phone
Seventy-one prior obligations closed - none open
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Abstract

This paper reports the current-evidence-complete Mathematics branch of the third clean-room reconstruction of Smithian Fold Theory (SFT). Starting only from the sixteen model-admitted Foundation receipts, it derives exact arithmetic and number structure; discrete mathematics; combinatorics; graph and network theory; algebraic structures; order and lattice structure; finite computational geometry and topology; exact probability and statistics; optimization; finite dynamical systems; logic and proof theory; and category, type and compositional structure. It then reconciles all Mathematics-owned V1/V2 observations through exact relation composition, Fold number theory, potential infinity, algebraic-balance certificates, native n-body recurrence, floored-fluid regularity, bounded prime-pair and Collatz censuses, the Riemann mirror boundary and self-similar convergence. Version 1.2 adds the complete Smithian Fold Scientific Calculator: an exact, proof-producing, cross-platform application whose ordinary calculator surface is bound to the admitted mathematical laws, whose local web interface works on computers and same-network phones, and whose prohibited outputs halt transactionally. No conventional mathematical axiom system, semantic numerical zero, negative quantity, irrational or imaginary proof value, floating-point proof quantity, completed infinity, ungenerated continuum, stochastic cause, fitted parameter, pretrained model or application result is admitted as a premise.

The twenty-two foundational claim grammars execute 9,984 generated candidate structures. Five calculator records execute a further 11,520 structures: the first is preserved as superseded adverse evidence after a parity defect was found, and claims 004 through 007 form the corrected active lineage. Across all twenty-seven registered Mathematics records, 21,504 candidate structures are preserved; the twenty-six current non-superseded claims account for 20,480. Every generated candidate receives an exact decision; each grammar has exactly one all-preserving survivor. Every claim passes minimality, named-shape uniqueness, a depth-independent base/successor certificate, false-premise, source-tamper, artifact-tamper and boundary controls, cryptographic sealing and implementation-distinct recomputation. The branch therefore contains twenty-seven depth-independent engine receipts with the supersession boundary explicit. The complete 763-entry V1/V2 source surface is reviewed for categorical ownership; 71 atomic Mathematics obligations are reconstructed or explicitly corrected, with none left open.

The contribution is not a relabelling of standard mathematics. Familiar terms enter only after each SFT derivation seals, as correspondence vocabulary. The primary objects are complete generated traces, exact held/whole parts, pair cells, held labels, canonical finite forms, relations and proof traces. Empty and identity cases use structural empty One rather than numerical zero; direction and opposition use held orientation rather than signed magnitude; uncertainty is a relation between deterministic support and observation rather than an imported random cause. Every prose claim is linked to machine-readable census, elimination, control, certificate and receipt evidence.

Keywords: Smithian Fold Theory; foundations of mathematics; exact arithmetic; scientific calculator; computational proof; exact rational enclosure; structural empty One; fail-closed computation; cross-platform software; discrete mathematics; combinatorics; graph theory; algebra; probability; proof theory; open computational science.

1. Central scientific claim

The exact claim of this paper is:

> *Within the SFT V3 Mathematics version 1.2 current-knowledge inventory, every one of the twenty-two mathematical-foundation obligations and every one of the five preserved calculator records has a depth-independent engine receipt. Claim 003 remains visibly superseded by the corrected 004 law; claims 004 through 007 form the active exact calculator lineage. The inventory contains no unclassified or frontier obligation at this evidence date and remains open to lawful future extensions.*

The inventory identity is `sha256:773faaad16dc29dc99fe2c401d8b15fe7d77000262bdb1ecfe2ce739056bf4d4`. It fixes the version 1.2 branch boundary and claim order before the paper is evaluated. Current-evidence completion means closure of these generated-finite mathematical kernels and calculator translations. It does not assert a completed infinite universe or permit a familiar named structure to inherit properties without its own generated witness. Applications in computation, physics, biology, engineering, Fold Protein, Fold Chess, Fold Go and Unison AI did not select any law in this paper. A new generated omission, counterexample or lawful extension must be registered and published in a later version rather than excluded by a permanent lock.

2. Standalone dependency foundation

This paper is standalone in exposition but not dependency-free in the scientific census. The prior Foundation branch supplies sixteen exact receipts, including operational occurrence, structural One, complete positive finite count, exact held/whole part coordinates, the minimal Fold, cross-partition equivalence, Fold assembly, finite form grammar, canonical identity, exact cast/Fold/Take, half-One, Fold dynamics, extended primitive-map uniqueness, root-bound replay, admission enforcement and the one-way measurement boundary. Those are cited as admitted dependencies, not copied assumptions. The published Foundation paper remains a separate work and is not modified by this paper.

The sole root theorem remains *there is no nothing*. Every dependency path terminates at its admitted receipt. Each Mathematics registration contains empty axiom and free-parameter tuples. Prior SFT versions, conventional theorems and correspondence names are excluded from derivational source manifests.

3. Exact mathematical constitution

The admitted domain contains structural One, complete positive finite generation traces, exact positive held/whole parts, canonical labels, fibres, branches, words, pair cells, relations, forms and proof traces. The domain does not contain semantic numerical zero, a negative quantity, an irrational or imaginary proof value, floating proof arithmetic, a completed infinite set or an ungenerated continuum. An empty selection, identity path or no-transition duration is the structural empty-One form. Opposed direction is a distinct held orientation with its positive remainder trace.

Python host integers, Boolean checks, empty containers and process status codes are implementation mechanics. They do not become SFT proof objects. The scientific sources explicitly return structural forms or exact positive part relations at the boundary where a conventional implementation might otherwise introduce zero, negative or floating values. The admission engine independently rejects registered axioms, free parameters, missing dependencies, incomplete censuses, multiple survivors, failed controls and source drift.

4. What forced, closed and independently validated mean

A declaration is not forced because it is elegant or familiar. Every claim specifies a product grammar of structural questions. Each coordinate supplies at least one explicitly rejected alternative and exactly one all-preserving alternative. The complete Cartesian product is generated in canonical order. Candidate identities, exact forms and trace hashes are recorded. Every candidate receives one survival decision, elimination reason and proof hash. Exactly one candidate must survive.

Closure then requires more than the finite run. Minimality shows that replacing any preserving coordinate loses a registered requirement and that no extra rule remains. Named-shape uniqueness shows that the full product contains only one all-preserving coordinate. A claim-specific structural One base and successor certificate extends the classification to every generated finite depth. This is depth-independent closure over finite generation, not a completed infinite object.

Four controls are mandatory. A false premise must be rejected; a changed official source identity must be detected; a missing, duplicated or additional survivor must fail; and an excluded object or answer-producing model must halt at the boundary. After the derivation is sealed, a separate Python process whose file hash is not part of the scientific implementation regenerates the literal candidate product, decision vector, unique survivor, closure flags and controls. Only the shared admission engine can add the receipt to the census.

5. Dependency order and executed census

Order	Claim	Candidate structures	Dimensions	Closure
1	`SFT-MATH-EXACT-ARITHMETIC-001`	256	8	depth-independent
2	`SFT-MATH-DISCRETE-001`	256	8	depth-independent

Order	Claim	Candidate structures	Dimensions	Closure
3	SFT-MATH-COMBINATORICS-001	256	8	depth-independent
4	SFT-MATH-GRAPH-NETWORK-001	512	9	depth-independent
5	SFT-MATH-ALGEBRA-001	512	9	depth-independent
6	SFT-MATH-ORDER-LATTICE-001	512	9	depth-independent
7	SFT-MATH-GEOMETRY-TOPOLOGY-001	1,024	10	depth-independent
8	SFT-MATH-PROBABILITY-STATISTICS-001	512	9	depth-independent
9	SFT-MATH-OPTIMIZATION-001	512	9	depth-independent
10	SFT-MATH-DYNAMICAL-SYSTEMS-001	1,024	10	depth-independent
11	SFT-MATH-LOGIC-PROOF-001	1,024	10	depth-independent
12	SFT-MATH-CATEGORY-TYPE-COMPOSITION-001	1,024	10	depth-independent
13	SFT-MATH-EXACT-RELATIONS-002	256	8	depth-independent
14	SFT-MATH-ORBIT-NUMBER-THEORY-002	256	8	depth-independent
15	SFT-MATH-LIMIT-CONTINUUM-002	256	8	depth-independent
16	SFT-MATH-ALGEBRAIC-BALANCE-002	256	8	depth-independent
17	SFT-MATH-BOUNDED-N-BODY-002	256	8	depth-independent
18	SFT-MATH-FLOORED-FLUID-REGULARITY-002	256	8	depth-independent
19	SFT-MATH-PRIME-PAIR-CENSUS-002	256	8	depth-independent
20	SFT-MATH-RIEMANN-MIRROR-002	256	8	depth-independent
21	SFT-MATH-COLLATZ-FINITE-CENSUS-002	256	8	depth-independent
22	SFT-MATH-SELF-SIMILAR-CONVERGENCE-002	256	8	depth-independent
23	SFT-MATH-SCIENTIFIC-CALCULATOR-003	1,024	10	depth-independent; superseded adverse evidence
24	SFT-MATH-SCIENTIFIC-CALCULATOR-004	1,024	10	depth-independent; current
25	SFT-MATH-SCIENTIFIC-CALCULATOR-005	8,192	13	depth-independent; current
26	SFT-MATH-SCIENTIFIC-CALCULATOR-006	1,024	10	depth-independent; current
27	SFT-MATH-SCIENTIFIC-CALCULATOR-007	256	8	depth-independent; current

The version 1.2 Mathematics evidence surface preserves **21,504** generated structures across twenty-seven registered records. The twenty-two foundational claims contribute 9,984; the calculator lineage contributes 11,520. Claim 003 is retained as the exact historical record superseded after the endpoint-parity defect was found; it is not counted as a current law. Claims 004 through 007 and the original twenty-two claims form the twenty-six current non-superseded results.

The finite product count reports representation classes executed by the registered claim grammars. It is not a count of every mathematical object producible by the laws. Generality is carried by the induction certificate stated for each claim below.

6. Derivation 1: Exact arithmetic and generated number structure

Claim identity: `SFT-MATH-EXACT-ARITHMETIC-001`

6.1 Question and necessity

Arithmetic is required before later branches can count arrangements, compare structures or assign exact support weights, but importing a conventional number field would violate the clean-room boundary.

The exact theorem statement is:

> *Every admitted arithmetic relation on generated positive finite traces and exact parts has one parameter-free structural kernel: disjoint junction for addition, complete pair-cell refinement for product, finite repeated composition for powers, common-refinement pairing for exact quotient and comparison, and held orientation for unmatched remainder; no semantic zero, negative, irrational, floating or completed-infinite value enters the law.*

Admitted dependencies: `SFT-FOUNDATION-COUNT-001`, `SFT-FOUNDATION-PART-001`, `SFT-FOUNDATION-PART-EQUIVALENCE-001`, `SFT-FOUNDATION-FORM-ENFORCEMENT-001`.

6.2 Generated grammar and exact boundary

Count supplies complete positive traces; Part supplies held/whole coordinates; Part Equivalence supplies common refinement and bijection. Exhausting the eight representation choices leaves one kernel that preserves every source identity without introducing a scale or forbidden value.

Generation rule: Generate the complete product of trace coverage, junction, pair-cell product, exact quotient, comparison, remainder orientation, finite generality and extra-rule status.

Exact grammar boundary: All arithmetic kernels constructible from admitted positive finite traces, exact held/whole parts, common refinement, canonical form identity and no imported scalar field.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:e5594c9df6b591e6d7bac2b5afb86bcb32a41d450fb06727426740b982a57500`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-EXACT-ARITHMETIC-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`coverage`	`partial-trace`	A partial trace omits an admitted generated occurrence.	`complete-trace` - The complete trace retains every generated occurrence once.
`junction`	`overlap-merge`	Overlap repeats a shared occurrence and loses disjoint provenance.	`disjoint-junction` - Disjoint junction preserves both complete positive traces.
`product`	`repeated-unlabelled`	Unlabelled repetition erases the two source coordinates.	`pair-cell-refinement` - Each generated left label is paired once with each generated right label.
`quotient`	`rounded-measure`	A rounded or decimal value is measured rather than exact derivational evidence.	`common-refinement-pairing` - A complete common refinement and bijection witnesses the exact part relation.
`comparison`	`borrowed-scalar-order`	A borrowed scalar order imports the answer-producing number model.	`complete-trace-pairing` - Pairing identifies equality and a held unmatched trace identifies orientation.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`difference`	`signed-magnitude`	A signed magnitude installs forbidden negative quantity.	`held-oriented-remainder` - A distinct held label records which trace retains the unmatched positive remainder.
`generality`	`bounded-answer-table`	A finite answer table has no depth-independent arithmetic certificate.	`base-successor-induction` - The One base and generated successor preserve each operation at every finite depth.
`addition`	`has-extra-rule`	An extra scale, constant or field is not supplied by the dependencies.	`no-extra-rule` - The kernel uses only admitted traces, parts, refinement, pairing and held labels.

6.3 Unique survivor, minimality and laws

> *The exact arithmetic kernel is complete positive-trace coverage with disjoint junction, pair-cell product, common-refinement quotient, complete pairing comparison, held-oriented remainder, base/successor generality and no extra rule.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- disjoint junction is associative on mutually disjoint generated traces.
- pair-cell product distributes over disjoint junction by generated cell provenance.
- pair-cell product composition is associative after canonical path flattening.
- exact quotient exists only with a complete refinement-pairing witness.
- comparison returns same-form or a held positive remainder orientation.

6.4 Operational witnesses and unfavorable checks

- `junction-associativity` - (a joined b) joined c and a joined (b joined c) retain the same canonical trace. Result: PASS.
- `pair-cell-cardinality` - Two generated cells paired with three generated cells produce the complete six labelled pair cells. Result: PASS.
- `product-associativity` - Canonical flattened pair paths are independent of binary grouping. Result: PASS.
- `exact-pairing` - Equal complete traces admit one-to-one and onto pairing; unequal traces do not. Result: PASS.
- `held-remainder` - A larger trace reports a positive held remainder instead of a negative value. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A partial trace omits an admitted generated occurrence. Result: PASS; receipt `sha256:33e3d5d241540fd337807b34e03db365ce48e9e3d0ea4d102d153a58df8199c6`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not a value; negative magnitude is replaced by held orientation; irrational, imaginary and floating proof values are excluded; completed infinity and an ungenerated continuum are excluded Result: PASS; receipt `sha256:19138ef235a0e43f75285e80856edccff9d19b3aced53986c83f5de9fa2f4fca`.

6.5 Depth-independent closure

Base: The structural One supplies the first nonempty trace and exact self-whole.

Successor: Appending one fresh generated occurrence preserves complete trace identity; junction appends it once, pair-cell refinement appends one labelled cell per opposite trace element, and pairing either extends with one mate or records the unmatched held remainder.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

6.6 Meaning and scientific consequence

This result changes the role of arithmetic. A number is not admitted as an unexplained occupant of a pre-existing field. Exact positive magnitude is the identity of a complete generated trace. Addition is the preservation of two disjoint traces in their junction; product is the complete provenance-retaining pair-cell refinement. An exact quotient is not a division oracle but the existence of a complete pairing over a common refinement. Comparison never requires a negative result: equality is complete pairing and difference is a held orientation together with the unmatched positive trace.

6.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: positive integers, rational numbers, addition, multiplication, division, order, signed difference. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

6.8 Exact exclusions and limitations

- semantic numerical zero is not a value.
- negative magnitude is replaced by held orientation.
- irrational, imaginary and floating proof values are excluded.
- completed infinity and an ungenerated continuum are excluded.

This law closes exact generated finite arithmetic and positive rational part relations. It does not admit a completed infinite set, continuum, irrational field or negative proof magnitude. Conventional notation may describe the result only in correspondence.

6.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:db8d9a5838f2f84cd04cd3425e0faa5232ec8afb6461f1a7fe62ab9f6263246f`.
- Derivation seal: `sha256:1254abf40a91886525ad8a846896b75a1023fd785907f233bcf817db2c744dc4`.
- Independent implementation: `sha256:9d0af694427b46f2b09ed218d83d0d87f30b16601b03c3694ac71738254bf164`.
- Independent certificate: `sha256:73b24906c1005bd7f42f36000f396bb5ec88f633a4a69b6159d5cc349bb3be2a`.
- External validation: `sha256:9a05361b10ccce837c8a32def2b24dd0ec714cb2f45d397a9e6b72fedec306a7`.
- Engine receipt: `sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbae71609cb05c4f40196418`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-EXACT-ARITHMETIC-001-28252bae62373d86.json`.

7. Derivation 2: Generated discrete objects, relations and induction

Claim identity: `SFT-MATH-DISCRETE-001`

7.1 Question and necessity

Combinatorics, graphs, algebra, logic and computation require exact collections and relations, but an imported set-theoretic universe would add axioms and completed infinities outside the SFT domain.

The exact theorem statement is:

> *Every admitted discrete object is a complete canonical generated finite collection or held selection; membership is retained occurrence, relations are held selections of complete pair-cell support, maps are total single-valued relations, sequences retain generated order, and general laws close by structural One/base-successor induction.*

Admitted dependencies: `SFT-FOUNDATION-COUNT-001`, `SFT-FOUNDATION-FORM-GRAMMAR-001`, `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-EXACT-ARITHMETIC-001`.

7.2 Generated grammar and exact boundary

Canonical form enforcement supplies identity; count supplies complete finite traces; exact arithmetic supplies lawful trace composition. Product enumeration of the eight representational questions forces the collection-selection-relation-map-induction kernel.

Generation rule: Generate the complete product of collection coverage, canonical identity, membership, relation support, map law, sequence order, induction closure and extra-constructor status.

Exact grammar boundary: All discrete structures generated from finite canonical One/Fold forms, exact positive traces, held selections, pair-cell relations and successor construction.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:0103778e8716a64ec9ed8d8b243af9af9bc7fed90360339b4be989a03d263302`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-DISCRETE-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`collection`	`partial-or-duplicated`	Omission or duplication changes the generated collection trace.	`complete-unique-trace` - Every generated member occurs once in the canonical collection trace.
`identity`	`presentation-alias`	Presentation names can merge distinct forms or split one form.	`canonical-form-identity` - Canonical construction traces supply exact member identity.
`membership`	`unrecorded-assertion`	An unrecorded assertion has no source-bound occurrence witness.	`retained-occurrence` - Membership retains the canonical member occurrence within the source trace.
`relation`	`free-pair-list`	A free pair list has no complete source-product boundary.	`held-pair-cell-selection` - A relation holds a generated selection from complete source pair-cell support.
`map`	`partial-or-multivalued`	A missing or repeated image does not define one image for every source member.	`total-single-valued` - Every source member has exactly one retained target mate.
`sequence`	`unordered-carrier`	An unordered carrier erases the generated predecessor relation.	`held-generation-order` - The canonical trace retains each predecessor-successor position.
`induction`	`sampled-depths`	Sampled depths do not establish the successor step.	`One-base-successor` - The One base and freshness-preserving successor cover every generated finite trace.
`addition`	`extra-constructor`	An extra constructor is not supplied by the Foundation grammar.	`no-extra-constructor` - All objects decompose into admitted forms, selections, pair cells and successors.

7.3 Unique survivor, minimality and laws

> *The discrete kernel is complete unique canonical collections with retained membership, pair-cell-selected relations, total single-valued maps, held sequence order, One/base-successor induction and no extra constructor.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- canonical membership is decidable for every generated finite collection.
- held selection never introduces an element outside its source collection.
- relation complement is a held complementary selection of the same complete pair support.
- map composition preserves total single-valuedness when interfaces agree.
- structural induction covers every generated finite collection trace.

7.4 Operational witnesses and unfavorable checks

- `unique-collection` - A generated collection retains each canonical member once. Result: PASS.
- `held-membership` - A held selection contains only members of its registered source. Result: PASS.
- `empty-One-selection` - A held-empty selection returns the empty One form, not numerical zero. Result: PASS.
- `relation-boundary` - Every retained relation pair belongs to the complete generated pair-cell support. Result: PASS.
- `total-map` - The example relation gives every source form exactly one image. Result: PASS.
- `fresh-successor` - A successor appends exactly one fresh canonical form. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission or duplication changes the generated collection trace. Result: PASS; receipt `sha256:bfcb72c0f0a5cfdb140547bb18d237268d4ae9231dbf57f6be92112190ed979a`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no completed infinite set; no ungenerated universal collection; no semantic numerical zero for the empty selection; no presentation alias may replace canonical form identity Result: PASS; receipt `sha256:fdcae55e3e8ce516575f15e1d490505002d0dffbdc3dc372c54ed70cd69b236e`.

7.5 Depth-independent closure

Base: The structural One is the first canonical generated member and the empty One marks a held-empty selection without numerical zero.

Successor: Given a complete unique canonical collection, append one fresh canonical form once; membership, relation pair support, map-image obligation and predecessor order extend by their local generated clauses.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

7.6 Meaning and scientific consequence

The theorem reconstructs the working core normally supplied by an ambient set universe. A collection has a finite generated boundary, canonical member identities and no unrestricted comprehension. Membership is a retained occurrence; a relation is a held selection from complete pair support; and a map is a relation that passes totality and single-valuedness. This makes every later discrete object auditable at its source.

7.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: finite set, subset, membership, relation, function, sequence, structural induction. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

7.8 Exact exclusions and limitations

- no completed infinite set.
- no ungenerated universal collection.
- no semantic numerical zero for the empty selection.
- no presentation alias may replace canonical form identity.

The theorem concerns generated finite objects. It does not install an ungenerated universe, completed infinite set or unrestricted comprehension principle.

7.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:41a82043fd37d0c054d490f9c426ad37c23d6bb8ea2e4f0e9a5af6fb8fe30285`.
- Derivation seal: `sha256:1d3b73c941e235c5ffe473ccafa9cd59b800ddd33d43e4feea7a6821ed8f45c9`.
- Independent implementation: `sha256:c808241fc28e1fa7bfbbea0e67fc594eab9b957575534f2f0b88af57def8dbae`.
- Independent certificate: `sha256:2f9da4efdc67b6f08074be8d3ddc47595444a9300fd00e507fb04c228686f44b`.
- External validation: `sha256:5dad446dc09d74caf9325f27b314d7f886275164df426cd561a687d9283cd109`.
- Engine receipt: `sha256:632de46c995329ed86f8397e8cf2cc493d5074027941a50069df66b4ac9e29bb`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-DISCRETE-001-632de46c995329ed.json`.

8. Derivation 3: Exact finite combinatorial generation

Claim identity: `SFT-MATH-COMBINATORICS-001`

8.1 Question and necessity

Counting arrangements by quoting familiar formulas would import their answers. The generated family itself must be primary so every count, symmetry reduction and recurrence remains inspectable.

The exact theorem statement is:

> *Every admitted finite combinatorial family is generated by an explicit held-choice recurrence over a complete canonical carrier, preserves construction order and provenance, removes duplicates only by canonical form identity, and counts the resulting complete trace; selections include the structural empty One form rather than numerical zero.*

Admitted dependencies: `SFT-FOUNDATION-FOLD-ASSEMBLY-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`.

8.2 Generated grammar and exact boundary

Discrete canonical collections provide the carrier and held selection; Fold words provide exact choice traces; arithmetic counts the complete result. Exhausting the eight assembly choices forces explicit held-choice recurrence and provenance-preserving class accounting.

Generation rule: Generate the complete product of carrier coverage, choice rule, provenance, duplication policy, symmetry identity, class accounting, recurrence closure and extra-formula status.

Exact grammar boundary: All arrangement, selection and finite classification families produced from a registered canonical finite carrier by held choice, remainder and successor recurrence.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:b5a4fe7ce476fb3f349f0ac30a1ccfbd35c2637505be5ef9c4022d90e65a403b`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-COMBINATORICS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`carrier`	`partial-carrier`	A partial carrier silently omits possible generated choices.	`complete-canonical-carrier` - Every distinct generated source member is present once.
`choice`	`sampled-choices`	Sampling does not exhaust the registered choice positions.	`held-choice-recurrence` - Each available member is held once at the next position and recursion continues on the exact remainder.
`provenance`	`erased-origin`	Erasing origin prevents reconstruction and duplicate diagnosis.	`complete-choice-trace` - Every result retains the ordered held-choice construction trace.
`duplicates`	`presentation-deduplication`	Presentation equality can merge structurally distinct results.	`canonical-identity-deduplication` - Only identical canonical construction traces are identified.
`symmetry`	`assumed-symmetry`	An assumed quotient may remove distinguishable held labels.	`generated-bijection-witness` - A declared label-preserving bijection is required before forms share a symmetry class.
`accounting`	`signed-inclusion-subtraction`	Signed subtraction imports negative quantities.	`disjoint-provenance-classes` - Each generated form is assigned once to a held disjoint class before counts are joined.
`generality`	`finite-size-table`	A table of small sizes does not force the next recurrence.	`carrier-successor-recurrence` - Adding one fresh carrier member partitions new forms by whether and where it is held.
`addition`	`imported-formula`	A stored factorial, binomial or inclusion formula selects the conventional answer.	`no-imported-formula` - Counts are identities of the complete generated output traces.

8.3 Unique survivor, minimality and laws

> *The combinatorial kernel is complete canonical carrier generation by held-choice recurrence with complete choice provenance, canonical deduplication, witnessed symmetry, disjoint class accounting, successor generality and no imported formula.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- every arrangement holds each carrier member exactly once.
- every selection is produced once by the fresh-member absent/held recurrence.
- canonical trace equality is the only duplicate-elimination rule.

- disjoint class ledgers conserve the complete generated family.
- enumerated count is the identity of the complete output trace.

8.4 Operational witnesses and unfavorable checks

- `arrangement-completeness` - Three distinct held labels generate six unique ordered arrangements. Result: PASS.
- `arrangement-coverage` - Every arrangement holds every carrier label exactly once. Result: PASS.
- `selection-recurrence` - Three held labels generate the structural empty One and every absent/held combination once. Result: PASS.
- `disjoint-accounting` - Parity-named witness classes partition the carrier without overlap or omission. Result: PASS.
- `fresh-extension` - Fresh-member recurrence creates a disjoint absent/held copy of every prior selection. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A partial carrier silently omits possible generated choices. Result: PASS; receipt `sha256:c1dbfffb55e34b1990cfc2d3bd521fd163d147f13c5598b09021959fcee3cfe3`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no random sampling as completeness evidence; no factorial or binomial answer table; no negative inclusion-exclusion proof value; no completed infinite combinatorial family Result: PASS; receipt `sha256:5e256307e37a27f3cb51ece4a149427a8131c602ec3eb31051eb5b10fdb3dced`.

8.5 Depth-independent closure

Base: A one-member carrier has one complete arrangement and two structural selection forms: empty One and held member.

Successor: For a fresh carrier member, arrangements partition by its held position and recurse on the prior remainder; selections partition into prior forms and the same forms with the fresh member held. These disjoint generated classes exhaust the successor carrier without an imported formula.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

8.6 Meaning and scientific consequence

Combinatorial values arise after, not before, generation. The arrangement and selection families are constructed by exhaustive held choices with each remainder retained. Their counts are identities of the resulting complete traces. Symmetry reduction requires an explicit label-preserving bijection, and overlap is handled by disjoint provenance classes instead of negative inclusion-exclusion proof quantities.

8.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: permutation, combination, recurrence, symmetry class, pigeonhole counting, inclusion-exclusion. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

8.8 Exact exclusions and limitations

- no random sampling as completeness evidence.

- no factorial or binomial answer table.
- no negative inclusion-exclusion proof value.
- no completed infinite combinatorial family.

The law closes the generative foundation of finite combinatorics. Named advanced enumerations require their own derived generators; none may be admitted by quoting a conventional closed-form answer.

8.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:7d7554f879a30b739c05c4f9aeba7c5812c4ceddea2a1d032a71ecb8fa10ba1a`.
- Derivation seal: `sha256:3177572ea372062f65a7969fdf4e498c86c9af558e37c048165e1d950cf1d942`.
- Independent implementation: `sha256:00ac1dfb12aa98fba2ec8b0ab7eae0b940cee7ca8c55b75fa07678864e9b4086`.
- Independent certificate: `sha256:ea22b95c87c9f336726c9df87d1e88fb38b1ad27519789c334a5052f5f864ae6`.
- External validation: `sha256:dbfc2711bbabde9be3bf4a4cc3c5de3cd62440c9fbeea20b2fceed6dcfb1275b`.
- Engine receipt: `sha256:52e1db94bb7865270bb3387c47c609c438de3338b542cc31d910b981e4250968`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-COMBINATORICS-001-52e1db94bb786527.json`.

9. Derivation 4: Exact finite graph and network structure

Claim identity: `SFT-MATH-GRAPH-NETWORK-001`

9.1 Question and necessity

Graphs are the minimal mathematics of explicit relations and paths. Their laws must follow from generated carriers and pair support rather than imported matrices, weights or probabilistic ensembles.

The exact theorem statement is:

> *Every admitted finite graph is a complete canonical node collection plus a held selection of generated ordered node-pair relations; paths retain every adjacent transition, reachability is a complete finite path witness, cycles retain return, trees are connected cycle-free forms, cuts are exact crossing-edge selections, and internal network conservation is complete ingress/egress pairing.*

Admitted dependencies: `SFT-FOUNDATION-FORM-GRAMMAR-001`, `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-COMBINATORICS-001`.

9.2 Generated grammar and exact boundary

Discrete mathematics supplies canonical carriers and relations; combinatorics supplies complete path-family generation. Exhausting nine structural choices leaves the relation/path/return/pairing graph kernel.

Generation rule: Generate the complete product of node coverage, edge provenance, orientation, path trace, connectivity, cycle return, cut support, internal balance and extra-network-rule status.

Exact grammar boundary: All finite graphs and networks generated from canonical finite form carriers, held pair-cell relations, complete path traces, return relations and exact positive flow-token pairings.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:447b14b12f08e4bcf5c21bd3b34a7e06cbccd3ad90d3669153c9d05306221bfd`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-GRAPH-NETWORK-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`nodes`	`partial-or-aliased-nodes`	Missing or aliased nodes change the graph carrier.	`complete-canonical-nodes` - Every generated node form occurs once with canonical identity.
`edges`	`free-endpoint-list`	A free list can contain ungenerated endpoints or duplicate relations.	`held-pair-cell-relation` - Edges are a held duplicate-free selection of complete node pair support.
`orientation`	`signed-edge-weight`	A signed weight imports negative magnitude and does not identify endpoints.	`held-endpoint-order` - The ordered endpoint labels retain orientation; complementary order represents reversal.
`path`	`endpoint-only`	Endpoints alone omit the intervening adjacency requirements.	`complete-adjacent-trace` - Every node and edge transition in the path is retained in order.
`connectivity`	`assumed-connected`	An assertion without a path omits the connecting structure.	`generated-path-witness` - Reachability requires a complete generated adjacent path.
`cycle`	`repeated-label-only`	A repeated label without a transition trace does not prove return.	`explicit-return-path` - A cycle is a nontrivial complete path whose terminal relation returns to a held prior node.
`cut`	`borrowed-scalar-threshold`	A scalar threshold imports an external coordinate.	`held-crossing-selection` - A held node selection forces exactly the edges with one held and one complementary endpoint.
`balance`	`unaccounted-net-value`	A net signed value hides unmatched tokens.	`ingress-egress-pairing` - Every incoming positive token has one outgoing mate and conversely.
`addition`	`extra-network-rule`	An extra weight, metric or random law is not supplied by the graph dependencies.	`no-extra-network-rule` - All graph properties are witnessed by carriers, relations, paths, returns, selections and pairings.

9.3 Unique survivor, minimality and laws

> *The graph/network kernel is complete canonical nodes with held ordered pair-cell edges, complete adjacent paths, witnessed reachability and return, held crossing cuts, ingress/egress pairing and no extra rule.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- path concatenation is lawful exactly when the interface node agrees.
- reachability is reflexive by the retained identity path and transitive by path composition.
- a tree is a connected generated graph with no nontrivial return path.
- a cut contains every and only relation crossing a held/complementary node partition.
- network balance preserves token identity rather than cancelling signed magnitudes.

9.4 Operational witnesses and unfavorable checks

- `valid-path` - The chain retains both required adjacent edges. Result: PASS.
- `missing-edge-control` - An endpoint sequence with a missing adjacency is not a path. Result: PASS.
- `reachability` - Generated path search reaches c from a and does not reach a from c in the directed chain. Result: PASS.
- `cycle-return` - The explicit c-to-a return distinguishes the cycle from the chain. Result: PASS.
- `cut-selection` - Holding the first chain node selects exactly its crossing edge. Result: PASS.
- `paired-balance` - Equal unique ingress and egress token traces witness internal balance. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Missing or aliased nodes change the graph carrier. Result: PASS; receipt `sha256:2c881c5d8c3f3ab8e894f11a4fce432fbbfa4e49226ddb271ea8023ed8e3228b`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ungenerated vertex or edge; no negative edge or flow quantity; no stochastic graph law; no infinite graph as a completed object Result: PASS; receipt `sha256:3f199f993e865f79dd3ff17ba48418bbfca2470b4a79c8a8697148409b3ba7be`.

9.5 Depth-independent closure

Base: One canonical node supplies the identity path and contains no nontrivial edge cycle.

Successor: Appending one fresh node generates exactly its ordered pair cells with prior nodes; any held edge selection extends paths locally, and connectivity, cycle, cut and balance witnesses update only through those new cells.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

9.6 Meaning and scientific consequence

Graph theory becomes the exact mathematics of generated relations. Vertices are canonical forms, edges are held ordered pair cells, and paths preserve every adjacency. Reachability, cycles and trees are not labels but path predicates. A cut is forced by a held/complementary node selection. Network conservation retains the identities of paired ingress and egress tokens instead of cancelling signed scalar totals.

9.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: directed graph, path, reachability, cycle, tree, cut, network flow. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

9.8 Exact exclusions and limitations

- no ungenerated vertex or edge.
- no negative edge or flow quantity.
- no stochastic graph law.
- no infinite graph as a completed object.

This law closes exact generated finite graph structure. Weighted geometry, random graphs, infinite graph limits and algorithmic graph complexity require later derived laws.

9.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:e9e22d5262250c7cb551e278121b29431ef59d58cb53b28c9188f2d912979d8c`.
- Derivation seal: `sha256:f1f74c52072189f08134293f1f55aaff63aeb73fc045d79ac09a9372554c7ad9`.

- Independent implementation: `sha256:82cc03abc10d06d9d288a85c6171bc60db065b82a074e9fc39745f5b1e59ebe4`.
- Independent certificate: `sha256:531d0c9a5ec7a97b8e97c4a1664d67af65d7fab17127bcf042e76a86386a9210`.
- External validation: `sha256:9f9e2483be55ff773cf570f6fc3aee074f68f44fde118492f152023767fef5c8`.
- Engine receipt: `sha256:b4072f93147c8cd9b7233fc96cfeadeb791210239ab4e4ff7dec0ed7462e6d2a`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-GRAPH-NETWORK-001-b4072f93147c8cd9.json`.

10. Derivation 5: Generated finite algebraic structures

Claim identity: `SFT-MATH-ALGEBRA-001`

10.1 Question and necessity

Algebra is forced when generated operations are composed and compared. The exhaustive table boundary prevents familiar algebraic classifications from supplying unproved laws.

The exact theorem statement is:

> *An admitted finite algebraic structure is a complete canonical carrier with a complete, single-valued, closed pair-cell operation relation. Identity, associative composition, reversible return mates, commutation, substructure and structure-preserving maps are admitted exactly when their exhaustive carrier witnesses pass; none is installed as an ungenerated axiom.*

Admitted dependencies: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`.

10.2 Generated grammar and exact boundary

Discrete pair support supplies all operation inputs; exact canonical equality tests results; form enforcement supplies carrier identity. The nine-way census leaves the witness-governed operation kernel.

Generation rule: Generate the complete product of carrier, operation coverage, closure, identity, association, return, property-admission, structure-map and extra-law status.

Exact grammar boundary: All finite operation structures generated from canonical carriers, complete pair-cell relations, canonical path composition, held reversal and exhaustive property witnesses.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:ccae1cb3b71ff2b6cffd4acead871b01722969c069b797ad61e9cb2050f09701`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-ALGEBRA-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`carrier`	`partial-or-aliased-carrier`	Omission or aliasing changes which operation inputs exist.	`complete-canonical-carrier` - Every generated carrier form occurs once with canonical identity.
`operation`	`partial-operation-sample`	A partial sample does not assign every generated input pair.	`complete-single-valued-table` - Every ordered carrier pair has exactly one retained result.
`closure`	`external-result`	An external result silently enlarges the registered carrier.	`carrier-closed-result` - Every result is a canonical member of the same carrier.
`identity`	`named-identity`	A name alone does not prove left and right preservation.	`exhaustive-identity-witness` - One carrier form preserves every member on both sides.
`association`	`assumed-parenthesis-law`	An assumed equation imports an algebraic axiom.	`complete-triple-witness` - Every carrier triple has equal canonical flattened composites.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`return`	`signed-inverse-value`	A signed magnitude imports negative quantity.	`unique-held-return-mate` - A retained complementary carrier mate returns on both sides to identity.
`properties`	`property-by-name`	A familiar structure name cannot supply commutation or return.	`property-by-exhaustive-witness` - Each property is registered only after every required carrier tuple passes.
`maps`	`arbitrary-carrier-map`	An arbitrary map can erase the operation relation.	`operation-preserving-map` - The complete image relation commutes with every source operation cell.
`addition`	`extra-algebraic-axiom`	An extra axiom is not forced by the registered operation traces.	`no-extra-algebraic-axiom` - All admitted properties have explicit generated witnesses.

10.3 Unique survivor, minimality and laws

> *The algebraic kernel is a complete canonical carrier and complete closed single-valued operation table, with identity, association, returns, special properties and maps admitted only by exhaustive witnesses.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- closure is exact membership of every operation result in the registered carrier.
- identity is a two-sided exhaustive preservation witness.
- association is equality of both complete composite traces for every carrier triple.
- return is a unique held mate relation and never a negative proof value.
- homomorphism is complete operation preservation, not resemblance of presentations.

10.4 Operational witnesses and unfavorable checks

- `closed-table` - Every complete input pair in the witness table has one carrier result. Result: PASS.
- `identity` - The structural One form preserves both witness carrier forms. Result: PASS.
- `association` - Both parenthesizations agree for every witness triple. Result: PASS.
- `return-mates` - Every witness form has one held two-sided return mate. Result: PASS.
- `commutation-is-conditional` - Commutation passes for the symmetric witness and fails for a lawful noncommuting table. Result: PASS.
- `structure-map` - The canonical self-map preserves every operation cell. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission or aliasing changes which operation inputs exist. Result: PASS; receipt `sha256:b0cb6ed9032c814991aae189ba084f58a0344948ecdeba10de30448360699e08`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported group, ring or field axioms; no negative or irrational carrier requirement; no property inferred from a conventional name; no completed infinite carrier Result: PASS; receipt `sha256:e7630fd960384d7d35b15f2e1c900d720e648b0dd23ec4cea7ac6ce9dadc209c`.

10.5 Depth-independent closure

Base: The One carrier has one pair cell, its result remains One, and identity and association coincide.

Successor: Adding one fresh carrier form generates exactly its left, right and self pair cells. A proposed extension is admitted only after results remain in the extended carrier and every newly formed identity, triple, return and map obligation is checked.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

10.6 Meaning and scientific consequence

The algebra theorem deliberately separates an operation kernel from named algebraic species. A complete closed operation table is forced first. Identity, association, return, commutation and homomorphism status are then admitted only if their complete carrier obligations pass. Thus the words group, monoid or commutative structure cannot import properties; they are correspondence classifications of an already sealed witness.

10.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: magma, monoid, group, commutativity, subalgebra, homomorphism. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

10.8 Exact exclusions and limitations

- no imported group, ring or field axioms.
- no negative or irrational carrier requirement.
- no property inferred from a conventional name.
- no completed infinite carrier.

This closes the finite algebraic kernel and its property tests. Particular richer algebraic species require their own generated carriers and independently witnessed additional operations.

10.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:c45fbdd44b40d4d5d007c619fbd6dd03554a43e980396d1d3158b5f9791b34a3`.
- Derivation seal: `sha256:ae92defb9a52942af89734332b791cb1921ae655499b56e808a6250f539d6396`.
- Independent implementation: `sha256:280231864879a04285ec75e17ecf961cb20c43bcf39c228fabf05b3ccd58e033`.
- Independent certificate: `sha256:b41bcba0ab04cfad3eb95cd55ae4286b9ea1c52748ae444b33e2330ec7b52bf4`.
- External validation: `sha256:5443f0a807dfeff488bfd29b7059fbbd086799f212e974d12867670b9f644654`.
- Engine receipt: `sha256:1f08bfaf9f602a3825dc75979070e05f9d51a7d01cdf3c34bab31c06849855bd`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-ALGEBRA-001-1f08bfaf9f602a38.json`.

11. Derivation 6: Exact finite order and conditional lattice structure

Claim identity: `SFT-MATH-ORDER-LATTICE-001`

11.1 Question and necessity

Order is needed for comparison, optimization, proof strength and topology. Retaining incomparability prevents a conventional scalar line from being imported where the Fold structure does not force one.

The exact theorem statement is:

> *An admitted finite order is a held carrier relation whose reflexive, antisymmetric and transitive cells are exhaustively witnessed. Totality is admitted only when every generated pair is comparable. Lower and upper bounds are complete relation selections; meet and join exist only as unique greatest-lower and least-upper witnesses, and monotone maps preserve every retained comparison.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-ALGEBRA-001`.

11.2 Generated grammar and exact boundary

Discrete relations supply the comparison cells; algebra supplies lawful composition; exact identity supplies antisymmetry. Exhausting nine choices forces witnessed partial order and conditional lattice operations.

Generation rule: Generate the complete product of carrier identity, relation provenance, partial-order laws, comparability, bound generation, meet uniqueness, join uniqueness, monotonicity and extra-order status.

Exact grammar boundary: All finite order structures generated from canonical carriers, held pair-cell relations, exhaustive relation closure, complete bound selections and unique extremal witnesses.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:06bc1ab2205eecb2fd980877791e2617c87d47e067be6162d904c903370cfeec`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-ORDER-LATTICE-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`carrier`	`presentation-carrier`	Presentation aliases do not fix exact ordered identity.	`canonical-carrier` - Every ordered form has canonical generated identity.
`relation`	`borrowed-scalar-comparison`	A scalar comparison imports an answer-producing order.	`held-pair-relation` - Each comparison is a retained generated carrier pair.
`order`	`named-order`	A name does not establish reflexivity, antisymmetry and transitivity.	`exhaustive-partial-order-witness` - Every required relation cell and composite is checked.
`comparability`	`assumed-totality`	Partial orders can contain incomparable forms.	`pairwise-totality-witness` - Totality is recorded only when every generated pair is comparable.
`bounds`	`selected-bound`	Selecting a familiar candidate can omit another held condition.	`complete-bound-selection` - Every carrier form is tested against every held target.
`meet`	`assumed-meet`	A lower bound need not be greatest or unique.	`unique-greatest-lower-witness` - Exactly one lower bound is above every other lower bound.
`join`	`assumed-join`	An upper bound need not be least or unique.	`unique-least-upper-witness` - Exactly one upper bound is below every other upper bound.
`maps`	`arbitrary-map`	An arbitrary map can reverse a retained comparison.	`comparison-preserving-map` - Every source relation cell maps to a target relation cell.
`addition`	`extra-order-axiom`	An extra totality or completeness axiom is not structurally forced.	`no-extra-order-axiom` - Only exhaustively witnessed order properties are admitted.

11.3 Unique survivor, minimality and laws

> *The order/lattice kernel is a canonical carrier with an exhaustively witnessed partial-order relation, conditional totality, complete bound selections, uniquely witnessed meet/join and monotone maps.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- partial order requires reflexive, antisymmetric and transitive relation closure.
- incomparability is retained and never forced into a borrowed total scale.
- meet and join are partial operations unless unique extremal witnesses exist.
- a finite lattice is exactly an admitted partial order with meet and join for every generated pair.
- monotonicity is preservation of every registered relation cell.

11.4 Operational witnesses and unfavorable checks

- `partial-order` - The diamond relation passes all three partial-order obligations. Result: PASS.
- `incomparability-retained` - The diamond retains left and right as incomparable. Result: PASS.
- `conditional-totality` - The explicit chain is total while the diamond is not. Result: PASS.
- `meet` - The two incomparable diamond arms have structural One as their unique meet. Result: PASS.
- `join` - The two incomparable diamond arms have whole as their unique join. Result: PASS.
- `monotone-identity` - The canonical identity map preserves every diamond comparison. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Presentation aliases do not fix exact ordered identity. Result: PASS; receipt `sha256:8b99f164faa8fd4a22432325f9d4c0afd5c8882ed01e4d35781cbb3544f51254`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported real-number line; no assumed total order; no lattice completeness without generated meets and joins; no completed infinite chain Result: PASS; receipt `sha256:484dd6f1f271f8054f6f3e5e74696abdbe274c36cb08b223628218b599f5da79`.

11.5 Depth-independent closure

Base: The One carrier has its identity relation and One is both unique lower and upper bound of itself.

Successor: Adding one fresh canonical form generates all comparisons to prior forms. Order closure, comparability, bounds and extremal uniqueness are rechecked only for pairs and composites touching the fresh form.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

11.6 Meaning and scientific consequence

Order is relational, not borrowed from a numerical line. Reflexivity, antisymmetry and transitivity are exhausted over the carrier. Incomparability is positive retained information and is not forced into a total ranking. Bounds are complete selections. A meet or join exists only when the respective greatest-lower or least-upper witness is unique, so lattice structure remains conditional on generated evidence.

11.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: partial order, total order, poset, meet, join, lattice, monotone map. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

11.8 Exact exclusions and limitations

- no imported real-number line.
- no assumed total order.
- no lattice completeness without generated meets and joins.
- no completed infinite chain.

This theorem closes finite generated orders. It does not assert totality, lattice status or completeness for a carrier unless the corresponding finite witness is present.

11.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:fb4627050dca9b64d628dfc773d8de1d30656f40689d1cb8a0ae8dd3105940bd`.
- Derivation seal: `sha256:d4421e8aaf34c239bda0fcacfe2a6aa770fac55ec3afe50c0ba5085d83946527`.
- Independent implementation: `sha256:e0b750be248ae4a9da52c952d8988d01fa236ccdd6a994063a12f297d2088a8f`.
- Independent certificate: `sha256:f1b8cbb06e153828d4410413baf3b6af6f075df5a5f3f297a1f6dee4d1c51bce`.
- External validation: `sha256:f2b2f4bbab7729985711dbdff1cbc25cd7d16f0423a9752f8abdba8caebdd136`.
- Engine receipt: `sha256:d0e20f40782f51cc85c1f572b624570e65b51bbb80135a5e8a6b3cf69f71a5f6`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-ORDER-LATTICE-001-d0e20f40782f51cc.json`.

12. Derivation 7: Exact finite geometry and topology for computation

Claim identity: `SFT-MATH-GEOMETRY-TOPOLOGY-001`

12.1 Question and necessity

Algorithms need incidence, locality, paths, neighborhoods and continuity, but do not require importing a real-coordinate continuum. The finite structural form is the exact computationally required kernel.

The exact theorem statement is:

> *Computational geometry is forced as canonical finite cells plus an acyclic held boundary-incidence relation; dimension is retained boundary depth, adjacency is shared incidence, and distance is a shortest generated path with self-distance represented by empty One rather than numerical zero. Computational topology is a complete generated open-family closed under generated joins and pairwise intersections; continuity is exact inverse-image preservation.*

Admitted dependencies: `SFT-FOUNDATION-PART-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`.

12.2 Generated grammar and exact boundary

Graphs supply paths and incidence; orders supply joins and intersections; exact parts supply held/whole selection. The ten-dimensional census forces the finite incidence-and-open-family kernel.

Generation rule: Generate the complete product of cell carrier, incidence, dimension, adjacency, path distance, open-family closure, connectedness, continuity, deformation and extra-geometric status.

Exact grammar boundary: All finite incidence complexes, path geometries and finite open families generated from canonical forms, held boundary relations, exact path traces, selections, joins and intersections.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:be96c202d0f654cb38088fbd1c4afc210f949ce35b52525cdc7d3e45ad189ed0`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-GEOMETRY-TOPOLOGY-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`cells`	`coordinate-samples`	Samples from a borrowed coordinate field omit structural provenance.	`canonical-generated-cells` - Every cell is an exact canonical generated form.
`incidence`	`assumed-containment`	Assumed containment has no exact face-to-cell witness.	`held-acyclic-boundary` - Every face relation is retained and boundary descent forbids return.
`dimension`	`borrowed-coordinate-count`	A coordinate count imports an ambient space.	`boundary-chain-depth` - Dimension is the longest retained face-descent trace.
`adjacency`	`metric-threshold`	A threshold imports a free scale.	`shared-incidence-witness` - Cells are adjacent exactly when they share a generated face.
`distance`	`floating-metric`	A floating metric imports precision and possibly irrational values.	`shortest-generated-path` - Distance is the complete least-length path trace; identity is empty One.
`opens`	`named-neighborhoods`	Named neighborhoods need not satisfy closure.	`generated-open-family-closure` - Empty One, whole, generated joins and pairwise intersections remain in the family.
`connectedness`	`visual-continuity`	A presentation cannot establish structural connection.	`no-disjoint-open-separation` - No two nonempty disjoint admitted opens jointly exhaust the carrier.
`continuity`	`epsilon-scale`	An imported scale is neither forced nor needed in the finite grammar.	`inverse-open-preservation` - Every target open has an admitted source inverse image.
`deformation`	`untracked-shape-change`	An untracked change can erase incidence.	`reversible-incidence-trace` - Each elementary change retains a lawful reverse and preserves registered incidence invariants.
`addition`	`continuum-or-extra-metric`	A continuum or metric field is outside the generated computational need.	`no-ambient-continuum` - The law uses only finite generated cells, paths and open families.

12.3 Unique survivor, minimality and laws

> *The geometry/topology kernel is finite canonical cells with held acyclic incidence, boundary-depth dimension, shared-face adjacency, exact shortest paths, closed finite open families, inverse-image continuity and reversible incidence-preserving deformation.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- cell dimension is determined by complete acyclic boundary descent.
- path distance retains a shortest generated transition trace and uses empty One at identity.
- finite topological closure is exhaustively decidable from the registered open family.
- continuity is exact inverse-image preservation of every admitted target open.

- geometric equivalence requires a reversible invariant-preserving transformation trace.

12.4 Operational witnesses and unfavorable checks

- `incidence` - Every witness face and cell belongs to the finite carrier. Result: PASS.
- `dimension-depth` - The witness triangle has one more boundary layer than its edge. Result: PASS.
- `adjacency` - The two witness edges are adjacent exactly through their shared q face. Result: PASS.
- `path-distance` - Shortest path retains a-to-b-to-c and identity returns empty One. Result: PASS.
- `finite-topology` - The complete two-point selection family is closed. Result: PASS.
- `continuity` - The identity map preserves every witness open by inverse image. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Samples from a borrowed coordinate field omit structural provenance. Result: PASS; receipt `sha256:602d7acdc1da5b0de53c819a78f381ccac9858cd0bba3688edd642448a2ff375`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ungenerated continuum; no irrational or floating proof coordinate; no free metric threshold; no completed infinite topological family Result: PASS; receipt `sha256:2133cc61a907c35bdc264f3d462617d553589263450769cb5527f812c4268469`.

12.5 Depth-independent closure

Base: One point is a canonical cell; its open family consists of empty One and the whole point carrier.

Successor: Adding one fresh cell generates its possible boundary incidences, shared-face adjacencies and path edges; adding one fresh point generates exactly the open selections needed for closure checks. Every new boundary, path, intersection, join and inverse image is then exhaustively tested.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

12.6 Meaning and scientific consequence

The theorem derives exactly the geometry and topology required by finite computation without an ambient continuum. Geometry is incidence, boundary depth, shared-face adjacency and shortest retained path. Topology is a generated family of opens closed under the operations available at the finite boundary. Continuity is inverse-open preservation, while deformation requires a reversible incidence-preserving trace. Smooth or real-coordinate descriptions may later correspond to approximations but do not select this kernel.

12.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: incidence complex, dimension, adjacency, metric, topology, continuity, homotopy. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

12.8 Exact exclusions and limitations

- no ungenerated continuum.

- no irrational or floating proof coordinate.
- no free metric threshold.
- no completed infinite topological family.

This closes geometry and topology where finite computation requires them. Smooth, differential, measure and continuum correspondences are not premises and require separate generated approximation laws if used.

12.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:36d95c3b1ee446c17b5b4e4f609e10ce1f679f164c720eebd8e482ab68873eaa`.
- Derivation seal: `sha256:7afff182127d9b93f9705212b997801eafd5a3629fd2a8a56753ce674832d0ad`.
- Independent implementation: `sha256:27f1aa8512b811cfef6ccf01fb51ebe844c79f99414e59d4783bc293980bb132`.
- Independent certificate: `sha256:8b8595f6ef14fa36929ff546752015da701e4a49af5213d54f0f6a1d4559ce68`.
- External validation: `sha256:35658ed3991ba98df705a53a75bacc5c7e813bde5f0505e5d20b549ed8cf3dd5`.
- Engine receipt: `sha256:a04f7de0ede6e2348896cf16e59981eebb5391198e498a1ec546b68a4a8d3a6e`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-GEOMETRY-TOPOLOGY-001-a04f7de0ede6e234.json`.

13. Derivation 8: Exact finite probability and statistics from Fold observation

Claim identity: `SFT-MATH-PROBABILITY-STATISTICS-001`

13.1 Question and necessity

Prediction and experiment require uncertainty and statistics, but a superdeterministic Fold model cannot import stochastic dynamics. Exact observation classes supply the required epistemic structure.

The exact theorem statement is:

> *Probability is an exact held-support/whole-support part relation over a complete deterministic generated microstate support; uncertainty records distinctions closed by an observation class and is not a causal random premise. Conditional weight is exact common-refinement restriction, independence is complete pair-cell factorization, and statistics are provenance-retaining finite trace summaries. The empty event is empty One, never numerical zero.*

Admitted dependencies: `SFT-FOUNDATION-PART-EQUIVALENCE-001`, `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`.

13.2 Generated grammar and exact boundary

Parts supply exact held/whole quantities; combinatorics supplies complete support; the measurement boundary isolates data. Nine structural choices force the observation-support account without a prior.

Generation rule: Generate the complete product of support coverage, event formation, weight, uncertainty status, conditioning, independence, statistics, empirical isolation and extra-random-law status.

Exact grammar boundary: All exact finite probability spaces and statistical traces generated from complete microstate support, held selections, observation classes, common refinement, exact parts and sealed empirical measurements.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:0702345917ea32913e5c17e13d86f16d7a885f34e09f8941c7e90be0963f286b`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-PROBABILITY-STATISTICS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`support`	`sampled-or-partial-support`	A partial support silently changes every derived weight.	`complete-generated-support` - Every registered deterministic microstate occurs once.
`event`	`named-outcome`	A name without retained membership does not identify support.	`held-support-selection` - An event is a duplicate-free held selection of exact support.
`weight`	`floating-propensity`	A floating propensity imports precision and a stochastic cause.	`exact-held-whole-part` - Weight is the exact held count over the complete support count.
`uncertainty`	`ontic-random-cause`	A causal random source is neither generated nor required.	`closed-observation-distinction` - Uncertainty records which deterministic microstate distinctions observation does not retain.
`conditioning`	`renormalized-guess`	A guessed renormalization can omit common support.	`exact-common-refinement` - Restrict to the complete held condition and count its exact event intersection.
`independence`	`uncorrelated-assertion`	An assertion does not prove factorized joint support.	`pair-cell-factorization` - Complete joint support is the exact product and event weights multiply by bijection.
`statistics`	`opaque-estimator`	An opaque scalar erases source rows and tie structure.	`provenance-retaining-summary` - Every summary is recomputable from the complete finite trace and retains ties.
`empirical`	`target-informed-fit`	Using target rows during derivation allows them to select the law.	`post-seal-blind-check` - Natural measurements enter only after formal sealing under registered isolation.
`addition`	`extra-random-parameter`	A seed distribution or prior is a free parameter.	`no-extra-random-parameter` - All weights follow from support and observation without stochastic dynamics.

13.3 Unique survivor, minimality and laws

> *The probability/statistics kernel is complete deterministic support, held events, exact held/whole weights, observation-class uncertainty, common-refinement conditioning, pair-cell independence, transparent finite summaries and post-seal empirical checking without random parameters.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- event weight is an exact positive part relation; empty event is empty One.
- conditional weight is exact restriction to a nonempty held condition.
- independence is a bijective complete joint-support factorization.
- uncertainty belongs to an observation partition and does not alter deterministic microstate evolution.
- statistical summaries retain exact provenance and all tied alternatives.

13.4 Operational witnesses and unfavorable checks

- `exact-weight` - Two held states in four complete states have the exact one-over-two part. Result: PASS.
- `empty-event` - The empty event is structural empty One rather than numerical zero. Result: PASS.
- `conditional` - Conditioning on the two a-prefix states retains one selected state as one-over-two. Result: PASS.
- `independence` - The complete two-by-two pair support factorizes exactly. Result: PASS.
- `ties-retained` - An exact modal summary preserves both equally supported alternatives. Result: PASS.

- `observation-classes` - A prefix observation retains two exact two-state classes. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A partial support silently changes every derived weight. Result: PASS; receipt `sha256:646eb932b825e9c344756ec999063ca07df35072a81f697e0b591039c293685a`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ontic stochastic premise; no floating probability in proof evidence; no Bayesian or distributional prior parameter; no target data may select a formal law Result: PASS; receipt `sha256:6f88285aa8b2bb6873871ac8acd897d8ea1b6e120bfad06b20ca3c3f9ad50133`.

13.5 Depth-independent closure

Base: A One-state support has whole certainty; its held-empty event is represented by empty One.

Successor: Adding one fresh microstate extends whole support once and either extends or leaves each held event. Exact parts, observation classes, intersections and summaries update from that retained membership without a prior.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

13.6 Meaning and scientific consequence

Probability is compatible with superdeterminism because it is not installed as a random cause. The complete microstate support is deterministic; an observation closes some distinctions and retains others. Exact event weight is the held-support/whole-support part relation. Conditioning is exact restriction to a held common refinement, and independence is a bijective product-support fact. Statistics remain recomputable from all source rows and preserve ties. Natural data may test a sealed consequence but cannot choose it.

13.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: probability space, conditional probability, independence, random variable, statistic, sampling. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

13.8 Exact exclusions and limitations

- no ontic stochastic premise.
- no floating probability in proof evidence.
- no Bayesian or distributional prior parameter.
- no target data may select a formal law.

This closes exact finite probability and descriptive statistics. Inferential claims about natural data remain empirical claims and require preregistered blind validation through the empirical engine.

13.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:45f28c927c6994a7817ce2fd870441d41ee99e030be73063970674b092e6bab4`.

- Derivation seal: `sha256:9ad2c0aa860b03d0fe11c390e7e1fada396f48565fd89d055f06fec28b32ed67`.
- Independent implementation: `sha256:cdb15cdfaa90a02a1222dcbb0e8000e6ff93fddf67172c0ee26db74d236a17a9`.
- Independent certificate: `sha256:d80527e03481be10b16d4765742d717ebcefaa9f1830e2c1c37d6e40869a62e9`.
- External validation: `sha256:39850e79359dbd3ae8e41fe032474bcf197212f6ac0214cdfd90605919561761`.
- Engine receipt: `sha256:2a86d644881cdacb757327d5aaca01de41c3458c043980c66a16f5084cdfb45b`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-PROBABILITY-STATISTICS-001-2a86d644881cdacb.json`.

14. Derivation 9: Exact finite optimization and retained optima

Claim identity: `SFT-MATH-OPTIMIZATION-001`

14.1 Question and necessity

Search and decision require lawful selection among alternatives. Exact dominance and tie retention prevent a familiar score, tolerance or traversal order from selecting the result.

The exact theorem statement is:

> *An admitted optimization problem is a complete canonical candidate carrier, a source-bound exact feasibility relation and a witnessed preference relation. Solutions are the complete undominated held selection; uniqueness is claimed only for a singleton, while ties and incomparable Pareto alternatives remain explicit. Constraints, decomposition and approximation are exact trace relations without floating objectives, negative penalties or free tuning parameters.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`.

14.2 Generated grammar and exact boundary

Orders supply witnessed preference and incomparability; graphs supply search relations and decomposable interfaces; exact arithmetic supplies part bounds. Nine choices force the complete undominated kernel.

Generation rule: Generate the complete product of candidate coverage, feasibility, preference, optimum selection, tie retention, constraint provenance, decomposition, approximation and extra-objective status.

Exact grammar boundary: All finite optimization instances generated from canonical candidates, exact predicates, held preference relations, complete dominance checks, interface-preserving decomposition and exact positive part bounds.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:a15ea2f582138e72734b4421a5d69ae92089f444f70597f7539532587a30ddbe`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-OPTIMIZATION-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`candidates`	`sampled-candidates`	Sampling cannot prove that a better generated candidate was not omitted.	`complete-canonical-candidates` - Every registered candidate form occurs once.
`feasibility`	`implicit-constraint`	An implicit constraint has no candidate-level check trace.	`source-bound-verdicts` - Every candidate receives one exact verdict from each registered constraint.
`preference`	`floating-score`	A floating score imports scale and precision choices.	`witnessed-order-relation` - Every preference is an exact retained candidate pair.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`solution`	`first-found`	First-found depends on traversal presentation.	`complete-undominated-selection` - Every feasible candidate is compared and all undominated forms are retained.
`ties`	`silent-tie-break`	A silent tie-break adds a free preference.	`retained-equivalence-class` - All jointly optimal or incomparable surviving forms remain explicit.
`constraints`	`penalty-parameter`	A tunable penalty is a free parameter and can select the answer.	`exact-membership-predicate` - Each constraint is a source-bound exact allowed-form relation.
`decomposition`	`assumed-separability`	Shared constraints can invalidate separate optima.	`exact-interface-factorization` - Subproblems compose only when all shared interface relations are retained.
`approximation`	`decimal-error`	A decimal error imports rounding and a scale.	`exact-positive-part-bound` - The candidate/optimum relation is a generated exact part certificate.
`addition`	`extra-objective-or-tie-rule`	An added score or tie rule is not forced by the instance.	`no-extra-objective` - Only registered feasibility and preference relations select solutions.

14.3 Unique survivor, minimality and laws

> *The optimization kernel is complete canonical candidates, source-bound feasibility, exact preference, complete undominated selection, retained ties, exact constraints and interfaces, and positive-part bounds without an added objective.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- optimization may eliminate a candidate only with a retained feasible dominator witness.
- all undominated candidates survive and uniqueness is a separately checked singleton property.
- Pareto comparison retains incomparability across exact criteria.
- decomposition is lawful only with complete shared-interface factorization.
- approximation is an exact generated part relation rather than a rounded scalar.

14.4 Operational witnesses and unfavorable checks

- `feasible-selection` - The complete verdict relation retains exactly a and b. Result: PASS.
- `unique-optimum` - A witnessed a-over-b relation leaves a as the singleton optimum. Result: PASS.
- `ties-retained` - With no forced preference both feasible forms survive explicitly. Result: PASS.
- `constraint-membership` - The witness candidate is checked against every exact allowed set. Result: PASS.
- `pareto-incomparability` - Opposed exact criteria retain both incomparable forms. Result: PASS.
- `exact-bound` - Three generated cells against a two-cell optimum produce exact two-over-three. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Sampling cannot prove that a better generated candidate was not omitted. Result: PASS; receipt `sha256:4fca7bb33759009f032e930b863d7c4fce12d4ce19f11686801c02efdbba245f`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt

`sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.

- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no floating objective or tolerance; no negative penalty term; no tunable regularizer or tie-break parameter; no unenumerated candidate domain Result: PASS; receipt `sha256:40faa5eb68ab3bcadce5ee1937d3b89e679a8f1fa2ba12fca91be6318f92420c`.

14.5 Depth-independent closure

Base: A One-candidate feasible carrier has that candidate as its unique complete undominated selection.

Successor: Adding one fresh candidate generates its feasibility verdicts and every preference pair with prior feasible forms. It is either eliminated by a retained dominator, eliminates witnessed dominated forms, or joins the retained optimal class; no prior optimum is silently discarded.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

14.6 Meaning and scientific consequence

Optimization is forced as elimination by exact witness rather than maximization of an imported floating score. Every candidate and feasibility verdict is present. A candidate is removed only when a retained feasible dominator exists. The complete undominated selection is the solution; singleton status alone licenses uniqueness. Multiple optimal or Pareto-incomparable forms remain visible rather than being broken by an undeclared convention.

14.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: feasible set, objective, optimum, Pareto frontier, constraint, approximation ratio. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

14.8 Exact exclusions and limitations

- no floating objective or tolerance.
- no negative penalty term.
- no tunable regularizer or tie-break parameter.
- no unenumerated candidate domain.

This closes exact finite optimization structure. Complexity of finding the complete frontier and laws for particular objective families belong to the algorithms and complexity branches.

14.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:435743b4bc243523b8c405514232bdc88a02fd48764040fa26c914085ae0f813`.
- Derivation seal: `sha256:11d1d87d66d732bca3efc296de283e8e5e11ed9d291dd083c10f86d6801277e9`.
- Independent implementation: `sha256:19c3f41f797dbf136a28392518b1ef3ff431214a314099a6080c8c5ea04edd95`.
- Independent certificate: `sha256:9cb0790543dac4fb7c8ff8bce76f783ccb1c96e84176f8cf59c203060920bfa4`.
- External validation: `sha256:7f6345c8109261bf2f63108a2613767ff27fae5f9d411bace11a33cc9046ddb3`.
- Engine receipt: `sha256:3385414ceea0fd4212111e42bd6ce7d6d35556b7c97d66563813d5b8d1531849`.

- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-OPTIMIZATION-001-3385414ceea0fd42.json`.

15. Derivation 10: Exact generated finite dynamical systems

Claim identity: `SFT-MATH-DYNAMICAL-SYSTEMS-001`

15.1 Question and necessity

Computation and natural modelling both require change through state. Exact transitions expose the laws of time, recurrence and information loss without importing continuous equations or stochastic causes.

The exact theorem statement is:

> *An admitted dynamical system is a complete canonical state carrier with a held transition relation. A trajectory retains every state and transition; time is the positive transition trace and the identity trajectory is empty One. Fixed forms, returns, recurrence, basins, reversibility and stability are admitted only through exhaustive generated witnesses. Merged predecessors remain recoverable only when their exact records are retained.*

Admitted dependencies: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`.

15.2 Generated grammar and exact boundary

Graph paths supply evolution; optimization supplies complete invariant and basin selection; canonical forms supply state identity. Ten choices force the finite transition-and-record dynamics kernel.

Generation rule: Generate the complete product of state coverage, transition provenance, trajectory trace, time, fixed forms, return, basin/recurrence, reversibility, stability and extra-dynamic status.

Exact grammar boundary: All finite dynamics generated from canonical state carriers, held transition relations, complete paths, return traces, reachability basins, retained predecessor records and registered finite perturbation classes.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:4a398993d220a557338f9d3f2e0592838df6bccbb7f1d20274106c1a406ad684`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-DYNAMICAL-SYSTEMS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`states`	`partial-or-aliased-state`	Omitted or aliased states change possible evolution.	`complete-canonical-states` - Every generated state occurs once with canonical identity.
`transitions`	`opaque-update-rule`	An opaque update hides which relation cells exist.	`held-state-relation` - Every allowed state transition is a retained generated pair.
`trajectory`	`endpoint-only`	Endpoints erase intervening states and cannot prove reachability.	`complete-transition-trace` - Every adjacent transition is retained in order.
`time`	`continuous-clock-parameter`	A free real clock imports scale and continuum.	`positive-transition-count` - Elapsed time is the exact trace of generated transitions; identity is empty One.
`fixed`	`visual-sameness`	Presentation similarity does not prove dynamical identity.	`retained-identity-transition` - A fixed form has an explicit self relation.
`return`	`periodic-assertion`	A period name without a trace omits return structure.	`explicit-first-return-trace` - A retained nonempty path returns to a prior canonical state.
`basin`	`sampled-neighborhood`	Sampling cannot establish reachability for the registered perturbation class.	`complete-reachability-selection` - Every registered state is exhaustively tested for reachability to the invariant form.
`reversibility`	`infer-unrecorded-predecessor`	A merged image does not identify which predecessor occurred.	`retained-predecessor-record` - The exact source-to-image distinction is held alongside the step.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`stability`	`epsilon-parameter`	A chosen epsilon imports a free scale.	`registered-perturbation-closure` - Every explicitly generated perturbation form satisfies the invariant reachability condition.
`addition`	`extra-force-or-random-law`	An added force, differential equation or random update is not supplied by the state relation.	`no-extra-dynamic-law` - All properties follow from exact finite transition structure.

15.3 Unique survivor, minimality and laws

> *The dynamical kernel is complete canonical states, held transitions, complete trajectories, transition-count time, witnessed fixed/return/basin structure, retained-record reversibility and registered perturbation stability.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- trajectory composition is lawful exactly at an equal canonical interface state.
- time is a positive transition trace and identity duration is empty One.
- fixed, cyclic and recurrent behavior require explicit transition witnesses.
- many-to-one evolution loses predecessor distinction unless a source record is retained.
- stability is quantified only over a complete registered finite perturbation class.

15.4 Operational witnesses and unfavorable checks

- `well-formed` - Every witness transition remains in the complete state carrier. Result: PASS.
- `trajectory` - The a-to-b-to-terminal trace retains both allowed transitions. Result: PASS.
- `time` - Two transitions are retained exactly and an identity trajectory returns empty One. Result: PASS.
- `fixed-form` - The terminal witness has an explicit identity transition. Result: PASS.
- `first-return` - A repeated canonical state yields its exact first return trace. Result: PASS.
- `predecessor-retention` - Both merged predecessors remain distinct and an exact retained ledger reverses them. Result: PASS.
- `registered-stability` - Every registered witness perturbation reaches terminal. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omitted or aliased states change possible evolution. Result: PASS; receipt `sha256:644913b363fe507365e0c82e18a6847f33d91874acd59dea32a28f49e1ed33b4`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no continuous real time premise; no differential equation imported as a law; no stochastic transition cause; no unrecorded predecessor reconstruction Result: PASS; receipt `sha256:5c5f8dd163823e87bd822f6397adc6f8723ee60d3aa2defdfb5cc33a3ff76649`.

15.5 Depth-independent closure

Base: One state supplies its identity trajectory; no transition time or predecessor ambiguity is introduced.

Successor: Adding one transition appends one exact state pair to every extended trajectory. Fixed, return, basin, predecessor and stability obligations change only for paths that use the new relation and are rechecked completely.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

15.6 Meaning and scientific consequence

Dynamics is state-transition structure before any differential equation. Time is the exact positive trace of transitions, with the identity trajectory represented structurally by empty One. Fixed forms and cycles require explicit returns. Basins and finite stability classes are exhaustive reachability selections. If distinct predecessors merge, reversal requires the precise predecessor record; the image alone cannot recreate a distinction it does not contain.

15.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: state space, trajectory, fixed point, cycle, attractor, reversibility, stability. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

15.8 Exact exclusions and limitations

- no continuous real time premise.
- no differential equation imported as a law.
- no stochastic transition cause.
- no unrecorded predecessor reconstruction.

This closes exact finite dynamical systems. Continuum differential models may later appear only as measured or finite approximation correspondences, never as premises of this theorem.

15.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:366c10101f60725e0e8bc8f29c974d4a7707b564125c8b7608d655e2092ad5a2`.
- Derivation seal: `sha256:972a41752e953bd0b4d12f5c3c7d8e68c2aa2107a5aacaf9e330a2278b1a3a51`.
- Independent implementation: `sha256:6c685056fb0e01ef2133db2ab947796fbade3bd1e246c56479bbac46055751c0`.
- Independent certificate: `sha256:c8e806f9f4e6b74927f0bfe79e2c5b4351113323176e59ecde786d4121290172`.
- External validation: `sha256:f93adcb26bb1c7baeec979059b40325d370b5f2f3d5f50cd6885a5159b10362f`.
- Engine receipt: `sha256:3e0e44b65fbe660fe2a29b9927b1870a40da25c4b0c786e2adaa3df9a2f7cabe`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-DYNAMICAL-SYSTEMS-001-3e0e44b65fbe660f.json`.

16. Derivation 11: Exact finite logic, inference and proof boundaries

Claim identity: `SFT-MATH-LOGIC-PROOF-001`

16.1 Question and necessity

Every branch depends on distinguishable valid and invalid derivations. The proof law makes admission mechanical while preventing conventional logical axioms or institutional authority from entering silently.

The exact theorem statement is:

> *An admitted proposition is a canonical distinguished form with a held orientation; denial is the retained complementary orientation, not a negative number. Conjunction retains joint proofs, disjunction retains a chosen generated branch and its alternatives, implication is a registered premise-to-conclusion transition, and a proof is a complete dependency trace of accepted rules. Consistency, soundness and completeness are machine-checkable only relative to the exact registered finite grammar and rule set.*

Admitted dependencies: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-ORDER-LATTICE-001`.

16.2 Generated grammar and exact boundary

Canonical forms supply proposition identity; held fibres supply opposed orientations; discrete relations supply rules; order supplies proof-strength relations. Ten choices force the replayable finite proof kernel.

Generation rule: Generate the complete product of proposition identity, denial, conjunction, disjunction, implication, inference provenance, consistency, soundness, grammar completeness and extra-logical status.

Exact grammar boundary: All propositions and finite proofs generated from canonical forms, two held orientations, joint and alternative constructions, registered inference transitions and complete dependency traces.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:c3171696c5a96a3fa85bb809773cca0fc438e1d7196b6657aa3f94ce06e1e622`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-LOGIC-PROOF-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`proposition`	`presentation-sentence`	A sentence presentation can alias or hide the judged form.	`canonical-distinguished-form` - The exact generated form is the proposition identity.
`denial`	`negative-truth-value`	A negative magnitude is outside the admitted domain.	`complementary-held-orientation` - Denial retains the same form with the other exact fibre orientation.
`conjunction`	`single-side-proof`	One side cannot establish the joint claim.	`joint-complete-proof` - Both constituent proof traces are retained together.
`disjunction`	`unidentified-choice`	An unidentified choice erases which alternative is supported.	`held-alternative-with-support` - The held branch and complete generated alternative family are retained.
`implication`	`truth-table-import`	A borrowed table installs conventional logical values.	`proof-carrying-transition` - A registered rule maps exact premises to an exact conclusion.
`inference`	`conclusion-only`	A bare conclusion erases its premises and rules.	`complete-dependency-trace` - Every step retains available premises, rule identity and conclusion.
`consistency`	`assumed-consistent`	An assertion does not search for opposed admitted orientations.	`no-opposed-admission` - No form and its complementary orientation are jointly admitted.
`soundness`	`authority-or-name`	Authority cannot replace validation against registered rules.	`rule-and-witness-preservation` - Every accepted step matches a registered source-bound rule and preserves its witnesses.
`completeness`	`unbounded-global-claim`	A finite proof kernel cannot establish an ungenerated universal domain.	`registered-grammar-exhaustion` - Every proposition in the exact generated grammar has its declared decision evidence.
`addition`	`extra-logical-axiom`	An extra axiom is not supplied by the canonical form and rule traces.	`no-extra-logical-axiom` - All admitted inference is registered and witnessed.

16.3 Unique survivor, minimality and laws

> *The logic/proof kernel is canonical distinguished propositions, complementary held denial, joint and held alternative proof forms, proof-carrying implication, complete inference traces, explicit consistency and soundness, and completeness only over the exhausted registered grammar.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- denial changes held orientation while preserving proposition identity.
- conjunction and disjunction retain their complete source proof structure.
- an implication is admitted through a registered proof-carrying transition.
- a valid proof is replayable step by step from its premises and rule registry.
- completeness is always indexed by an explicit generated grammar boundary.

16.4 Operational witnesses and unfavorable checks

- `held-denial` - Denial preserves P while changing only its held orientation. Result: PASS.
- `joint-proof` - Conjunction retains both exact constituent proofs. Result: PASS.
- `held-disjunction` - Disjunction retains the supported branch and both alternatives. Result: PASS.
- `proof-replay` - The registered P-to-Q step replays from its available premise. Result: PASS.
- `tampered-rule-control` - Changing the rule identity invalidates the same conclusion trace. Result: PASS.
- `consistency` - P and Q are consistent, while P and its held denial are not. Result: PASS.
- `bounded-completeness` - Both registered proposition forms have explicit held decisions. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A sentence presentation can alias or hide the judged form. Result: PASS; receipt `sha256:6d035b68865a940e61e80880013f484689afa3fbfab68a65b3604a51f203a98c`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported truth-value arithmetic; no negative truth magnitude; no authority-based proof admission; no global completeness beyond a generated grammar Result: PASS; receipt `sha256:5dff4ae767ffc5d54e6bf8debb3ec71db9285d631376f5ba4b148be47b1cda93`.

16.5 Depth-independent closure

Base: One proposition form and its two orientations provide the first decidable grammar cell.

Successor: Adding one proposition or rule generates its complementary orientation, joint and alternative combinations, and all new premise tuples. Consistency and proof replay are rechecked for precisely those new forms and steps.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

16.6 Meaning and scientific consequence

Logic is reconstructed as distinguished canonical forms and proof-carrying transitions. Denial changes a held orientation; it is not negative truth magnitude. Conjunction retains both supporting traces and disjunction retains the supported alternative and the full alternative family. A proof is valid only by stepwise replay against registered rules. Soundness and completeness are therefore exact but explicitly relative to the generated grammar; no unbounded global completeness claim is smuggled into the result.

16.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: proposition, negation, conjunction, disjunction, implication, proof, soundness, completeness. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

16.8 Exact exclusions and limitations

- no imported truth-value arithmetic.
- no negative truth magnitude.
- no authority-based proof admission.
- no global completeness beyond a generated grammar.

This closes finite generated proof theory. Computability, self-reference, undecidability and incompleteness boundaries require the later formal-computation branch and are not claimed here.

16.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:5b549044ca00a139f499155286a06ed09af0735799cf371650de8fd8d336a66d`.
- Derivation seal: `sha256:849294dd48e23a2aefd379804094c01132234adad33b937a3164b3c73f8230bf`.
- Independent implementation: `sha256:636f912fdacfb994061d4018dcbfa82b20c5f99693af0f3f5e9be0ee84b13a42`.
- Independent certificate: `sha256:dfeb338769fcf4b551277891fd7c5576f15ce79e533addbb15e6e4eed87a288a`.
- External validation: `sha256:d3cc342c7a7bf4b50450ae303b15dd250328f761236f869150456ba1550c3cb9`.
- Engine receipt: `sha256:3d366d57b26d002cb27a16e74b022a0856e133c616b7da6a83233c84bdd50813`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-LOGIC-PROOF-001-3d366d57b26d002c.json`.

17. Derivation 12: Forced category, type and compositional structure

Claim identity: `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`

17.1 Question and necessity

Every later science must compose local derivations without losing interfaces or provenance. The forced path kernel supplies this language while keeping category and type concepts downstream of Fold structure.

The exact theorem statement is:

> *Canonical Fold forms act as objects and proof-carrying transitions as arrows. Identity is the empty-One return, composition is forced exactly by equal interfaces, and association follows from canonical complete path flattening. Types are source-bound predicates over generated carriers; products are complete joint pair support and sums*

> *retain held alternative labels. Functors preserve interfaces, identity and composition, while naturality is equality of the two complete composite traces.*

Admitted dependencies: `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ALGEBRA-001`, `SFT-MATH-LOGIC-PROOF-001`.

17.2 Generated grammar and exact boundary

Graphs supply paths, algebra supplies operation association, and logic supplies proof-carrying transitions. Ten choices force exact interfaces, identities, flat composition, types and preservation structures.

Generation rule: Generate the complete product of objects, arrows, identity, composition, association, types, product/sum structure, functoriality, naturality and extra-composition status.

Exact grammar boundary: All finite compositional structures generated from canonical objects, proof-carrying arrows, exact interface matching, empty-One returns, flattened paths, source-bound type predicates and held joint/alternative forms.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:61559f26b9f1269604ea49148c48493a1565fa04b140255215af667d0a20aa20`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-CATEGORY-TYPE-COMPOSITION-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`objects`	`presentation-objects`	Presentation names can alias distinct structures.	`canonical-Fold-objects` - Every object is an exact canonical generated form.
`arrows`	`endpoint-only-arrow`	Endpoints alone erase the transformation proof.	`proof-carrying-transition` - An arrow retains source, target and complete transition trace.
`identity`	`assumed-neutral-map`	A named neutral map has no return witness.	`empty-One-return` - The zero-step structural return preserves the same object without numerical zero.
`composition`	`presentation-concatenation`	Textual concatenation can join unequal interfaces.	`exact-interface-match` - The first target and second source must be the same canonical object.
`association`	`borrowed-category-axiom`	Importing an axiom would reverse the derivation order.	`canonical-path-flattening` - Both groupings retain the same ordered elementary transition trace.
`types`	`informal-class-name`	A class name need not decide carrier membership.	`source-bound-form-predicate` - Every generated carrier form receives one exact membership verdict.
`constructors`	`borrowed-set-constructor`	A borrowed constructor imports its ambient universe.	`joint-pairs-and-held-alternatives` - Products are complete pair support; sums retain left/right held provenance.
`functor`	`arbitrary-renaming`	Renaming can break interfaces and composites.	`identity-and-composition-preservation` - Object and arrow maps retain sources, targets, returns and composites.
`naturality`	`diagram-by-appearance`	A drawn square cannot prove both routes equal.	`equal-complete-paths` - Both interface-compatible composite routes have the same canonical trace.
`addition`	`extra-composition-axiom`	An extra universal property is not supplied by the registered traces.	`no-extra-composition-axiom` - Only generated interfaces, paths and preservation witnesses are admitted.

17.3 Unique survivor, minimality and laws

> *The compositional kernel is canonical Fold objects, proof-carrying arrows, empty-One identity, exact-interface composition, path-flattening association, source-bound types, joint products, held sums, preserving functors and equal-path naturality.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- identity preserves source and target and contributes no nonidentity transition.

- composition exists exactly at a canonical interface match.
- associativity is equality of flattened complete transition traces.
- products preserve both coordinates and sums preserve the held source alternative.
- functoriality and naturality are finite replayable preservation witnesses.

17.4 Operational witnesses and unfavorable checks

- `interface-composition` - f then g composes from A to C with both traces retained. Result: PASS.
- `identity` - Empty-One return preserves the interface and adds no elementary transition. Result: PASS.
- `association` - Both groupings of f, g and h flatten to the same complete path. Result: PASS.
- `type-predicate` - The witness predicate retains exactly its admitted carrier forms. Result: PASS.
- `product-sum` - Joint products and held sums retain both source coordinates. Result: PASS.
- `functor-interface` - The identity transport preserves the witness arrow interface. Result: PASS.
- `naturality-path` - Two independently assembled routes with the same elementary trace commute. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Presentation names can alias distinct structures. Result: PASS; receipt `sha256:0b1412200b24f35693db81fac968726a12b5f6d4a14d7866a33dfdfa597ffe8a`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported category axioms; no ungenerated universe of all objects; no impredicative type constructor; no universal property without an exhaustive finite witness Result: PASS; receipt `sha256:b3ab5d7231f463ee56b4759ebf5e5a9ec0c2a2f15ff312709eed57a7ba33281d`.

17.5 Depth-independent closure

Base: One canonical object supplies its empty-One identity arrow and the first singleton type.

Successor: Adding one object or arrow generates its identities, every exact interface-compatible composite, product and sum cells, type verdicts and preservation squares. Closure follows by appending its trace to prior flat paths.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

17.6 Meaning and scientific consequence

Composition is the final mathematical unifier of the branch. Canonical forms are objects and complete proof-carrying transitions are arrows. Equal interfaces force lawful composition; empty-One return supplies identity; and associativity is equality of flattened elementary paths. Types are exact predicates over a registered carrier. Products retain joint coordinates, sums retain held origin, and functorial or natural structure is admitted by replayable path-preservation witnesses.

17.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: category, object, morphism, identity, composition, type, product, sum, functor, natural transformation. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

17.8 Exact exclusions and limitations

- no imported category axioms.
- no ungenerated universe of all objects.
- no impredicative type constructor.
- no universal property without an exhaustive finite witness.

This closes the finite compositional kernel. Any stronger universal construction must be separately generated and exhaustively witness its claimed mapping property inside an explicit grammar.

17.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:8962e0d82e8291e69ae4b1477518316402aea5b3e663f08b418e88bec287d135`.
- Derivation seal: `sha256:756d7d6dce776fc1fb4b3906e2203e4e2da4e1ded6cf733eb041334eb7cb5c02`.
- Independent implementation: `sha256:ced0eda6a320565b205b93618d604af7b494f3ad001044d71c87df0f4c281f90`.
- Independent certificate: `sha256:1d380a6b0d0ce9d32cfdff52fae121315257e3e561da9379ab5509c835680dad`.
- External validation: `sha256:fc7b4ce671b53f4d62da07705c4fc5d92b0195d5218a30f2a40a773a2fb7c981`.
- Engine receipt: `sha256:47873a83a90e29fa22d065ff13aa0152edb6212414e35cc0435e8eff2c9ea5a8`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-CATEGORY-TYPE-COMPOSITION-001-47873a83a90e29fa.json`.

18. Derivation 13: Exact ratio, separation, reciprocal, threshold and cycle composition

Claim identity: `SFT-MATH-EXACT-RELATIONS-002`

18.1 Question and necessity

V1/V2 separately registered ratio, separation, threshold, beat and joint-cycle laws that the broad arithmetic claim did not enumerate atomically.

The exact theorem statement is:

> *Exact generated relations force positive ratios, shorter-part separation, reciprocal return to the One, the unique m-fold holding complement, local Fold separation transport, telescoping relative views, and least-common-return composition for every supplied finite family of exact cycles.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`.

18.2 Generated grammar and exact boundary

Common refinement forces exact ratios; held orientation preserves a positive remainder; Fold supplies its own transport count; exhaustive returns force the least common return.

Generation rule: Exhaust the provenance, relation, orientation, threshold, transport, composition and generality choices.

Exact grammar boundary: Exact positive generated parts, held orientation and supplied positive finite cycle families.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:8ea91d0dcba9fca561d79bb2a7cc1dcc43af1caa1861db2123c98f81a5366bc0`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-EXACT-RELATIONS-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`provenance`	`borrowed-field`	A borrowed field imports the result.	`generated-parts` - Both terms retain generated part provenance.
`relation`	`rounded-scalar`	Rounding destroys exact identity.	`exact-ratio` - Common refinement forces the positive ratio.
`orientation`	`signed-gap`	Signed magnitude is outside the domain.	`held-shorter-gap` - A held side records orientation and the shorter positive part records separation.
`threshold`	`selected-angle`	A selected angle adds a parameter.	`unique-complement` - The m-fold complement is the unique (m less One)-of-m part.
`transport`	`fitted-rate`	A fitted rate is not derived.	`fold-factor` - One Fold transports an uncast local gap by the generated fibre count.
`composition`	`sampled-return`	Sampling cannot prove joint return.	`least-common-return` - The least common multiple is the first shared return.
`generality`	`fixed-table`	A fixed table has no successor certificate.	`finite-family-induction` - Adding one cycle composes its period by one further least-common return.
`addition`	`extra-rule`	An extra rule violates zero-parameter closure.	`no-extra-rule` - Only admitted part and cycle operations occur.

18.3 Unique survivor, minimality and laws

> *The unique exact relation kernel is generated-part ratio, held shorter separation, unique complement threshold, Fold-factor transport and least-common-return composition with finite-family induction.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- a positive ratio composed with its reciprocal returns the One.
- relative views telescope by exact ratio composition.
- the first common return of finite periods is their iterated least common multiple.
- count times the corresponding equal part reconstructs the One.

18.4 Operational witnesses and unfavorable checks

- `reciprocal` - The ratio three-of-two composed with two-of-three returns the One. Result: PASS.
- `count-measure` - Eight equal one-of-eight parts reconstruct the One. Result: PASS.
- `telescoping` - The exact views five-of-three and three-of-two compose to five-of-two. Result: PASS.
- `threshold` - The binary holding complement is the unique one-of-two part. Result: PASS.
- `joint-return` - Periods two, three and four first return together at twelve. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A borrowed field imports the result. Result: PASS; receipt `sha256:f4a735a031ef37a2a648b155e7a8b2f3c18457f33a93f23ab569081113e28c09`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt

sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: PASS; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

18.5 Depth-independent closure

Base: One exact cycle returns after its own generated period.

Successor: Composing one more positive finite cycle replaces the current joint period by its least common return with the new period.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

18.6 Meaning and scientific consequence

This reconstruction joins the relation laws that V1/V2 stated separately. Ratio is a common-refinement identity; separation is a positive shorter part with held orientation; reciprocal composition returns the One; the m-fold threshold is its unique complement; and finite cycles return jointly at the least common period. The result is stronger than a table because adding one cycle extends the same return certificate.

18.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: ratio, distance, reciprocal, threshold, least common multiple, beat period. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

18.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

The cycle theorem applies to supplied positive finite exact periods and never constructs a completed infinite family.

18.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:57cdb2201f91bc9edd24a9a610305d92426d3d47ee69b2f4c70283343738c3d2`.
- Derivation seal: `sha256:516d208341bde6e161b32070c8051f089f2d0f11ccc3748c75680c6c4e4d1dc6`.
- Independent implementation: `sha256:9b3d9ce6df6053f3f2a7b6935480a45bd0a5daaca31c76fb3beaeadba9f79530`.
- Independent certificate: `sha256:2ac8d699989a021d04fe3f921d3858795423d28863d83fcfb06542408a254a36`.
- External validation: `sha256:3d32599f5a7999ee642da792bdd7e0fbde43faf5e6a8a73468ed802b380396b1`.
- Engine receipt: `sha256:8141546a120b73aa7cac0cd843728bdf3122d1bf3c0048c132760c82fd103238`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-EXACT-RELATIONS-002-8141546a120b73aa.json`.

19. Derivation 14: Fold orbit number theory and exact prime-spectrum structure

Claim identity: `SFT-MATH-ORBIT-NUMBER-THEORY-002`

19.1 Question and necessity

The orbit/order identity, prime-period division, cyclotomic tiling and finite Artin census were explicit prior Mathematics results.

The exact theorem statement is:

> *For every reduced part over an odd positive finite denominator, Fold orbit length is exactly the multiplicative order of the binary generator; reduced residues partition into equal cyclotomic orbits, prime periods divide one-less-than-prime, and a power-of-two factor supplies exactly its counted transient depth.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`.

19.2 Generated grammar and exact boundary

Fold return of p/q requires a first k with p times 2^k congruent to p; reduction cancels p and leaves exactly the definition of the binary multiplicative order.

Generation rule: Exhaust denominator domain, reduction, transition, period, partition, transient, generality and extra-rule choices.

Exact grammar boundary: All supplied positive finite reduced rational parts, with odd cores and counted binary factors.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:e0aad69a9214fa26f8480e30923fd5a5399c6cbcc15ac6ec0a5ecad661f97921`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-ORBIT-NUMBER-THEORY-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`domain`	`arbitrary-real`	An arbitrary real imports infinite information.	`finite-rational` - Generated parts have finite whole numerator and denominator traces.
`reduction`	`untracked-factor`	An untracked common factor duplicates state identity.	`lowest-terms` - The reduced denominator fixes the orbit class.
`transition`	`sampled-map`	A sampled map is not the Fold.	`double-and-cast` - Fold doubles the residue and casts complete denominators.
`period`	`observed-cycle`	An observed cycle has no arithmetic identity.	`multiplicative-order` - Return is exactly the first binary power congruent to the One residue.
`partition`	`overlap-cover`	Overlapping cycles duplicate residues.	`cyclotomic-tiling` - Disjoint equal-length orbits tile all reduced residues.
`transient`	`eternal-even-core`	A binary factor cannot remain under repeated Fold.	`counted-binary-depth` - Each Fold removes one binary denominator factor before the odd cycle.
`generality`	`bounded-prime-list`	A bounded list is not the theorem.	`definition-identity` - The Fold return predicate and multiplicative-order predicate are the same remainder identity for every supplied denominator.
`addition`	`extra-number-law`	An imported number-theory theorem would select the answer.	`no-extra-rule` - Only Fold, whole remainder and exact counting occur.

19.3 Unique survivor, minimality and laws

> *Fold return and binary multiplicative order are one exact remainder identity; odd reduced residues tile into cyclotomic orbits and each binary denominator factor contributes one transient Fold.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- orbit length equals the multiplicative order of the binary generator modulo the reduced odd denominator.
- the reduced-residue count is the product of orbit length and orbit count.
- for a prime denominator the period divides one less than that prime.
- binary valuation is the exact transient depth to the odd recurrent core.

19.4 Operational witnesses and unfavorable checks

- `orders` - The independently counted orders for denominators three, five, seven, nine and eleven are two, four, three, six and ten. Result: PASS.
- `all-reduced-ranks` - Every reduced rank through odd denominator ninety-nine has orbit length equal to its denominator order. Result: PASS.
- `prime-division` - Each tested prime period divides one less than the prime. Result: PASS.
- `orbit-tiling` - Reduced residues for denominator seventy-three tile into eight disjoint orbits of length nine. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An arbitrary real imports infinite information. Result: PASS; receipt `sha256:fea1540fb44cc68fbfc743b66ed93b6afee3da4b0032929ab3b8cc8ce3b8c976`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: PASS; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

19.5 Depth-independent closure

Base: The first reduced odd denominator above the One executes one complete finite orbit.

Successor: For any next supplied odd denominator the finite residue permutation independently partitions into cycles; adjoining one binary factor prepends exactly one transient Fold.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

19.6 Meaning and scientific consequence

Number theory appears here as Fold dynamics, not as imported doctrine. Returning p/q after k Folds and requiring the binary generator to have order k modulo the reduced odd denominator are literally the same remainder identity. That identity forces the cyclotomic partition, prime-period divisor law and exact binary transient depth. The Artin and asymptotic prime questions remain visible boundaries rather than inherited claims.

19.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: multiplicative order, Fermat period, cyclotomic coset, primitive root, prime spectrum. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

19.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

The theorem identifies the exact orbit skeleton. It does not claim an asymptotic prime-counting law or prove Artin's conjecture.

19.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:beec43280b54374b36d3da29ec4ca906cd28cee4decfcb7cc49d6435fbc499f4`.
- Derivation seal: `sha256:8c00984dae257d972ebc6ed4e059a3c958f43e2c29b39f9775f78883dd376086`.
- Independent implementation: `sha256:66273410b8c810940888e97f3fd03d86752ca8ca1f450ae3cf1fcfa9b5af5142`.
- Independent certificate: `sha256:b47923bcf2b3c784ecd0689286d35a111a91faa6f86ec1c1c8971414772bcc6d`.
- External validation: `sha256:1ff8387a5a11bb9bd5bf273921e948e63554192e6cd284b33ef50734a2254174`.
- Engine receipt: `sha256:8725bce36fe77923288b99b1a978ef3bf84cac9ffe1b172fd67bd221527d19ff`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-ORBIT-NUMBER-THEORY-002-8725bce36fe77923.json`.

20. Derivation 15: Potential infinity, exact refinement and the continuum boundary

Claim identity: `SFT-MATH-LIMIT-CONTINUUM-002`

20.1 Question and necessity

V1/V2 registered continuum-limit, potential-infinite, continuum-hypothesis and finite-inventory statements that require one exact object boundary.

The exact theorem statement is:

> *Every generated rung is a finite exact positive part with a counted successor; refinement can continue without a greatest finite depth, exact rational sequences can carry monotone bounds and convergence certificates, and no completed infinity, unbounded denominator or continuum is admitted as a proof object.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-ORDER-LATTICE-001`.

20.2 Generated grammar and exact boundary

Fold assembly supplies a base and successor but never an already-completed unbounded collection. Exact convergence therefore lives in replayable rational bounds, not an imported limit value.

Generation rule: Exhaust object, successor, refinement, convergence, continuum, cardinality, generality and extra-rule choices.

Exact grammar boundary: Generated positive finite depth traces and exact rational bounds at every supplied finite stage.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:b146824353456f9812e707aca84e39c4f1c7aaf50beda6e2d64bab7a2f94090b`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-LIMIT-CONTINUUM-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`object`	`completed-infinity`	A completed unbounded totality is not generated.	`finite-stage` - Every admitted stage has a complete finite trace.
`successor`	`largest-stage`	No generated rule supplies a largest finite stage.	`always-next` - Appending one Fold supplies the next finite stage.
`refinement`	`vanishing-zero`	Numerical zero is outside the domain.	`positive-rung` - Every refinement remains a positive exact part.
`convergence`	`limit-as-value`	An ungenerated limit cannot enter as a value.	`nested-rational-bounds` - Exact nested rational bounds certify approach and error.
`continuum`	`ambient-real-line`	An ambient continuum imports forbidden objects.	`unbounded-process-boundary` - The continuum is correspondence for an unbounded refinement process.
`cardinality`	`completed-comparison`	Cardinality of a nonobject has no SFT witness.	`finite-bound-count` - Every reached bound has an exact generated count.
`generality`	`fixed-depth`	One fixed depth does not close the successor law.	`base-successor` - The positive-rung property is preserved by every next finite refinement.
`addition`	`extra-limit-rule`	A limit oracle adds an axiom.	`no-extra-rule` - Only exact parts, bounds and successor traces occur.

20.3 Unique survivor, minimality and laws

> *Potential infinity is the never-terminal generated successor process; every reached object is finite and positive, convergence is an exact nested-bound certificate, and completed infinity or continuum is outside the admitted object language.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- every dyadic rung reaches the One in exactly its counted depth.
- every finite depth has a lawful next depth.
- exact finite-difference sequences may converge by shrinking positive rational error bounds.
- no completed continuum cardinality claim is admitted without a generated object.

20.4 Operational witnesses and unfavorable checks

- `finite-rungs` - Every generated dyadic rung through depth fourteen is positive and returns after its counted depth. Result: PASS.
- `quadratic-curvature` - The centered second difference of x squared is exactly two at every tested rational spacing. Result: PASS.
- `cubic-convergence` - The forward second-difference curvature of x cubed at the One has exact error six times the spacing, halving at each refinement. Result: PASS.
- `finite-closure` - The first five dyadic rungs plus the boundary rung reconstruct the One. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A completed unbounded totality is not generated. Result: PASS; receipt `sha256:579bf8f8ad064ae312c49bd3194500e06ac9a4dff8dc68b6498b2dcc8c1d9233`.

- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

20.5 Depth-independent closure

Base: The One is the complete finite base and its first refinement is a positive exact part.

Successor: Appending one generated Fold depth replaces a positive one-of-b^k rung by the positive one-of-b^(k+1) rung and records one additional return step.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

20.6 Meaning and scientific consequence

The theorem distinguishes an indefinitely extendable procedure from an already completed infinite object. Every reached rung is finite, exact and positive, and every rung has a lawful successor. Convergence is carried by nested rational bounds and shrinking exact errors. The continuum can describe that open-ended refinement at correspondence, but cannot enter the derivation as an ungenerated container or answer-producing cardinality.

20.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: potential infinity, constructive limit, continuum, continuum hypothesis, finite difference convergence. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

20.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

This is a theorem about the SFT object language and exact finite certificates; it does not deny the utility of continuum correspondence outside derivation.

20.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:3e1480da83b7864de42150a1dfe2616b954a893ff1f782267ae8de12e497682b`.
- Derivation seal: `sha256:b4d6ada858b868c1e51dbc259fafd1672b962526efc836f55e2d8e4958729e94`.
- Independent implementation: `sha256:cb2b5e3aa8dfc0485cffc29b26c677a7abad0c1e2a8cec704353005ff760820f`.
- Independent certificate: `sha256:cd1fbdbe8cc78f9e9a0ef75dc32657fab1b1173c5a44cb9c1e871dfd05321f3c`.

- External validation: `sha256:5873d50654bda8941184c01fe4fabee288a50208f9b899a81f529dfbf2c05ef6`.
- Engine receipt: `sha256:a32c6d6ee891f97acadd53cbec922bbcad5a6ad8a928264bb654b4caaf997098`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-LIMIT-CONTINUUM-002-a32c6d6ee891f97a.json`.

21. Derivation 16: Positive polynomial-balance certificates for algebraic magnitudes

Claim identity: `SFT-MATH-ALGEBRAIC-BALANCE-002`

21.1 Question and necessity

The V1 algebraic-magnitude engine and V2 Vieta cross-check require an explicit representation consistent with the prohibition on irrational proof values.

The exact theorem statement is:

> *A magnitude not itself admitted as an irrational value may be identified by two positive-coefficient polynomial sides, an exact rational bracket in which their order swaps, and a generated bisection trace that narrows that bracket without ever importing the root as a proof value.*

Admitted dependencies: `SFT-MATH-ALGEBRA-001`, `SFT-MATH-ORDER-LATTICE-001`.

21.2 Generated grammar and exact boundary

Exact arithmetic evaluates two positive polynomial sides; exact order supplies a bracket; repeated common refinement supplies a replayable enclosure without adding the missing scalar field.

Generation rule: Exhaust representation, coefficients, enclosure, refinement, identity, reconstruction, generality and extra-rule choices.

Exact grammar boundary: Positive polynomial-side evaluations and exact rational brackets at supplied finite refinement depth.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:143a3ac73bd3e1ce5da20a137c0b51e4aad53f7d86e2a29717a115408881616d`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-ALGEBRAIC-BALANCE-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`representation`	`irrational-scalar`	The scalar is forbidden as a proof value.	`balance-certificate` - The object is the equality condition between two positive sides.
`coefficients`	`signed-polynomial`	Negative coefficients import signed magnitude.	`positive-separated-sides` - Terms are moved to two positive sides with held orientation.
`enclosure`	`decimal-guess`	A decimal guess is not exact.	`rational-order-swap` - Exact endpoint comparisons bracket the balance.
`refinement`	`floating-iteration`	Floating iteration loses exact provenance.	`exact-bisection` - Each midpoint is an exact rational part.
`identity`	`approximate-equality`	Tolerance cannot define the algebraic object.	`polynomial-balance` - The defining identity is exact equality of the two sides.
`reconstruction`	`unverified-root`	An asserted root has no certificate.	`bracket-trace` - The full nested bracket trace reconstructs every decision.
`generality`	`named-radicals`	A radical list omits higher-degree forms.	`finite-polynomial-induction` - Each supplied finite polynomial pair is evaluated by the same exact trace law.
`addition`	`extra-field`	An algebraic field is not supplied by Foundation.	`no-extra-rule` - Only arithmetic, order and exact parts occur.

21.3 Unique survivor, minimality and laws

> *The unique admitted algebraic-magnitude representation is a positive polynomial-balance identity with exact rational order-swap bracket and replayable bisection trace; the irrational root is never a proof value.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- separating signed terms into two positive sides preserves the defining balance.
- an endpoint order swap plus exact bisection yields nested rational enclosures.
- Vieta relations provide an implementation-distinct coefficient cross-check for a bracketed root family.

21.4 Operational witnesses and unfavorable checks

- `square-balance` - The positive sides x squared and two swap order across seven-of-five and three-of-two. Result: `PASS`.
- `exact-midpoints` - Twenty bisections of that bracket retain exact rational endpoints and an order swap. Result: `PASS`.
- `vieta-rational` - The polynomial with roots one-of-two, one-of-three and one-of-six has exact sum One and exact pair sum eleven-of-thirty-six. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: The scalar is forbidden as a proof value. Result: `PASS`; receipt `sha256:98b59157c376c2de4e3224bf91fa0fa031bc85c840cfea93fe45c1599a773101`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

21.5 Depth-independent closure

Base: One exact rational bracket with opposite side order is a complete first enclosure.

Successor: The exact midpoint retains the order swap in exactly one half-bracket unless it is itself the exact balance, so the certificate narrows at every supplied finite step.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

21.6 Meaning and scientific consequence

The no-irrational rule does not erase algebraic geometry or spectral magnitude. It changes the proof object. Instead of storing an irrational scalar, SFT stores the exact equality of two positive polynomial sides and a rational bracket whose order reverses. Exact bisection narrows the enclosure, while Vieta identities cross-check whole root families. The object is its replayable balance certificate, never a rounded decimal.

21.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: algebraic number, root isolation, bisection, Vieta identities, Eudoxan magnitude. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

21.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

The certificate identifies an algebraic balance to any supplied finite rational enclosure depth; it does not admit the incommensurable magnitude as a scalar proof object.

21.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:e12152b418d417621346edd4eabd39be17b1dabe6741c85682aa51290b3c8736`.
- Derivation seal: `sha256:5e3dbc95200564d41c0418f508c6fb35eaadfd1a0077f49a9faae4aa1eaeaf14`.
- Independent implementation: `sha256:e9a7d54155ece0dcb7d43f8027e3f531eb4b2f2f67d5433cd29e4a82b4b4a32a`.
- Independent certificate: `sha256:d309c7ff8f819486d4e24671c0e8a7587328436e35e6b0f0343f9d37b6618202`.
- External validation: `sha256:e2d92d98d73d97f48673bbc2181cdf7a54b3db60afd4e0ef58bd2d4022d5efff`.
- Engine receipt: `sha256:ba975fdfe0c8946cb713eabcd9f48e77284941ec08fc257b9d3bb614b701540d`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-ALGEBRAIC-BALANCE-002-ba975fdfe0c8946c.json`.

22. Derivation 17: Exact recurrence of every finite componentwise Fold configuration

Claim identity: `SFT-MATH-BOUNDED-N-BODY-002`

22.1 Question and necessity

The V1/V2 three-body and general n-body results were explicitly restricted to Fold-built bounded-denominator configurations.

The exact theorem statement is:

> *Every supplied positive finite tuple of reduced rational Fold states has a finite exact joint support; after its transient binary depths, the componentwise Fold configuration recurs with period equal to the least common multiple of the odd-core component periods.*

Admitted dependencies: `SFT-MATH-ORBIT-NUMBER-THEORY-002`, `SFT-MATH-EXACT-RELATIONS-002`.

22.2 Generated grammar and exact boundary

Each component has a finite transient and odd-core cycle. Their Cartesian product is finite and deterministic; exact return occurs at the iterated least common period.

Generation rule: Exhaust state domain, component rule, denominator custody, support, recurrence, period, generality and extra-dynamics choices.

Exact grammar boundary: Finite tuples whose dynamics is exactly componentwise Fold on generated rational parts.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:ba9434365f40862645ffa648cf1c71c99e659011a45ccdb7f7745a13425a4d6d`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-BOUNDED-N-BODY-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`state_domain`	`continuum-coordinates`	Continuum coordinates are outside the object language.	`finite-rational-tuple` - Each component has a finite exact trace.
`component_rule`	`imported-force-law`	An imported force law changes the system.	`componentwise-fold` - Each component advances only by Fold.
`denominator`	`unbounded-drift`	Fold cannot introduce a new denominator factor.	`conserved-odd-core` - The reduced odd denominator core is invariant.
`support`	`uncounted-space`	An uncounted space cannot establish recurrence.	`finite-product-support` - The joint support is the finite product of component supports.
`recurrence`	`chaos-label`	A label does not defeat a finite deterministic return certificate.	`exact-return` - The joint tuple is replayed exactly.
`period`	`selected-period`	A selected period is not forced.	`least-common-period` - The first joint return is the least common component return.
`generality`	`three-body-table`	One tuple size is not the theorem.	`tuple-successor` - Appending one component extends the period by one least-common return.
`addition`	`external-dynamics`	External gravitational dynamics is a different claim.	`no-extra-rule` - The theorem contains only componentwise Fold.

22.3 Unique survivor, minimality and laws

> *All finite componentwise Fold tuples are eventually recurrent; their recurrent joint period is the iterated least common multiple of component odd-core periods.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- Fold preserves each reduced odd denominator core.
- a finite product of finite deterministic component orbits is finite.
- the first joint recurrent return is the least common component period.

22.4 Operational witnesses and unfavorable checks

- `three-cycle` - The tuple one-, two- and four-of-seven returns after three componentwise Folds. Result: PASS.
- `four-cycle` - The four ranks over denominator five rotate and return after four Folds. Result: PASS.
- `joint-period` - Components over denominators three, five and seven return jointly after twelve Folds. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Continuum coordinates are outside the object language. Result: PASS; receipt `sha256:44a51a0546406e551657eda4a8017def7dcf9eb869401c2fb7045d7e03f36db6`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.

- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

22.5 Depth-independent closure

Base: One rational Fold component is transient-to-periodic by its denominator decomposition.

Successor: Appending one component preserves finite product support and replaces the recurrent period by its least common return with the new component period.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

22.6 Meaning and scientific consequence

The n-body theorem is exact for the model V1/V2 actually names: a finite tuple of rational states advanced componentwise by Fold. Each component has a finite transient and odd-core cycle; their product is finite and its first joint return is the least common period. This does not swap in continuum Newtonian dynamics after the fact. It demonstrates precisely how the SFT native dynamics makes every finite tuple replayable and recurrent.

22.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: finite dynamical system, three-body choreography, n-body recurrence, joint period. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

22.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

This theorem is not a claim about arbitrary continuum Newtonian or relativistic n-body differential equations; it closes the native SFT componentwise Fold model stated by the prior record.

22.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:cdbbb4b8d5888934ecf54a47c059ec62c8dcc975af4bd9c5d0e50509cdfb2aec`.
- Derivation seal: `sha256:847c6bb6921ac26739ce553890f204b1a5b39c3cd085a008554db882434fc93d`.
- Independent implementation: `sha256:a631e8eb34441bc501703dcc24f92b1c5949aa0d93b9b58769a2463326d81a8c`.
- Independent certificate: `sha256:1d3d888f14db2f03b13c38a139422f63da47c5f5753dfd7483c543aec36280d3`.
- External validation: `sha256:456731eecc5f2e6ff6ffaebf17f952836f77a5c93554fc3f4d8d140a5a90b481`.
- Engine receipt: `sha256:e355bc5aa649df8fe2daf21a3e0ae56d5582680edd7f867a751bb954fca224a4`.

- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-BOUNDED-N-BODY-002-e355bc5aa649df8f.json`.

23. Derivation 18: Depth-independent discrete-gradient bound for floored Fold fluids

Claim identity: `SFT-MATH-FLOORED-FLUID-REGULARITY-002`

23.1 Question and necessity

The prior Navier-Stokes and CFD statements require the exact native-lattice regularity theorem and an explicit boundary against the distinct continuum Millennium grammar.

The exact theorem statement is:

> *At every supplied positive finite Fold depth k, an exact lattice with minimum spacing one-of-b^k and velocity separation at most the One has discrete gradient and vorticity magnitude bounded by b^k; at depth five the exact bound is thirty-two, so no finite-depth Fold-lattice blow-up is expressible.*

Admitted dependencies: `SFT-MATH-GEOMETRY-TOPOLOGY-001`, `SFT-MATH-LIMIT-CONTINUUM-002`.

23.2 Generated grammar and exact boundary

Generated geometry supplies a positive minimum edge; the One ceiling supplies the maximum positive velocity gap; exact quotient forces their reciprocal-floor bound.

Generation rule: Exhaust lattice, spacing, velocity, derivative, bound, conservation, generality and continuum-scope choices.

Exact grammar boundary: Exact finite Fold lattices with positive minimum spacing and bounded positive velocity differences.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:b0c0b232760703024bf387b6944b8719cca03fcc99d223697cc4e1d8c8c51639`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-FLOORED-FLUID-REGULARITY-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`lattice`	`continuum-points`	An ungenerated continuum has no smallest spacing.	`generated-grid` - Every site and edge is generated.
`spacing`	`vanishing-length`	Numerical zero is outside the domain.	`positive-floor` - Minimum spacing is one exact rung.
`velocity`	`unbounded-speed`	Unbounded speed violates the One ceiling.	`one-bounded-gap` - Velocity separation is at most the One.
`derivative`	`limit-derivative`	A continuum limit changes the claim.	`edge-quotient` - The discrete gradient is exact gap over exact edge spacing.
`bound`	`inserted-cap`	An inserted cap would be an engineering parameter.	`derived-reciprocal-floor` - The maximum quotient is forced by One over the minimum rung.
`conservation`	`floating-rescale`	Floating rescaling cannot prove exact transport.	`edge-ledger` - Every transferred part retains source and destination identity.
`generality`	`depth-five-only`	One depth is only an example.	`depth-successor` - At k plus One the reciprocal-floor bound multiplies by the Fold count.
`scope`	`continuum-millennium`	The classical continuum problem is not the same grammar.	`fold-lattice-regularity` - The theorem is exactly the native finite-depth Fold lattice.

23.3 Unique survivor, minimality and laws

> *The native Fold-lattice gradient bound is b^k at depth k, attained by a One velocity gap across one minimum edge; the depth-five value is thirty-two and every finite-depth field remains bounded.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- positive minimum spacing prevents a vanishing denominator.
- One-bounded velocity separation divided by one-of-b^k is at most b^k.
- the successor depth multiplies the bound by the generated Fold count.

23.4 Operational witnesses and unfavorable checks

- `depth-five` - The exact reciprocal of the depth-five binary floor is thirty-two. Result: PASS.
- `depth-table` - Every depth one through fourteen has reciprocal-floor bound two-to-depth. Result: PASS.
- `mass-ledger` - An exact transport split preserves a sample whole mass ledger. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An ungenerated continuum has no smallest spacing. Result: PASS; receipt `sha256:8afa8db26073931de8036d7069b515191f55870645ca256d0c09fba76b1fee7d`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: PASS; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

23.5 Depth-independent closure

Base: At the first Fold depth, the positive floor is one-of-b and the edge-gradient bound is b.

Successor: Refining one Fold divides the floor by b while preserving the One velocity ceiling, so the exact finite bound is multiplied by b.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

23.6 Meaning and scientific consequence

Regularity is forced from two structural bounds: velocity separation cannot exceed the One and lattice spacing cannot fall below its declared positive rung. Their exact quotient is b^k at depth k, giving 32 at depth five. The number is therefore not an artificial CFD stabilizer. The paper keeps the categorical distinction between this depth-independent Fold-lattice theorem and the separate classical continuum Millennium problem.

23.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: discrete fluid, vorticity bound, Navier-Stokes, finite-volume lattice, regularity. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

23.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.

- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

This closes SFT floored-lattice regularity. It does not claim a proof of the classical continuum Navier-Stokes existence-and-smoothness problem.

23.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:80e3b092ab9de1ffc5d8cdff13dc5f12f42cd26c5f0cd360366228db1b277135`.
- Derivation seal: `sha256:18460e9edef12ead62e6409f1a92e8cc5a6e8dec68cde2e1c44e36aa76d20ff2`.
- Independent implementation: `sha256:e51e73101b8bd2730f2aa03837b17acfbd5adc8c861034f703dba1b45359d6ec`.
- Independent certificate: `sha256:49330684436f47d6548aa71275bc43bef3c45ca37528b6d574ddf24fb4d430d2`.
- External validation: `sha256:4b299d88a28e233a46659148ceb5fe0662ef1714e84a8e23302ea7be4765abc2`.
- Engine receipt: `sha256:6fb3f40f5631367f23bad6cda58c87b335d33fd212a7c9d37068d8b920ad65e0`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-FLOORED-FLUID-REGULARITY-002-6fb3f40f5631367f.json`.

24. Derivation 19: Exact bounded Goldbach and twin-prime census

Claim identity: `SFT-MATH-PRIME-PAIR-CENSUS-002`

24.1 Question and necessity

V2 Step 278 registered exact bounded counts; an unrestricted Goldbach or twin-prime claim was not machine-established and is explicitly excluded.

The exact theorem statement is:

> *Exhaustive exact whole-number enumeration proves that every even whole from four through ten thousand has at least one prime complementary pair, that this range contains exactly four-thousand-nine-hundred-ninety-nine tested evens with no failure, and that exactly two-hundred-five twin-prime pairs have upper member at most ten thousand.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`.

24.2 Generated grammar and exact boundary

Generated whole arithmetic supplies the interval, complements and divisor certificates; exhaustive enumeration supplies exact finite coverage without an imported prime table.

Generation rule: Exhaust range, primality, pair generation, complement, coverage, twin adjacency, boundary and extra-theorem choices.

Exact grammar boundary: Positive finite whole-number census with even inputs four through ten thousand inclusive.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:c101ca3690ea2f183271329f9aa7671c7441160017c056df669fa3d70748debc`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in claims/SFT-MATH-PRIME-PAIR-CENSUS-002/candidate_census.json.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
range	sampled-evens	Sampling cannot close the declared range.	all-declared-evens - Every even input in the frozen range is executed.
primality	borrowed-table	A borrowed table hides verification.	trial-certificate - Each prime status is independently decided by complete divisor search to the square bound.
pairs	one-guessed-pair	A guess omits alternatives and failures.	all-complements - Every positive complementary split is generated.
complement	signed-sum	Signed arithmetic is unnecessary.	positive-junction - Both prime parts join exactly to the even whole.
coverage	spot-check	Spot checks do not prove the bounded census.	exhaustive-census - All 4,999 declared evens receive a verdict.
twins	approximate-density	A density estimate is not an exact count.	exact-gap-two - Every prime pair at whole gap two is counted once.
boundary	unrestricted-conjecture	The finite run cannot prove an unrestricted conjecture.	explicit-ten-thousand - The exact boundary is sealed into the claim.
addition	external-answer	An external published count cannot enter generation.	no-extra-rule - Only exact arithmetic and complete enumeration occur.

24.3 Unique survivor, minimality and laws

The exhaustive 4 through 10,000 census contains 4,999 even inputs, zero Goldbach failures and exactly 205 twin-prime pairs with upper member at most 10,000.

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- every declared even input is partitioned by all positive complementary pairs.
- prime status is decided by a complete finite divisor certificate.
- the finite boundary is part of the theorem and cannot be erased.

24.4 Operational witnesses and unfavorable checks

- `goldbach-range` - Every even whole from four through ten thousand has a generated prime complement pair. Result: PASS.
- `even-count` - The declared range contains exactly four thousand nine hundred ninety-nine even inputs. Result: PASS.
- `spot-counts` - Twelve has one unordered prime pair and one hundred has six. Result: PASS.
- `twin-count` - The complete twin-prime count through ten thousand is two hundred five. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Sampling cannot close the declared range. Result: PASS; receipt `sha256:f0d8ec988eaf6a70e795c33155c0ec30332c2558947a51a3fcc305cdcdadb26c`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.

- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

24.5 Depth-independent closure

Base: The first declared even whole, four, is the junction of two and two.

Successor: The census successor advances to the next even whole, exhausts every complementary split and records either a prime witness or an explicit failure.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

24.6 Meaning and scientific consequence

This is a genuine exhaustive empirical proof over a frozen finite mathematical population. Every one of 4,999 even wholes is generated, every complement is checked and every prime status is independently recomputed. The result has zero failures and the twin census is exactly 205. Its strength comes from total declared coverage, not from pretending that a boundary of 10,000 proves an unrestricted conjecture.

24.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: Goldbach census, twin primes, primality certificate, finite exhaustive proof. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

24.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

This is an exact finite theorem through 10,000, not an unrestricted proof of Goldbach's conjecture or the twin-prime conjecture.

24.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:a5a78fe935ed3c85b51d8123cbae2d6fc773a1d384e74f1ea0379ac7effb151c`.
- Derivation seal: `sha256:670c35bc2174d10dd88d35ee6892f424ce2a6c039ea35b2d1c22652511398f80`.
- Independent implementation: `sha256:50dd6ed96c8f0d57441b24b3d3a605e43c57146cb5c82112a757a01058a7ba03`.
- Independent certificate: `sha256:ef0331a6ee13bc25a810cc2bebef35fb073d1d91c695a60e6963b02fe9b1d572`.
- External validation: `sha256:a3643b08d7428ba160534292675aba56bd828fa8b80dd84f938a492633b42fc1`.
- Engine receipt: `sha256:52fcc090c336360dcdfd4b95919ffa5926d9d8fcd430ff6bb53d7d18cf5db1d6`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-PRIME-PAIR-CENSUS-002-52fcc090c336360d.json`.

25. Derivation 20: Unique half-One reflection axis and Riemann correspondence boundary

Claim identity: `SFT-MATH-RIEMANN-MIRROR-002`

25.1 Question and necessity

V1 XII-2 records the prime skeleton and half-One mirroring while expressly leaving complex zero location outside the framework; V2 Step 132 requires the fixed-axis result and its adverse logical control.

The exact theorem statement is:

> *The exact complement involution on positive parts has one and only one self-partner, the half-One; every other admitted part forms a distinct complementary pair. This forces the unique SFT reflection axis corresponding to the classical zeta functional equation's one-half symmetry, while complex zero location remains outside the no-imaginary proof language.*

Admitted dependencies: `SFT-MATH-EXACT-RELATIONS-002`, `SFT-FOUNDATION-HALF-ONE-001`.

25.2 Generated grammar and exact boundary

Exact complement is an involution. Solving self-complementarity by positive junction forces two equal parts to be the One, whose unique first Fold split is half-One.

Generation rule: Exhaust domain, involution, fixed point, off-axis pairing, prime tie, complex boundary, generality and extra-analytic choices.

Exact grammar boundary: Exact positive rational parts under complement within the One; no complex analytic continuation object.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:89db3b94da1ed133bcb1ef1b66c505b79d2f35dd5623b8b7d4c6cac95f3616a2`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-RIEMANN-MIRROR-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`domain`	`complex-plane`	Imaginary values are not admitted proof objects.	`positive-parts` - The involution acts on exact parts of the One.
`involution`	`imported-functional-equation`	An external equation cannot derive the Fold law.	`one-complement` - The map is exact complement within the One.
`fixed_point`	`selected-half`	Selecting one-half would not prove uniqueness.	`unique-self-complement` - Exact self-complement forces two equal parts to reconstruct the One.
`off_axis`	`unpaired-values`	Complement is total on the declared part domain.	`distinct-pairs` - Every nonfixed part has one distinct complementary partner.
`prime_tie`	`zeta-import`	The analytic zeta object is not an SFT input.	`orbit-prime-skeleton` - Prime correspondence comes from independently derived Fold orders.
`zero_location`	`claim-classical-rh`	Symmetry alone cannot locate every classical complex zero.	`explicit-language-boundary` - Only the symmetry axis is claimed inside SFT.
`generality`	`bounded-denominator-only`	A fixed denominator table does not prove the involution law.	`exact-part-identity` - The equality p=One-p has the unique exact solution two-p=One for every admitted part.
`addition`	`extra-analysis`	Complex analysis is outside the registered dependencies.	`no-extra-rule` - Only complement, equality and prior orbit number theory occur.

25.3 Unique survivor, minimality and laws

> *Half-One is the unique fixed axis of exact complement; every other positive part is paired. This closes the SFT Riemann-mirroring claim and explicitly does not assert classical complex zero location.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- complement composed twice is identity.
- self-complementarity uniquely forces the half-One.
- reflection symmetry does not by itself imply that all points of an invariant set lie on the fixed axis.

25.4 Operational witnesses and unfavorable checks

- `fixed-half` - Half-One is equal to its exact complement. Result: PASS.
- `off-axis-pairs` - Quarter/three-quarter and one-third/two-thirds are distinct complement pairs. Result: PASS.
- `symmetry-control` - A symmetric two-point set can lie off the fixed axis, so symmetry alone cannot prove zero location. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Imaginary values are not admitted proof objects. Result: PASS; receipt `sha256:47986d182ec9dcb7e128ba21b6556b7f1a6cd9e96b2a99e5d6aef6ae62254001`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: PASS; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

25.5 Depth-independent closure

Base: The Foundation half-One certificate supplies the self-complementary witness.

Successor: Every newly generated non-half part is paired with its exact complement and neither becomes a second fixed point.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

25.6 Meaning and scientific consequence

The complement involution has exactly one fixed point: the half-One. That is the forced SFT mirror corresponding to the one-half axis of the classical functional equation. An unfavorable control constructs an off-axis symmetric pair, proving that symmetry alone cannot locate all members of a symmetric set. This preserves the strongest detailed V1 statement: prime-orbit structure and the mirror close, while classical complex zero location does not.

25.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: Riemann functional symmetry, critical line, reflection involution, Riemann hypothesis boundary. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

25.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

This theorem does not prove the classical Riemann hypothesis; it proves the SFT arithmetic prime skeleton and unique reflection-axis correspondence stated in the stronger detailed V1 record.

25.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:3b19a68ec191cac8e3039f672c938c16213165cc8ca33c752c1be7b1a15136c0`.
- Derivation seal: `sha256:100e6f21f2a33c3c374d80f8d96672c9351cf1678caed4084e3535e6c0b3d89a`.
- Independent implementation: `sha256:743035dbc946735329c39331a72b4bff57f4ff7fbd4f850703367923ebca7f6a`.
- Independent certificate: `sha256:e66aa5200f0ef07b92629cd5e0b9b18c210ab9b371a1522e100ed681e7200870`.
- External validation: `sha256:dee17f0c705cb0506767a9ed5f9bc61294d4d624f8bbc1ddcbdf8cb82cbd285c`.
- Engine receipt: `sha256:27847629dbd2fb1d1c2f14d05017f8fb9395482d6e6ca001a24622e55364dfc4`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-RIEMANN-MIRROR-002-27847629dbd2fb1d.json`.

26. Derivation 21: Exact bounded Collatz census and contraction-control correction

Claim identity: `SFT-MATH-COLLATZ-FINITE-CENSUS-002`

26.1 Question and necessity

V2 Step 277 contains an exact bounded census and a separate contraction rationale. V3 must reproduce the former and mechanically reject the latter where it is false.

The exact theorem statement is:

> *Every positive whole start from the One through one-hundred-thousand reaches the 1-4-2 cycle under the declared Collatz transition; start twenty-seven takes exactly one-hundred-eleven steps. The exact census also rejects the prior constant-three-quarter pointwise contraction shortcut: a lawful odd macrostep may rise, so only the bounded execution result is admitted.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`.

26.2 Generated grammar and exact boundary

Exact whole arithmetic supplies the transition and full finite traces. Exhaustive execution proves the declared bound; the adverse start three disproves a constant pointwise contraction premise.

Generation rule: Exhaust range, transition, parity, trace, cycle, contraction, boundary and extra-theorem choices.

Exact grammar boundary: Positive whole starts one through one hundred thousand under the exact conventional Collatz transition.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:018b733207ce8a309044f81d99745b01d33beae74730ced7f35aa611a3899a38`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-COLLATZ-FINITE-CENSUS-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`range`	`sampled-starts`	Sampling cannot close the declared range.	`all-declared-starts` - Every start in the frozen bound is executed.
`transition`	`fold-analogy`	An analogy is not the conventional Collatz map.	`exact-collatz` - Even starts halve and odd starts map to three-times plus One.
`parity`	`assumed-even`	Parity must be checked from exact whole arithmetic.	`computed-parity` - Every transition computes its parity exactly.
`trace`	`terminal-only`	A terminal result without its path cannot be replayed.	`complete-step-trace` - Every successor is counted until the cycle.
`cycle`	`zero-sink`	Zero is not in the declared positive domain.	`one-four-two` - The terminal recurrent class is the exact 1-4-2 cycle.
`contraction`	`constant-three-quarters`	The shortcut is false: the odd macrostep from three reaches five after removing all binary factors.	`executed-variable-descent` - No pointwise contraction premise enters; the exact trace decides the bounded result.
`boundary`	`unrestricted-collatz`	A finite census cannot establish the unrestricted conjecture.	`explicit-one-hundred-thousand` - The sealed maximum is part of the claim.
`addition`	`external-sequence`	A published trajectory cannot enter the execution.	`no-extra-rule` - Only exact whole operations drive the census.

26.3 Unique survivor, minimality and laws

> *The complete starts One through 100,000 census has zero failures, start 27 takes 111 steps, the terminal cycle is 1-4-2, and the constant-3/4 pointwise contraction premise is rejected.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- every odd three-times-plus-One result is even.
- every declared start has a finite replay trace to the terminal cycle.
- the claim boundary remains finite and does not become the unrestricted Collatz conjecture.
- a false contraction heuristic cannot serve as proof even when the bounded census passes.

26.4 Operational witnesses and unfavorable checks

- `odd-even` - Every positive odd start through ten thousand one maps to an even three-times-plus-One result. Result: PASS.
- `bounded-census` - Every start through one hundred thousand reaches the One. Result: PASS.
- `twenty-seven` - Start twenty-seven takes exactly one hundred eleven steps. Result: PASS.
- `contraction-reject` - The accelerated odd step from three reaches five and therefore is not a pointwise contraction. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Sampling cannot close the declared range. Result: PASS; receipt `sha256:f0d8ec988eaf6a70e795c33155c0ec30332c2558947a51a3fcc305cdcdadb26c`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.

- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: PASS; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

26.5 Depth-independent closure

Base: The One lies on the declared 1-4-2 recurrent cycle.

Successor: For each next positive whole start, execute the exact transition until an already certified cycle member appears and retain the full prefix.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

26.6 Meaning and scientific consequence

All starts through 100,000 execute and reach the 1-4-2 cycle; start 27 takes 111 steps. Crucially, V3 also tests the explanation attached to the old count. Start three maps to five after the first required halving, so a constant three-quarter pointwise contraction is false. The engine retains the exact census and records the failed shortcut, showing how clean-room reconstruction can strengthen a result without rewriting its evidential history.

26.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: Collatz map, hailstone sequence, finite census, cycle detection. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

26.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

The result is exact for starts through 100,000. The unrestricted Collatz conjecture remains outside this finite certificate.

26.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:b23dfdfbf1bd53b640539bdc1dbdd900c5786659755b5cb5622bbe362cc8081f`.
- Derivation seal: `sha256:e57a27f1b95bade1301c14d5bfc05434c7c63751e8d32cf64aad2e29f8958be8`.
- Independent implementation: `sha256:9d533f192814a4720b16e34cfe3c49acbe02de80dce91c0d60b1be9c01c02421`.
- Independent certificate: `sha256:83e49654996c1fc161e8b8d3abfe7e78342759ed823d227809aa1f80e9034eda`.
- External validation: `sha256:e52f71a74ebea2b23714d471b78148eed57a6163e4b3ca9f522c20bb0e82a101`.
- Engine receipt: `sha256:3bb3a9fe4790150369defc5b2dc8568fed100b688dbb6a9c1d5cbd40c4f9a045`.

- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-COLLATZ-FINITE-CENSUS-002-3bb3a9fe47901503.json`.

27. Derivation 22: Exact self-similarity, chaotic rate and convergent-series laws

Claim identity: `SFT-MATH-SELF-SIMILAR-CONVERGENCE-002`

27.1 Question and necessity

V1/V2 separately registered Lyapunov/entropy antilogs, covering, scale depth, unit power, exact convergence rate and bounded accumulated separation.

The exact theorem statement is:

> *Fold forces binary local separation expansion, one closed distinction per step, the unique unit exponent under rank doubling, exact m^d support growth, and rational convergence certificates whose tails shrink by a generated factor without importing logarithms or irrational limit values.*

Admitted dependencies: `SFT-MATH-EXACT-RELATIONS-002`, `SFT-MATH-LIMIT-CONTINUUM-002`.

27.2 Generated grammar and exact boundary

Fold supplies the only scale successor and predecessor merge; exact support recurrence and rational tail comparison force the registered rates without analytic functions.

Generation rule: Exhaust scale action, separation, information, power exponent, support, series, generality and extra-function choices.

Exact grammar boundary: Exact generated finite scales, supports and rational partial sums.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:bb9aec7646934eff006d7d0d640b9a605b82eb46f40bc3b18374c5e37238e7e6`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-SELF-SIMILAR-CONVERGENCE-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`scale`	`continuous-rescale`	A continuous scale group is not generated.	`fold-doubling` - The scale successor is the Fold count.
`separation`	`fitted-lyapunov`	A fitted rate is not a theorem.	`exact-binary-expansion` - Before cast, local separation is multiplied by the Fold count.
`information`	`logarithmic-value`	A logarithm is outside exact proof arithmetic.	`one-closed-label` - One Fold merge closes exactly one binary predecessor distinction.
`power`	`selected-exponent`	A selected exponent is a parameter.	`unit-self-similar` - Rank doubling and count halving uniquely select the One exponent in the enumerated exponent grammar.
`support`	`binary-only`	The successor law is not tied to one label count.	`m-to-depth` - An m-label successor multiplies support by m at every depth.
`series`	`irrational-limit`	The limit cannot enter as a proof value.	`rational-tail-bound` - Exact partial sums and a shrinking rational tail prove finiteness.
`generality`	`finite-table`	A depth table lacks a recurrence.	`base-successor` - Every law carries a structural base and exact successor.
`addition`	`extra-analytic-function`	Logarithm or exponential would add a model.	`no-extra-rule` - Only exact ratios, counts and recurrences occur.

27.3 Unique survivor, minimality and laws

> *The exact Fold rate is binary separation expansion and one closed label per step; unit rank power is the unique Fold-self-similar exponent, m-labelled support is m^d, and decreasing rational terms close convergence without importing their limit as a value.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- local uncast separation multiplies by the Fold count.
- one predecessor label is closed per Fold merge.
- rank doubling paired with count halving selects the unit power among the generated nonnegative whole exponent forms.
- m-labelled successor support obeys the base/successor count m^d.
- a geometric rational tail bound proves finite accumulated separation.

27.4 Operational witnesses and unfavorable checks

- `chaotic-rate` - The separation two-of-thirty-five advances to four-of-thirty-five before cast. Result: PASS.
- `support` - Ternary support has three, nine and twenty-seven states at depths one, two and three. Result: PASS.
- `unit-power` - Only exponent One among whole exponents One through four gives exact halving when rank doubles. Result: PASS.
- `gap-formula` - The exact gap formula is positive and strictly decreases through depth fourteen. Result: PASS.
- `finite-bracket` - The first eleven exact gap terms sum below eleven-of-sixty. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A continuous scale group is not generated. Result: PASS; receipt `sha256:e3868d8a6f435a5025d3289a24b18b2bef31ca89fa78f2d5992aadece72362ad`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: PASS; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

27.5 Depth-independent closure

Base: At depth One an m-labelled successor has m states and one Fold closes one binary distinction.

Successor: Appending one label multiplies support by m; applying one Fold multiplies an uncast local gap by b and closes one predecessor label; one further rational term preserves the tail bound.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

27.6 Meaning and scientific consequence

One Fold doubles an uncast local gap and closes one binary predecessor label. The same successor forces m^d support, selects the unit rank exponent from the declared whole-exponent grammar, and gives exact rational convergence rates. Finite accumulated separation is proved by partial sums and a rational tail bracket; no logarithm, exponential or irrational limit value has to be smuggled into the proof language.

27.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: Lyapunov antilog, KS entropy, power law, covering number, convergent series. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

27.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

The unit exponent is unique within the explicitly enumerated positive whole exponent grammar; natural-system correspondence is tested only after the formal seal in owning empirical branches.

27.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:a10ca3b6a55f0437152c829d3f7362cb209922b9d65a00cea34005c6929aa238`.
- Derivation seal: `sha256:3b5b67ae9a24eac5e74f0b60feb92e8f66224134eb397f911876b3e37eec91d1`.
- Independent implementation: `sha256:06101b52023de1f7ae1ec661673cf97b543f66fa9d46b2071557fc41fe6b828a`.
- Independent certificate: `sha256:ace930a00105674c923aa54768e10d198ce879d8b3cf8828c92b2eb5bd27363d`.
- External validation: `sha256:fad1024f01bba59f703064314f4b493e5386d5811e1d3e2d5ed01e848184bf1c`.
- Engine receipt: `sha256:0abf8d557108f37f4d9cfe735db3fd6e0d79a3d9405129f0a40ae86fb606d5b1`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SELF-SIMILAR-CONVERGENCE-002-0abf8d557108f37f.json`.

28. Version 1.2 extension: the Smithian Fold Scientific Calculator

The calculator is not a conventional floating-point calculator carrying SFT labels. It is an executable translation of the admitted Mathematics value law. Familiar keys and notation form a human boundary; every accepted expression is parsed into exact SFT runtime types, every operation is routed through one exact evaluator, and every result that would require a prohibited scalar halts before it can enter the answer, memory or history. The browser surface contains no arithmetic implementation.

Five claim records are included because transparent correction is part of the scientific result. Claim 003 passed its declared grammar but a later operational audit found reversed odd/even endpoints in one alternating arctangent enclosure. Its artifacts and receipt remain preserved as superseded adverse evidence. Claim 004 re-derived the core with corrected endpoint parity and an exact tangent-composition check. Claim 005 extended that core to the declared scientific-calculator language and stateful application. Claim 006 completed the calculator-first app, law explorer, resource controls and active-file coverage. Direct manual testing then exposed a blank deprecated desktop widget on macOS and usability failures on phones. Claim 007 held the claim-006 evaluator immutable and forced the accessible standards-rendered local web adapter.

28.1 Exact traced SFT-native scientific calculator

Claim identity: `SFT-MATH-SCIENTIFIC-CALCULATOR-003`

Publication status: `superseded_adverse_evidence`. The original machine receipt is preserved, but this claim is not the active calculator law. Its later-discovered parity defect is the reason claim 004 was derived independently rather than mutating claim 003.

WHY. A calculator makes the Mathematics laws directly inspectable, but it may not smuggle the conventional signed real or complex number model back into the proof domain through its user interface.

DERIVATION. Exhausting the ten representation and execution choices leaves one compositional machine: exact entered parts, empty-One, held orientations, admitted arithmetic recurrences, rational balance enclosures, typed orthogonal fibres, mandatory halts and complete traces. Each supported scientific function is then a finite composition of those structures, not a new axiom or fitted parameter.

CHECK. Execute all 1,024 generated semantics, require the sole all-preserving survivor, run exact arithmetic, root, combinatorial, transcendental, orthogonal, parser, trace and halt witnesses, run four adverse controls, and independently regenerate both the product and operational examples.

Exact statement:

> *A scientific-calculator interface is admitted in SFT only when every entered decimal is translated character-by-character to an exact rational part, displayed zero is structural empty-One, subtraction uses held orientation, all algebraic and transcendental non-rational results remain exact rational enclosures with replayable bounds, orthogonal values remain typed Fold fibres, every operation is traced, and every undefined or uncertified operation halts.*

Dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-MATH-ALGEBRAIC-BALANCE-002`, `SFT-MATH-LIMIT-CONTINUUM-002`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`.

Grammar boundary: All calculator semantics formed from the ten declared binary representation and execution choices, with exact finite rational recurrences and a declared finite resource bound.

Generation rule: Generate the complete product of input-value, empty, orientation, exact-operation, non-rational, transcendental, orthogonal, domain, trace and extra-rule choices.

The complete Cartesian product contains 1,024 candidate forms and has completeness identity `sha256:e6ff9fd1094ac119184dc42e13e19c44ecdbed05f9aa48a7d074d81408645e3c`. Every coordinate is shown below; the candidate census and elimination receipt preserve every product member and decision.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`input`	`host-float` - A host float loses the exact entered part before proof begins.	`character-exact-part` - Characters generate the exact counted numerator, denominator and exponent.
`empty`	`numeric-zero` - Numerical zero is not an admitted SFT value.	`structural-empty-One` - The empty-One form records complete cancellation or empty magnitude.
`orientation`	`negative-scalar` - A negative scalar imports a forbidden signed field.	`held-fibre-orientation` - One of two generated held labels records orientation with a positive magnitude.
`operations`	`host-arithmetic-oracle` - A host arithmetic oracle erases the generated Fold construction.	`generated-exact-kernels` - Junction, pairing, refinement and recurrence execute the operation exactly.
`nonrational`	`irrational-scalar` - An irrational scalar is outside the proof domain.	`rational-balance-enclosure` - An exact rational bracket and polynomial balance certify the result without importing a scalar.
`transcendental`	`black-box-library-value` - A black-box value supplies no Fold derivation or remainder certificate.	`finite-rational-recurrence-bound` - A finite exact recurrence and positive tail bound enclose the function value.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`orthogonal`	`imaginary-scalar` - An imaginary scalar is not an admitted proof value.	`typed-orthogonal-fibre-pair` - Two exact Fold fibres retain real and orthogonal coordinates compositionally.
`domain`	`nan-infinity-continuation` - NaN or infinity silently continues beyond the admitted domain.	`explicit-lawful-halt` - The machine halts before returning an uncertified value.
`trace`	`answer-only` - An answer without provenance cannot be replayed or independently checked.	`complete-proof-resource-trace` - Input translation, operations, result and counted resources are retained.
`addition`	`has-extra-rule` - A fitted constant or answer-selecting rule is not supplied by the dependencies.	`no-extra-rule` - The calculator composes only already-admitted exact structures and proofs.

Unique survivor and exact result:

> *The unique admitted calculator semantics uses exact character-generated rational parts, structural empty-One, held orientation, generated exact kernels, certified rational enclosures, finite rational recurrences with explicit remainder bounds, typed orthogonal fibres, mandatory domain halts, complete proof/resource traces and no extra rule.*

Replacing any forced coordinate by its enumerated alternative loses a registered requirement; adding an undeclared rule violates the no-extra-rule boundary. The unique product survivor therefore closes the declared grammar by minimality and named-shape uniqueness. Its operational laws are:

- decimal and scientific notation are exact rational boundary encodings, never floating proof values.
- complete cancellation returns structural empty-One and unmatched remainder retains held orientation.
- every exact operation composes admitted arithmetic, combinatorial and order structures.
- every non-rational result is an exact rational enclosure carrying its generating recurrence and bound.
- an unclosed enclosure, exhausted resource bound or invalid domain halts without a result.
- orthogonal composition is a typed two-fibre product and introduces no imaginary scalar.
- decimal output is a downstream display projection and never proof evidence.

Executed witnesses:

- `exact-decimal` - 0.1 plus 0.2 is the exact rational part 3/10, not a host floating result. Result: PASS.
- `empty-cancellation` - One substituted from One yields structural empty-One. Result: PASS.
- `held-remainder` - One substituted by three yields counter-held positive magnitude two. Result: PASS.
- `root-balance` - The exact sqrt(2) bracket has lower square below two and upper square at or above two. Result: PASS.
- `combinatorial-recurrence` - Factorial and selection recurrence reproduce exact finite arrangements. Result: PASS.
- `transcendental-enclosure` - exp, ln, sin and circle operations return ordered rational enclosures with certificates. Result: PASS.
- `orthogonal-composition` - The square of the unit orthogonal fibre is counter-held real One with empty orthogonal fibre. Result: PASS.
- `domain-halt` - Division by structural empty-One halts without returning NaN or infinity. Result: PASS.
- `complete-trace` - A parsed calculation retains boundary translation, operation and terminal records. Result: PASS.
- `scientific-notation` - Scientific notation is translated exactly without a floating intermediate. Result: PASS.

Adverse controls:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A host float loses the exact entered part before proof begins. Result: PASS; receipt `sha256:f1cae5eb43509a204e1458eacacacc8b1011649fb3e24e2004e089a9a431f8d5`.

- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero, NaN and infinity are not admitted values; negative, irrational and imaginary scalars are replaced by typed Fold structures; binary floating-point and opaque host-library answers are excluded from proof; uncertified roots, series, singularities and exhausted resource bounds halt; decimal projections are correspondence displays only Result: PASS; receipt `sha256:bdef05251b3846ffaedc057704bd0da3f30f3d8ddb10ea21de6236ba416c59f5`.

Depth-independent base: One entered digit is translated to an exact counted part; the structural empty-One and structural One supply cancellation and identity bases.

Depth-independent successor: Appending a generated digit refines the exact denominator by one counted tenfold composition; appending an operator composes two already-certified values, and appending one recurrence term retains the previous exact sum plus one exact part while replacing the remainder by a proven smaller positive bound.

Exact exclusions:

- semantic numerical zero, NaN and infinity are not admitted values.
- negative, irrational and imaginary scalars are replaced by typed Fold structures.
- binary floating-point and opaque host-library answers are excluded from proof.
- uncertified roots, series, singularities and exhausted resource bounds halt.
- decimal projections are correspondence displays only.

Limitation: Closure is for the declared calculator grammar and finite resource bounds. A displayed decimal is not proof evidence. Enclosures report their exact bounds; requests that cannot close inside the declared bound halt rather than returning a guess. Additional named functions remain lawful extensions only when they reduce to admitted exact recurrences and pass the same engine route.

Evidence identities:

- Source manifest: `sha256:5e8d18b5f8595c1ba6292a3bb736436d28fcc4813cd47aa6bc32744c300d8ac0`.
- Derivation seal: `sha256:8c6a64e9113e32d23e6bf5c08a1fad24692788433b322c7d4b55f7828f81ee18`.
- Independent implementation: `sha256:0a52eca4b5694848230b3671ed2b0c5ff4b2d170f394686aee38e98a60f8f8b5`.
- Independent certificate: `sha256:f26d7f99675fa8d4bd035214d0817ca65309d23fe4d4bb53b71b907fa9a46b22`.
- External validation: `sha256:0fa836f9219c8cae0ba1b24cb0fea4e2fbd0c530ed93fd996f4551c44a5e78e6`.
- Engine receipt: `sha256:f95ac235ac96429b36b10e8331478f110ed074e98a338c6f015d3f8874dc787c`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SCIENTIFIC-CALCULATOR-003-f95ac235ac96429b.json`.

28.2 Exact traced SFT-native scientific calculator with corrected enclosure parity

Claim identity: `SFT-MATH-SCIENTIFIC-CALCULATOR-004`

Publication status: `independently_replicated`; current versioned calculator lineage.

WHY. A calculator makes the Mathematics laws directly inspectable, but it may not smuggle the conventional signed real or complex number model back into the proof domain through its user interface.

DERIVATION. Exhausting the ten representation and execution choices leaves one compositional machine: exact entered parts, empty-One, held orientations, admitted arithmetic recurrences, rational balance enclosures, typed orthogonal fibres, mandatory halts and complete traces. Each supported scientific function is then a finite composition of those structures, not a new axiom or fitted parameter.

CHECK. Execute all 1,024 generated semantics, require the sole all-preserving survivor, run the exact arithmetic, root, combinatorial, transcendental, orthogonal, parser, trace and halt witnesses, verify alternating-series endpoint parity and the exact tangent-composition identity, run four adverse controls, and independently regenerate both the full product and operational certificates.

Exact statement:

> *A scientific-calculator interface is admitted in SFT only when every entered decimal is translated character-by-character to an exact rational part, displayed zero is structural empty-One, subtraction uses held orientation, all algebraic and transcendental non-rational results remain exact rational enclosures with replayable bounds, orthogonal values remain typed Fold fibres, every operation is traced, and every undefined or uncertified operation halts.*

Dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-MATH-ALGEBRAIC-BALANCE-002`, `SFT-MATH-LIMIT-CONTINUUM-002`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`.

Grammar boundary: All calculator semantics formed from the ten declared binary representation and execution choices, with exact finite rational recurrences and a declared finite resource bound.

Generation rule: Generate the complete product of input-value, empty, orientation, exact-operation, non-rational, transcendental, orthogonal, domain, trace and extra-rule choices.

The complete Cartesian product contains 1,024 candidate forms and has completeness identity `sha256:e6ff9fd1094ac119184dc42e13e19c44ecdbed05f9aa48a7d074d81408645e3c`. Every coordinate is shown below; the candidate census and elimination receipt preserve every product member and decision.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`input`	`host-float` - A host float loses the exact entered part before proof begins.	`character-exact-part` - Characters generate the exact counted numerator, denominator and exponent.
`empty`	`numeric-zero` - Numerical zero is not an admitted SFT value.	`structural-empty-One` - The empty-One form records complete cancellation or empty magnitude.
`orientation`	`negative-scalar` - A negative scalar imports a forbidden signed field.	`held-fibre-orientation` - One of two generated held labels records orientation with a positive magnitude.
`operations`	`host-arithmetic-oracle` - A host arithmetic oracle erases the generated Fold construction.	`generated-exact-kernels` - Junction, pairing, refinement and recurrence execute the operation exactly.
`nonrational`	`irrational-scalar` - An irrational scalar is outside the proof domain.	`rational-balance-enclosure` - An exact rational bracket and polynomial balance certify the result without importing a scalar.
`transcendental`	`black-box-library-value` - A black-box value supplies no Fold derivation or remainder certificate.	`finite-rational-recurrence-bound` - A finite exact recurrence and positive tail bound enclose the function value.
`orthogonal`	`imaginary-scalar` - An imaginary scalar is not an admitted proof value.	`typed-orthogonal-fibre-pair` - Two exact Fold fibres retain real and orthogonal coordinates compositionally.
`domain`	`nan-infinity-continuation` - NaN or infinity silently continues beyond the admitted domain.	`explicit-lawful-halt` - The machine halts before returning an uncertified value.
`trace`	`answer-only` - An answer without provenance cannot be replayed or independently checked.	`complete-proof-resource-trace` - Input translation, operations, result and counted resources are retained.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`addition`	`has-extra-rule` - A fitted constant or answer-selecting rule is not supplied by the dependencies.	`no-extra-rule` - The calculator composes only already-admitted exact structures and proofs.

Unique survivor and exact result:

> *The unique admitted calculator semantics uses exact character-generated rational parts, structural empty-One, held orientation, generated exact kernels, certified rational enclosures, finite rational recurrences with explicit remainder bounds, typed orthogonal fibres, mandatory domain halts, complete proof/resource traces and no extra rule.*

Replacing any forced coordinate by its enumerated alternative loses a registered requirement; adding an undeclared rule violates the no-extra-rule boundary. The unique product survivor therefore closes the declared grammar by minimality and named-shape uniqueness. Its operational laws are:

- decimal and scientific notation are exact rational boundary encodings, never floating proof values.
- complete cancellation returns structural empty-One and unmatched remainder retains held orientation.
- every exact operation composes admitted arithmetic, combinatorial and order structures.
- every non-rational result is an exact rational enclosure carrying its generating recurrence and bound.
- an unclosed enclosure, exhausted resource bound or invalid domain halts without a result.
- orthogonal composition is a typed two-fibre product and introduces no imaginary scalar.
- decimal output is a downstream display projection and never proof evidence.

Executed witnesses:

- `exact-decimal` - 0.1 plus 0.2 is the exact rational part 3/10, not a host floating result. Result: PASS.
- `empty-cancellation` - One substituted from One yields structural empty-One. Result: PASS.
- `held-remainder` - One substituted by three yields counter-held positive magnitude two. Result: PASS.
- `root-balance` - The exact sqrt(2) bracket has lower square below two and upper square at or above two. Result: PASS.
- `combinatorial-recurrence` - Factorial and selection recurrence reproduce exact finite arrangements. Result: PASS.
- `orthogonal-composition` - The square of the unit orthogonal fibre is counter-held real One with empty orthogonal fibre. Result: PASS.
- `domain-halt` - Division by structural empty-One halts without returning NaN or infinity. Result: PASS.
- `complete-trace` - A parsed calculation retains boundary translation, operation and terminal records. Result: PASS.
- `scientific-notation` - Scientific notation is translated exactly without a floating intermediate. Result: PASS.
- `alternating-parity-certificate` - Odd arctangent partial counts supply upper bounds; even counts supply lower bounds, with the next exact part closing the opposite endpoint. Result: PASS.
- `exact-angle-composition` - Two exact tangent doublings give 120/119 and subtraction of atan(1/239) gives tangent One, certifying the quarter-turn identity without decimal target selection. Result: PASS.
- `corrected-circle-enclosure` - The circle constant has ordered exact rational endpoints carrying the corrected alternating parity and exact angle-composition trace. Result: PASS.

Adverse controls:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A host float loses the exact entered part before proof begins. Result: PASS; receipt `sha256:f1cae5eb43509a204e1458eacacacc8b1011649fb3e24e2004e089a9a431f8d5`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt

`sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.

- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero, NaN and infinity are not admitted values; negative, irrational and imaginary scalars are replaced by typed Fold structures; binary floating-point and opaque host-library answers are excluded from proof; uncertified roots, series, singularities and exhausted resource bounds halt; decimal projections are correspondence displays only Result: `PASS`; receipt `sha256:bdef05251b3846ffaedc057704bd0da3f30f3d8ddb10ea21de6236ba416c59f5`.

Depth-independent base: One entered digit is translated to an exact counted part; the structural empty-One and structural One supply cancellation and identity bases.

Depth-independent successor: Appending a generated digit refines the exact denominator by one counted tenfold composition; appending an operator composes two already-certified values, and appending one recurrence term retains the previous exact sum plus one exact part while replacing the remainder by a proven smaller positive bound.

Exact exclusions:

- semantic numerical zero, NaN and infinity are not admitted values.
- negative, irrational and imaginary scalars are replaced by typed Fold structures.
- binary floating-point and opaque host-library answers are excluded from proof.
- uncertified roots, series, singularities and exhausted resource bounds halt.
- decimal projections are correspondence displays only.

Limitation: Closure is for the declared calculator grammar and finite resource bounds. A displayed decimal is not proof evidence. Enclosures report their exact bounds; requests that cannot close inside the declared bound halt rather than returning a guess. Additional named functions remain lawful extensions only when they reduce to admitted exact recurrences and pass the same engine route. The earlier 003 receipt is preserved as adverse evidence because its arctangent endpoint parity was reversed; this 004 law supersedes it and does not depend on it.

Evidence identities:

- Source manifest: `sha256:8969c3c7e5c60d8f5978f45f734e109ba530f00f838bf32d9d9beb938adbf104`.
- Derivation seal: `sha256:4fea624cffb8780f8e96d0a85eccb81c8fb609c1e712983a339a6f3f9b57d0d9`.
- Independent implementation: `sha256:5d2c4b11b1cb3ba9a7ba092af4d6d622ac73caf25b682f3bfb32ad7a000e6983`.
- Independent certificate: `sha256:500be40f0488c5003ebf92841385cec6f50ef8405b48e0d02288c7d7726f56c9`.
- External validation: `sha256:da76fedc1a4a70de18e6a51ab3f10065f18e5f04e09cf810ee913a4d11149c1f`.
- Engine receipt: `sha256:2d243dbd08260c483e79009067051239f1304ad3998006c1ff82c301062ba28b`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SCIENTIFIC-CALCULATOR-004-2d243dbd08260c48.json`.

28.3 Complete exact SFT scientific calculator and cross-platform learning app

Claim identity: `SFT-MATH-SCIENTIFIC-CALCULATOR-005`

Publication status: `independently_replicated`; current versioned calculator lineage.

WHY. A 1-to-1 scientific calculator must be usable like the familiar desktop calculator, not merely expose a narrow proof API. Accessibility cannot weaken the exact SFT value law or hide its certificates.

DERIVATION. Claim 004 supplies the admitted exact core. Exhausting the three extension choices forces complete declared function coverage, explicit session state and a cross-platform evidence-bearing app. Every new function is then reduced to exact arithmetic, rational recurrence, certified enclosure, typed fibre composition or a mandatory domain halt; UI controls map one-for-one to those operations.

CHECK. Execute all 8,192 combined semantics, require one survivor, replay every core witness, exercise ordinary equals, general powers, all function families, three angle modes, state and memory, terminal execution and the complete GUI control map; run adverse controls and independently regenerate the product and operational examples.

Exact statement:

> *The admitted exact calculator core extends uniquely to the declared mainstream scientific-calculator surface: terminal equals execution; editable expressions; exact and certified general powers, roots, logarithmic, exponential, trigonometric, inverse-trigonometric, hyperbolic, combinatorial and statistical operations; radian, degree and gradian modes; retained answer, memory and history; and one cross-platform desktop application exposing decimal projection, exact evidence, proof trace and SFT learning guidance. No interface action may bypass the same exact fail-closed evaluator.*

Dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-MATH-ALGEBRAIC-BALANCE-002`, `SFT-MATH-LIMIT-CONTINUUM-002`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`, `SFT-MATH-SCIENTIFIC-CALCULATOR-004`.

Grammar boundary: The complete thirteen-coordinate product formed by the ten exact core semantics plus the declared mainstream scientific function manifest, session interaction law and standard-library desktop evidence surface.

Generation rule: Generate the complete product of input-value, empty, orientation, exact-operation, non-rational, transcendental, orthogonal, domain, trace and extra-rule choices. Extend that complete product by declared scientific-function coverage, stateful interaction, and cross-platform application/evidence-surface choices.

The complete Cartesian product contains 8,192 candidate forms and has completeness identity `sha256:081b1dcd0e2431b4d583b8dbfa65cccf17502ee1d570b06609baf708f2878903`. Every coordinate is shown below; the candidate census and elimination receipt preserve every product member and decision.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`input`	`host-float` - A host float loses the exact entered part before proof begins.	`character-exact-part` - Characters generate the exact counted numerator, denominator and exponent.
`empty`	`numeric-zero` - Numerical zero is not an admitted SFT value.	`structural-empty-One` - The empty-One form records complete cancellation or empty magnitude.
`orientation`	`negative-scalar` - A negative scalar imports a forbidden signed field.	`held-fibre-orientation` - One of two generated held labels records orientation with a positive magnitude.
`operations`	`host-arithmetic-oracle` - A host arithmetic oracle erases the generated Fold construction.	`generated-exact-kernels` - Junction, pairing, refinement and recurrence execute the operation exactly.
`nonrational`	`irrational-scalar` - An irrational scalar is outside the proof domain.	`rational-balance-enclosure` - An exact rational bracket and polynomial balance certify the result without importing a scalar.
`transcendental`	`black-box-library-value` - A black-box value supplies no Fold derivation or remainder certificate.	`finite-rational-recurrence-bound` - A finite exact recurrence and positive tail bound enclose the function value.
`orthogonal`	`imaginary-scalar` - An imaginary scalar is not an admitted proof value.	`typed-orthogonal-fibre-pair` - Two exact Fold fibres retain real and orthogonal coordinates compositionally.
`domain`	`nan-infinity-continuation` - NaN or infinity silently continues beyond the admitted domain.	`explicit-lawful-halt` - The machine halts before returning an uncertified value.
`trace`	`answer-only` - An answer without provenance cannot be replayed or independently checked.	`complete-proof-resource-trace` - Input translation, operations, result and counted resources are retained.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`addition`	`has-extra-rule` - A fitted constant or answer-selecting rule is not supplied by the dependencies.	`no-extra-rule` - The calculator composes only already-admitted exact structures and proofs.
`scientific_surface`	`partial-basic-function-subset` - A basic subset is not the requested scientific calculator.	`complete-declared-scientific-surface` - Arithmetic, rational/non-rational powers, roots, logs, exponentials, circular/inverse/hyperbolic functions, combinatorics, exact statistics and typed orthogonal operations are all routed through certified kernels.
`session`	`stateless-expression-only` - A stateless expression evaluator lacks terminal equals, answer, memory and history behavior.	`equals-answer-memory-history` - The terminal equals key executes; Ans, memory and ordered history retain explicit typed state without altering laws.
`application`	`terminal-only-opaque-answer` - A terminal-only answer surface is not a clean general-user calculator app and hides evidence.	`cross-platform-app-trace-and-guide` - One standard-library desktop app provides editable display, keypad, angle selector, memory, history, exact detail, proof trace and embedded SFT guidance.

Unique survivor and exact result:

> *The unique expanded calculator is the exact core plus the complete declared scientific-function surface, terminal equals/Ans/memory/history state, RAD/DEG/GRAD translation, and one cross-platform desktop app whose buttons and keyboard invoke only the same fail-closed evaluator. Every exact result retains its Fold type; every non-rational result remains a certified rational enclosure; decimal output is display only.*

Replacing any forced coordinate by its enumerated alternative loses a registered requirement; adding an undeclared rule violates the no-extra-rule boundary. The unique product survivor therefore closes the declared grammar by minimality and named-shape uniqueness. Its operational laws are:

- decimal and scientific notation are exact rational boundary encodings, never floating proof values.
- complete cancellation returns structural empty-One and unmatched remainder retains held orientation.
- every exact operation composes admitted arithmetic, combinatorial and order structures.
- every non-rational result is an exact rational enclosure carrying its generating recurrence and bound.
- an unclosed enclosure, exhausted resource bound or invalid domain halts without a result.
- orthogonal composition is a typed two-fibre product and introduces no imaginary scalar.
- decimal output is a downstream display projection and never proof evidence.
- a rational exponent is counted root followed by counted composition; a certified exponent uses exp of the certified product with ln.
- degree and gradian inputs translate through the independently certified circle enclosure before circular evaluation.
- inverse circular and hyperbolic operations are exact recurrence compositions with explicit domain halts.
- memory, answer and history retain values but never mutate operation law or proof certificates.
- terminal and desktop interfaces are two views of one evaluator and cannot create an untraced answer.

Executed witnesses:

- `exact-decimal` - 0.1 plus 0.2 is the exact rational part 3/10, not a host floating result. Result: PASS.
- `empty-cancellation` - One substituted from One yields structural empty-One. Result: PASS.
- `held-remainder` - One substituted by three yields counter-held positive magnitude two. Result: PASS.
- `root-balance` - The exact sqrt(2) bracket has lower square below two and upper square at or above two. Result: PASS.
- `combinatorial-recurrence` - Factorial and selection recurrence reproduce exact finite arrangements. Result: PASS.

- `orthogonal-composition` - The square of the unit orthogonal fibre is counter-held real One with empty orthogonal fibre. Result: PASS.
- `domain-halt` - Division by structural empty-One halts without returning NaN or infinity. Result: PASS.
- `complete-trace` - A parsed calculation retains boundary translation, operation and terminal records. Result: PASS.
- `scientific-notation` - Scientific notation is translated exactly without a floating intermediate. Result: PASS.
- `alternating-parity-certificate` - Odd arctangent partial counts supply upper bounds; even counts supply lower bounds, with the next exact part closing the opposite endpoint. Result: PASS.
- `exact-angle-composition` - Two exact tangent doublings give 120/119 and subtraction of atan(1/239) gives tangent One, certifying the quarter-turn identity without decimal target selection. Result: PASS.
- `corrected-circle-enclosure` - The circle constant has ordered exact rational endpoints carrying the corrected alternating parity and exact angle-composition trace. Result: PASS.
- `ordinary-equals-calculation` - A user may enter 1+1= and receive exact forward-held two. Result: PASS.
- `general-power-surface` - Rational and certified non-rational exponents both return exact rational enclosures. Result: PASS.
- `angle-mode-correspondence` - DEG sin(30) encloses exact half-One and DEG atan(One) encloses exact forty-five. Result: PASS.
- `hyperbolic-base` - cosh(empty-One) returns an enclosure containing structural One. Result: PASS.
- `exact-statistics` - Mean and population variance retain exact rational parts. Result: PASS.
- `memory-answer-history` - Stored answer, recalled memory and ordered history execute as typed session state. Result: PASS.
- `discoverable-desktop-surface` - The app exposes the ordinary keypad, scientific functions, equals, memory, history evidence and SFT guide. Result: PASS.

Adverse controls:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A host float loses the exact entered part before proof begins. Result: PASS; receipt `sha256:f1cae5eb43509a204e1458eacacacc8b1011649fb3e24e2004e089a9a431f8d5`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero, NaN and infinity are not admitted values; negative, irrational and imaginary scalars are replaced by typed Fold structures; binary floating-point and opaque host-library answers are excluded from proof; uncertified roots, series, singularities and exhausted resource bounds halt; decimal projections are correspondence displays only; no GUI button, memory action, angle mode or display conversion may bypass the exact evaluator; no host random generator, floating transcendental library, silent complex promotion or hidden calculator state; no claim to functions outside the explicit feature manifest without a new versioned extension Result: PASS; receipt `sha256:627845b4d6543d567c0cd5487a9f699ffe05b60347b66670929301ef27028896`.

Depth-independent base: One entered digit is translated to an exact counted part; the structural empty-One and structural One supply cancellation and identity bases.

Depth-independent successor: Appending a generated digit refines the exact denominator by one counted tenfold composition; appending an operator composes two already-certified values, and appending one recurrence term retains the previous exact sum plus one exact part while replacing the remainder by a proven smaller positive bound. Each additional named function is admitted only when its finite expression tree terminates in already certified kernels; each session action appends a typed state record; each GUI action maps to exactly one parser token or state transition.

Exact exclusions:

- semantic numerical zero, NaN and infinity are not admitted values.
- negative, irrational and imaginary scalars are replaced by typed Fold structures.
- binary floating-point and opaque host-library answers are excluded from proof.
- uncertified roots, series, singularities and exhausted resource bounds halt.
- decimal projections are correspondence displays only.
- no GUI button, memory action, angle mode or display conversion may bypass the exact evaluator.
- no host random generator, floating transcendental library, silent complex promotion or hidden calculator state.
- no claim to functions outside the explicit feature manifest without a new versioned extension.

Limitation: Closure covers the explicit feature manifest documented by the app and README, not every proprietary button ever shipped on every calculator. Additional functions remain open lawful extensions. Physical-unit conversion tables and random-number generation are excluded because they require separately registered empirical units or a deterministic Fold-randomness law rather than an opaque host oracle.

Evidence identities:

- Source manifest: `sha256:682bc8e380e0b70795ab3c5868cd2cf8ca71c65b26cab9624d9491d54205c990`.
- Derivation seal: `sha256:2fd07cdc7dc1f857c209c3814c709f68ab21def91f3dee750f097614add757d9`.
- Independent implementation: `sha256:a5ed2ee9a44574a28530382f7b74c0b0f39446a683fd36979342248f574c5828`.
- Independent certificate: `sha256:1e9b190d79b01ebaa1a13c76556fd3d0ad80e9bac8e7cf65570bd0e1e07b1db2`.
- External validation: `sha256:24d777eb642d44e737201e495de6c64aa30ad0c484913f11be86cf405b9561a9`.
- Engine receipt: `sha256:75b0b5df0397c0aeaba035272561fd0e62949b0ef75580a153c8a109e84a06bb`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SCIENTIFIC-CALCULATOR-005-75b0b5df0397c0ae.json`.

28.4 Fully realised SFT scientific calculator and complete Mathematics law explorer

Claim identity: `SFT-MATH-SCIENTIFIC-CALCULATOR-006`

Publication status: `independently_replicated`; current versioned calculator lineage.

WHY. Exact Mathematics is most useful when any person can calculate with it without surrendering the derivation chain. Familiarity, complete current-knowledge reach and machine-inspectable evidence must therefore coexist.

DERIVATION. The twenty-four dependencies fix every current Mathematics law and the corrected exact calculator. Exhausting the ten completion distinctions eliminates an API-only tool, prohibited host values, partial functions, omitted branches, opaque output, overwhelming presentation, platform lock-in, unreduced large-angle exhaustion, silent resource failure and incomplete tests. The only surviving form is the familiar exact app plus its progressively disclosed law explorer.

CHECK. Enumerate all 1,024 completion forms; require one all-admitted survivor; execute ordinary, prohibited-value, certified nonrational, large-angle, result-chain, current-census, proof-output, visible-control, advanced-discovery and exact coverage witnesses; pass four adverse controls; independently regenerate the product, replay all twenty-four Mathematics families and rerun the declared active file coverage gate.

Exact statement:

> *The admitted exact calculator extends uniquely to a familiar, progressively disclosed and cross-platform application that evaluates the complete declared scientific expression language, exposes every currently registered predecessor Mathematics family, emits engine-comparable but explicitly non-admission proof output, halts at every exact value/domain/resource boundary, and is completely statement-and-branch tested.*

Dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ALGEBRA-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-GEOMETRY-TOPOLOGY-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`, `SFT-MATH-EXACT-RELATIONS-002`, `SFT-MATH-ORBIT-NUMBER-THEORY-002`, `SFT-MATH-LIMIT-CONTINUUM-002`, `SFT-MATH-ALGEBRAIC-BALANCE-002`, `SFT-MATH-BOUNDED-N-BODY-002`, `SFT-MATH-FLOORED-FLUID-REGULARITY-002`, `SFT-MATH-PRIME-PAIR-CENSUS-002`, `SFT-MATH-RIEMANN-MIRROR-002`, `SFT-MATH-COLLATZ-FINITE-CENSUS-002`, `SFT-MATH-SELF-SIMILAR-CONVERGENCE-002`, `SFT-MATH-SCIENTIFIC-CALCULATOR-004`, `SFT-MATH-SCIENTIFIC-CALCULATOR-005`.

Grammar boundary: The complete ten-coordinate binary product of application completions over the immutable claim-005 calculator and the exact set of twenty-four registered predecessor Mathematics claims.

Generation rule: Hold all twenty-four admitted predecessor Mathematics claims fixed, then exhaust the ten independent completion choices for interaction, value law, scientific surface, branch census, proof output, accessibility, portability, periodic scale, fail-closed resources and complete testing.

The complete Cartesian product contains 1,024 candidate forms and has completeness identity `sha256:1bcc8b271c6aaea78b627a79be19833dafce9c2f98d4df0ce66e0b2eb8938b21`. Every coordinate is shown below; the candidate census and elimination receipt preserve every product member and decision.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`interaction`	`expression-api-only` - An API alone is not a familiar calculator.	`familiar-keypad-and-result-chaining` - Digits, keypad, terminal equals, Ans chaining, memory, clear-entry, all-clear and backspace are all explicit.
`value_law`	`projected-host-scalars` - Host zero, negative, irrational, imaginary or floating proof values violate the admitted Mathematics boundary.	`exact-SFT-runtime-types` - Every answer is empty-One, a held positive rational, a certified rational enclosure or typed Fold fibres.
`scientific_surface`	`partial-function-subset` - A partial subset cannot realise the declared scientific calculator.	`complete-declared-expression-language` - Every documented arithmetic, power, root, circular, hyperbolic, logarithmic, combinatorial, statistical and typed-fibre function is routed through an exact kernel.
`mathematics_census`	`calculator-only-scalar-view` - A scalar-only view omits registered structured and theorem-level Mathematics families.	`all-current-predecessor-families` - All twenty-four predecessor claims have a one-to-one structured/scalar translation and local enumeration replay.
`proof_output`	`opaque-friendly-number` - A projected number without exact evidence hides the calculation law.	`typed-certificate-trace-and-resources` - Exact type, rational parts, certificate, operation trace, resources, dependencies and official receipt reference remain inspectable.
`accessibility`	`all-evidence-visible-at-once` - Dumping proof detail onto first-time users makes ordinary calculator use inaccessible.	`familiar-first-progressive-disclosure` - The standard calculator is the default; proof, functions, learning and law exploration open on demand.
`portability`	`platform-specific-or-heavy-runtime` - A platform-specific or container-dependent app blocks independent access.	`standard-library-mac-Windows-Linux` - One Python-standard-library application plus installed command and three operating-system launchers provides the same evaluator.
`periodic_scale`	`unreduced-direct-series-only` - A direct series alone exhausts resources on ordinary large circular arguments.	`certified-whole-turn-reduction` - Large angles subtract an exact whole count of certified turns before rational recurrence.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`resource_boundary`	`unbounded-or-silent-failure` - Unbounded construction or NaN/infinity hides an unclosed computation.	`counted-early-check-and-mandatory-halt` - Digits, exponents and counted operations are checked early and every invalid or unclosed expression halts without a value.
`testing`	`passing-examples-with-gaps` - Passing examples do not execute every implementation path.	`complete-statement-and-branch-coverage` - Every statement and branch in the declared active inherited and completion implementation is executed, alongside CLI, GUI, launcher, control and law-replay tests.

Unique survivor and exact result:

> *The unique completion is the familiar cross-platform app whose ordinary input invokes only exact SFT runtime types; whose advanced view exposes the complete current Mathematics census and engine-comparable proof; whose large circular inputs are certified by whole-turn reduction; and whose declared active implementation records no missing statement or branch.*

Replacing any forced coordinate by its enumerated alternative loses a registered requirement; adding an undeclared rule violates the no-extra-rule boundary. The unique product survivor therefore closes the declared grammar by minimality and named-shape uniqueness. Its operational laws are:

- human boundary notation is translated before evaluation and never becomes a prohibited proof scalar.
- every button, keyboard action, memory transition and terminal expression uses one exact evaluator.
- a local Mathematics replay can reproduce a registered census but cannot issue or impersonate an engine admission receipt.
- progressive disclosure changes presentation only and cannot change the exact retained answer.
- large circular input is equivalent after subtraction of any generated whole number of complete turns.
- the current expression census is one-to-one with the registered predecessor catalog and halts if that catalog changes without a translation.
- complete implementation coverage is necessary but remains distinct from engine forcing and independent validation.

Executed witnesses:

- `ordinary-use` - 1+1= returns exact forward-held two. Result: PASS.
- `prohibited-value-translation` - Correspondence zero and negative are empty-One and counter-held positive magnitude. Result: PASS.
- `nonrational-certificate` - sqrt(2) remains a replayable rational enclosure. Result: PASS.
- `large-angle-closure` - A million-radian sine closes after certified whole-turn reduction. Result: PASS.
- `familiar-result-chain` - 2 followed by +3 returns readable five through retained Ans. Result: PASS.
- `complete-predecessor-census` - The exact dependency set contains twenty-four unique current Mathematics predecessors. Result: PASS.
- `engine-comparable-not-admission` - Proof output contains exact checks and explicitly denies issuing an engine admission. Result: PASS.
- `familiar-visible-controls` - Every ordinary calculator control is present once. Result: PASS.
- `advanced-function-discovery` - Aggregate and typed-fibre operations remain discoverable without crowding the keypad. Result: PASS.
- `complete-coverage` - The declared active implementation has no missing statement or branch. Result: PASS.

Adverse controls:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An API alone is not a familiar calculator. Result: PASS; receipt `sha256:ac04d355305d1827450c11aba9b90a9b2ae9bf5aeb23b9fcccd3efc6acf7c1a2`.

- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no numerical zero, negative magnitude, irrational scalar, imaginary scalar or floating proof value; no fitted scientific parameter, host transcendental answer, random oracle, NaN or infinity; no GUI or terminal path may bypass the exact evaluator; no calculator replay is labelled an admission receipt; no claim to future Mathematics families not present in the exact predecessor census; no proprietary calculator-specific surface is claimed beyond the declared expression language Result: PASS; receipt `sha256:eefb6e9e942e0a51c61bd23f963e704839a3ef7cb8507308620dae96ab4f04fc`.

Depth-independent base: Claim 005 supplies one exact scalar calculation and one desktop evidence surface; the twenty-four fixed predecessors supply the complete current Mathematics catalog.

Depth-independent successor: Adding one expression, interaction, platform view or Mathematics family is lawful only when it maps to an admitted kernel, retains exact typed evidence, adds a generated census coordinate or registered translation, and introduces no uncovered execution branch.

Exact exclusions:

- no numerical zero, negative magnitude, irrational scalar, imaginary scalar or floating proof value.
- no fitted scientific parameter, host transcendental answer, random oracle, NaN or infinity.
- no GUI or terminal path may bypass the exact evaluator.
- no calculator replay is labelled an admission receipt.
- no claim to future Mathematics families not present in the exact predecessor census.
- no proprietary calculator-specific surface is claimed beyond the declared expression language.

Limitation: Closure covers the exact current predecessor catalog, declared scientific expression language and application behavior. It does not claim that a calculation is a new theorem, that every future Mathematics discovery is already translated, or that every proprietary calculator button is required. Lawful future additions remain open versioned extensions.

Evidence identities:

- Source manifest: `sha256:033235dcc6970ac7ca58141f8497d4e43edf6cb15ce2d7ba31d2d84e7b9e6aca`.
- Derivation seal: `sha256:e14a12e172dab6232491cb9591dc4b82928760b721e3a99aaabba68a2857b054`.
- Independent implementation: `sha256:89732fb0c989727f66e8f370ac133fd2472ba0faf7e4af904f8f01e915eab6be`.
- Independent certificate: `sha256:1a5a1a0a29eae41f6822d1f1184279361ad4058e9b1afdb83baa9089c6a3a62e`.
- External validation: `sha256:d2eaedf0621c01b32b0e669c6428e62b622057e53b21bb5fb1773ab1d5ace28f`.
- Engine receipt: `sha256:addeb561438c9913313824ba43eb8fceccb53c574f5a9853b3c9c1fac4e4e4b7`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SCIENTIFIC-CALCULATOR-006-addeb561438c9913.json`.

28.5 Accessible SFT-native scientific calculator application

Claim identity: `SFT-MATH-SCIENTIFIC-CALCULATOR-007`

Publication status: `independently_replicated`; current versioned calculator lineage.

WHY. A proof-producing calculator is accessible only when the ordinary surface remains familiar and the complicated distinctions stay available underneath. A painted, responsive and state-isolated surface is therefore part of the application boundary.

DERIVATION. Claim 006 fixes every mathematical operation. Exhausting the eight presentation distinctions eliminates the blank deprecated widget path, arbitrary keypad reflow, conventional prohibited projections, duplicate arithmetic, shared visitor state, computer-only reach, heavy dependencies and geometry-only testing. The remaining adapter changes no law.

CHECK. Enumerate all 256 adapters; require one all-admitted survivor; execute ordinary, cancellation, negative-halt, exact-certificate, fresh-session, complete-control, standard-layout, responsive-contract, single-kernel and complete-coverage witnesses; then independently start the application as a subprocess and replay its HTTP boundary.

Exact statement:

> *The admitted exact calculator extends to one dependency-free standards-rendered application whose familiar calculator notation never selects the mathematics: displayed zero retains empty-One semantics; negative-result attempts halt transactionally; certified intervals and orthogonal fibres remain typed exact objects; every page has an independent session; the conventional standard pad remains structurally coherent; scientific functions are progressively disclosed; and the same application is reachable on macOS, Windows, Linux and same-network phones.*

Dependencies: `SFT-MATH-SCIENTIFIC-CALCULATOR-006`.

Grammar boundary: The complete eight-coordinate binary product of presentation adapters over the immutable claim-006 controller; no coordinate may alter arithmetic, value construction, expression parsing, proof law or engine protocol.

Generation rule: Hold the immutable claim-006 evaluator fixed and exhaust eight independent presentation completions: rendered surface, keypad organisation, value boundary, kernel route, session state, network reach, dependency boundary and validation.

The complete Cartesian product contains 256 candidate forms and has completeness identity `sha256:06ce9a083d2cd80333cfb1ab854033ad171d986829472dd33b7ef0006536ec1e`. Every coordinate is shown below; the candidate census and elimination receipt preserve every product member and decision.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`rendered_surface`	`deprecated-widget-geometry-only` - Mapped geometry did not ensure painted controls on the target macOS runtime.	`standards-rendered-browser-surface` - The installed browser paints semantic HTML controls through a local standard-library service.
`keypad_organisation`	`flattened-function-grid` - Arbitrary reflow destroys the familiar four-column calculator grammar.	`standard-pad-plus-scientific-panel` - Memory, standard four-column keys and scientific functions retain separate coherent organisations.
`value_boundary`	`conventional-prohibited-projection` - Displaying a negative or irrational-style decimal silently reimports a prohibited scalar.	`familiar-notation-with-SFT-halt-and-types` - Zero is familiar empty-One notation, negative-result attempts halt, and non-scalar results remain exact typed certificates.
`kernel_route`	`client-side-duplicate-arithmetic` - A second arithmetic implementation could diverge from the admitted evaluator.	`server-side-immutable-claim-006-controller` - Every visible action reaches the same admitted controller and the browser performs no arithmetic.
`session_state`	`shared-global-visitor-state` - One visitor or test can contaminate another visitor's landing value.	`fresh-independent-page-session` - Every page load receives a fresh retained controller identity while its own history and memory persist.
`network_reach`	`desktop-loopback-only` - A computer-only default blocks ordinary same-network phone use.	`same-network-default-with-private-option` - The default binds to the local network and displays its phone address; private mode remains explicit.
`dependency_boundary`	`heavy-GUI-or-container-runtime` - A heavy platform runtime violates the accessibility requirement.	`Python-standard-library-and-installed-browser` - HTTP service, exact controller and page assets require no third-party runtime or container.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`validation`	`hidden-widget-dimensions-only` - Widget dimensions did not detect the observed blank window.	`API-render-responsive-adverse-and-complete-coverage` - The corrected route checks HTTP behavior, semantic rendering, mobile structure, prohibited results, session isolation and complete active-file coverage.

Unique survivor and exact result:

> *The unique completion is a familiar calculator-first local web application with a coherent standard pad, optional scientific and proof panels, fresh per-page state, same-network default access and one immutable exact SFT evaluation route.*

Replacing any forced coordinate by its enumerated alternative loses a registered requirement; adding an undeclared rule violates the no-extra-rule boundary. The unique product survivor therefore closes the declared grammar by minimality and named-shape uniqueness. Its operational laws are:

- presentation notation may describe an admitted value but cannot create a new value type.
- displayed 0 denotes structural empty One and does not admit a numerical-zero proof object.
- a result containing any counter-held scalar or bound halts before answer, memory or history mutation.
- a certified rational interval is a typed certificate and is never displayed as one irrational decimal scalar.
- each browser page owns one fresh controller state and cannot inherit another page's tested answer.
- the standard four-column keypad grammar is invariant under phone-width reflow.
- touch-keypad actions never focus the editable field or summon the phone keyboard.
- the browser performs interaction and presentation only; all evaluation remains in claim 006.

Executed witnesses:

- `ordinary-result` - 1+1 displays familiar exact 2. Result: PASS.
- `empty-one-display` - 1-1 displays familiar 0 while the evaluator retains empty One. Result: PASS.
- `negative-result-halt` - 1-4 halts without retaining history. Result: PASS.
- `certificate-not-irrational` - sqrt(2) displays exact rational bounds without an irrational-style decimal. Result: PASS.
- `fresh-session` - A calculation in one page leaves the next page at 0. Result: PASS.
- `complete-key-partition` - All sixty-four controls occur exactly once across memory, standard and scientific organisations. Result: PASS.
- `standard-keypad` - The visible standard calculator is six complete four-key rows. Result: PASS.
- `responsive-contract` - The page declares phone disclosure, four-column standard layout and horizontal-overflow prevention. Result: PASS.
- `touch-focus-contract` - Phone keypad taps do not automatically open the keyboard. Result: PASS.
- `one-kernel-route` - The browser sends actions to the local API and defines no JavaScript arithmetic model. Result: PASS.
- `complete-active-coverage` - Every active adapter statement and branch is executed. Result: PASS.

Adverse controls:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Mapped geometry did not ensure painted controls on the target macOS runtime. Result: PASS; receipt `sha256:7695430e13d324c741370ca77b432f700184cdda88fded52e646b1bb7554aa87`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt

sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no edit to the frozen engine or immutable claim-006 source; no browser-side arithmetic or alternate evaluator; no displayed negative, irrational scalar, imaginary scalar, NaN or infinity; no counter-held result retained in answer, memory or history; no shared landing-state contamination between page sessions; no automatic expression focus on a touch-width calculator; no heavy GUI framework, container or network service beyond the user's local network Result: PASS; receipt sha256:b3a4601336eba4f37db8cf7851d8743b5a75a8be9e1793ed1f984b360c1cb4e7.

Depth-independent base: Claim 006 supplies the exact evaluator, controller, proof trace and complete scientific expression language; one fresh page begins with its familiar displayed 0 and empty history.

Depth-independent successor: Adding one interaction, viewport or visitor is lawful only when it preserves the standard keypad organisation, routes to the same exact controller, retains a distinct session and rejects every prohibited result transactionally.

Exact exclusions:

- no edit to the frozen engine or immutable claim-006 source.
- no browser-side arithmetic or alternate evaluator.
- no displayed negative, irrational scalar, imaginary scalar, NaN or infinity.
- no counter-held result retained in answer, memory or history.
- no shared landing-state contamination between page sessions.
- no automatic expression focus on a touch-width calculator.
- no heavy GUI framework, container or network service beyond the user's local network.

Limitation: Closure covers the current calculator presentation and declared browser-standard route. It does not claim every future browser or device has been physically tested. Manual review remains a separate publication gate, and lawful extensions remain open.

Evidence identities:

- Source manifest: sha256:ee871d9a0d471b7525254af0fed9dff8f54a5f9254e7e2cc642c5150284b1822.
- Derivation seal: sha256:fdb9d7eb5203c03e24df26187742b0650bfef090233f9f4fcbd46331a29b2fa6.
- Independent implementation: sha256:f554a6058ae2fc15ea6ff60ddbcfd95409601eac16cec182d3d737f9268b2f5e.
- Independent certificate: sha256:9576ee0eee57fe8d2ab6e7245012100ddfbbb25d4eefe3b1b3cd761952fa80fb.
- External validation: sha256:a1402278d191f89ea10fde58a863df1f12a30c8ec97f891892efdd0e587f487d.
- Engine receipt: sha256:feeb1879d97191edc8827be6da5421a2c78aba56ba2d3e0fe5dbf0882aac9867.
- Engine receipt path: receipts/engine/model_admitted/SFT-MATH-SCIENTIFIC-CALCULATOR-007-feeb1879d97191ed.json.

28.6 Exact value translation and fail-closed semantics

Familiar surface	Exact calculator meaning	Boundary behavior
0	structural empty One	displayable and usable as the empty/identity form; never stored as a conventional numerical-zero proof scalar

Familiar surface	Exact calculator meaning	Boundary behavior
positive integer or decimal	exact positive whole or rational Fold part, parsed character by character	no binary floating conversion is admitted
subtraction with a nonnegative result	exact trace pairing and retained forward remainder	cancellation displays `0` with empty-One semantics
subtraction whose result crosses below empty One	prohibited negative/counter-held numeric result	transactional `HALT`; prior answer, memory and history are restored
exact rational root or function result	exact Fold scalar	exact numerator/denominator retained
non-rational correspondence such as `sqrt(2)`	certified rational lower and upper bounds plus replay trace	no irrational decimal is promoted to a scalar
orthogonal correspondence	typed real and orthogonal Fold fibres	no imaginary scalar or silent complex promotion
undefined domain, singularity, exhausted resource or unclosed certificate	no admitted result	mandatory `HALT`; no NaN or infinity

This distinction is why `1-1` may visibly return `0` while `1-4` must halt. The first is a structural empty result. The second would require a prohibited scalar to be admitted as the calculator answer. Claim 007 snapshots the complete controller state before every action and restores that snapshot if answer, memory or any history row contains a counter-held scalar. The rejected expression and explanation remain visible, but the prohibited value does not.

28.7 Declared calculation language

The expression grammar accepts exact integers, decimal and scientific notation, parentheses, addition, subtraction, multiplication, division, powers, postfix factorial and percent, and optional terminal equals. It supports result chaining, `Ans`, memory clear/recall/store/add/subtract, ordered history, clear-entry, all-clear, backspace and explicit RAD, DEG and GRAD angle modes. Every button and typed expression reaches the same evaluator.

The declared scientific functions are:

```
abs recip sqrt cbrt root pow
sin cos tan asin acos atan
sinh cosh tanh asinh acosh atanh
exp ln log log10 log2
ncr npr complex conj
sum prod mean variance stddev
floor ceil gcd lcm mod hypot
```

Constants and retained state are `pi`, `tau`, `e`, `phi`, `empty`, `ans` and `mem`. Circle, exponential, logarithmic, inverse and hyperbolic operations are not delegated to opaque host transcendental functions. They produce exact rational enclosure objects with finite recurrence evidence. Large circular inputs undergo exact whole-turn reduction before enclosure, preventing magnitude from silently exhausting the recurrence. Operation limits and requested display precision are counted interface/resource boundaries; they do not tune a mathematical law.

28.8 One evaluator, three operating systems and same-network phones

The current public application uses only Python's standard library and an installed standards-compliant browser. The local server binds to the machine's local network by default, advertises the phone URL, and offers an explicit private mode. Each page load receives a fresh controller identity; one visitor cannot inherit another visitor's answer, memory or history. A request token and page-session identifier bind local actions, requests are size bounded, responses disable storage, and the content-security policy prevents external script or content dependencies.

The visible standard pad is always six rows of four keys. Memory controls are a separate five-key row and thirty-five scientific controls remain a separate seven-by-five organisation, giving sixty-four controls exactly once. On phone widths the standard pad appears first and scientific functions are disclosed by one button. Keypad taps do not focus the expression field or summon the software keyboard; the keyboard opens only when the user deliberately taps the editable expression. The layout was rendered and exercised at a 390-by-844 phone viewport without horizontal overflow.

Launch routes:

```
python3 -m sft.mathematics.calculator_browser
calculator_launchers/Launch Smithian Fold Calculator.command   # macOS
calculator_launchers/Launch Smithian Fold Calculator.bat       # Windows
calculator_launchers/launch-smithian-fold-calculator.sh         # Linux
```

28.9 Verification, coverage and direct-use evidence

Claim 006 records 1,869 executable statements and 696 branches across its declared active calculator implementation, all covered with no missing statement, branch or partial branch. Claim 007 records 293 executable statements and 86 branches across the active browser adapter, again with no missing statement, branch or partial branch. These are statement-and-branch coverage facts over the declared files, not a claim that every physical browser or device in existence has been tested.

The executed checks cover exact values; every operation and parser branch; invalid domains and resource exhaustion; expression-machine, session, memory and history integration; pure controller state; command-line execution; operating-system launchers; server API behavior; request bounds and authorization; fresh-session isolation; full control partition; semantic page structure; phone reflow and focus behavior; exact interval presentation; negative-result rollback; and implementation-distinct HTTP replay. Direct use supplied an additional unfavorable route: the first desktop widget visibly opened blank even though hidden geometry tests passed. That failure is preserved in the 007 false-premise control and is why the standards-rendered route, rather than the deprecated widget path, survives.

Representative end-to-end outcomes:

Input	Visible outcome	Exact significance
`1+1`	`2`	exact disjoint-junction result
`1-1`	`0`	structural empty One
`1-4`	`HALT`	prohibited negative result rejected transactionally, history unchanged
`0.1+0.2`	`3/10` or its familiar exact display	decimal characters translated to exact rational parts
`sqrt(2)`	certified rational interval	no `1.414...` scalar and no approximation glyph in proof output
`complex(2,3)`	real and orthogonal Fold fibres	typed composition, not an imaginary proof scalar

28.10 Proof output, engine boundary and lawful transfer

A calculator result carries its exact expression, typed result, rational numerator/denominator or enclosure bounds, certificate, complete operation trace, tokens read, operations executed, calculator claim dependency and official calculator receipt identity. That output is computational evidence that an admitted calculation law evaluated the given expression. It is not, by itself, a receipt admitting a new theorem or scientific claim.

A researcher may transfer the exact expression, result object and trace into a proposed claim package, but official admission still requires a registered statement and dependencies, complete candidate generation, uniqueness, minimality, adverse controls, independent validation, execution through the untouched `SFTAdmissionEngine` and the engine-issued receipt. The calculator cannot mint that receipt and does not call or modify the engine. Version 1.2 does not claim a one-click source-bound evidence-export/replay bridge; that would be a separate lawful extension. This boundary prevents a useful calculation tool from becoming a shadow admission route.

The frozen engine tree remains `ad30f4866c18b2adbade95a0b2de40d5caa61308`. The calculator work did not alter that tree. Claims 006 and 007 are immutable admitted dependencies at their recorded source identities; future calculator work must use a new claim and cannot rewrite either receipt.

29. Complete V1/V2 Mathematics reconciliation

The clean-room rule does not permit the new branch boundary to make an older result disappear. The audit reads all 356 V1 theorem-manifest rows and all 407 V2 numbered results as observational reconstruction requirements, decomposes composite rows into atomic scientific obligations and assigns each atomic obligation to exactly one categorical owner. Prior executables and answer artifacts never enter the V3 derivation runtime.

The Mathematics ledger identifies 71 owned atomic obligations across 29 V1 rows and 38 V2 steps. Every atom maps to one or more of the twenty-two foundational receipts and every mapping is closed. The remaining 327 V1 rows and 369 V2 steps are recorded in an exact exclusion identity as reviewed non-Mathematics entries, not silently ignored. The authoritative file is `census/mathematics_prior_obligations.json`.

Reconciliation result	Exact outcome
V1 rows reviewed	356
V2 numbered results reviewed	407
Mathematics atomic obligations	71
Closed Mathematics atomic obligations	71
Open Mathematics atomic obligations	0
Explicit corrections or invalidations retained	4

Four explicit reconciliation outcomes deserve front-page visibility because they demonstrate that V3 is neither a rubber stamp nor a historical rewrite. First, old signed and zero host storage is retained as implementation history but replaced at the admitted boundary by held orientation and structural empty One. Second, V1's detailed Riemann statement governs: the prime skeleton and unique reflection axis close, while an unrestricted complex-zero theorem does not arise from symmetry alone. Third, the prior constant-three-quarter Collatz shortcut fails the start-three control even though the complete 100,000-start census succeeds. Fourth, the depth-five fluid value 32 is derived as One divided by the positive floor, not installed as numerical damping. Physical interpretations of formal rational thresholds remain assigned to their empirical owning branches rather than feeding back into Mathematics.

30. Cross-derivation synthesis

The twenty-two foundational theorems form one dependency chain rather than a list of renamed fields. Exact arithmetic gives lawful trace junction, pair-cell product, refinement and comparison. Discrete mathematics turns those operations into canonical carriers, selections, relations, maps and induction. Combinatorics then generates complete families of choices; graph theory holds a relation from complete pair support and promotes it to path, cycle, cut and network structure. Algebra asks which complete operation relations close, associate, return and map. Order retains only witnessed comparison and derives conditional extremal operations.

Geometry and topology use graph paths, incidence and order closure to derive the finite structures computation actually requires. Probability uses complete combinatorial support and exact parts to quantify observation classes without altering deterministic state. Optimization uses order to preserve feasible undominated alternatives. Dynamics uses graph transitions to define trajectory, return, basin and record-dependent reversal. Logic then turns distinguished forms and relations into proof-carrying inference. Category, type and composition close the chain by showing how exact objects and transformations compose without losing interface provenance.

No result licenses backward importation. Because category-like composition is derived last, it cannot be an axiom used to select arithmetic. Because probability is downstream of complete deterministic support, it cannot install an ontic random cause in the earlier state law. Because geometry is finite incidence before continuum correspondence, an irrational coordinate cannot retroactively become foundational proof evidence.

The ten lineage reconstructions then make the archived named claims explicit. Relation composition supplies the beat and joint-return laws; the same remainder identity unifies Fold orbit and multiplicative order; exact finite successor traces separate potential infinity from a completed continuum; polynomial balance retains algebraic magnitude without irrational proof values; finite product support closes native n-body recurrence; reciprocal-floor geometry closes native fluid regularity; complete bounded censuses establish their declared populations; and adverse controls prevent Riemann symmetry or Collatz heuristics from being inflated beyond what the execution establishes.

31. Exact arithmetic without zero, negatives or irrational proof values

The prohibition on semantic numerical zero does not make empty, identity or no-transition cases inexpressible. It forces them to retain their structural origin. Empty selection is empty One; identical traces return same-form; an identity arrow is the empty-One return; identity duration is the one-state trajectory with no transition. These forms are typed structural records, not a scalar that can be silently substituted across domains.

Negative quantities are likewise unnecessary for exact comparison. If two traces pair completely they are the same form. Otherwise the unmatched positive occurrences are retained and a held label records which trace owns them. Network balance pairs positive token identities. Logical denial changes held orientation. Algebraic return uses a complementary carrier mate. No proof relies on subtracting a larger magnitude from a smaller one.

Irrational, imaginary and floating values are excluded from formal admission because the generated Foundation supplies exact positive parts and finite refinement, not an ungenerated completed field. This does not deny that conventional sciences record such descriptions. It establishes that they may enter only as correspondence, finite approximation or measured values after the law is sealed. A later approximation theorem must state its generated sequence, exact error relation, convergence boundary and measurement custody.

32. Probability, statistics and superdeterminism

The probability theorem is often the point at which a deterministic model is tempted to import an incompatible prior. SFT does not do so. A microstate is one exact generated form. The registered support contains every such form once. Observation is a total classification relation from microstates to retained observation labels. States with the same image form an observation class. Uncertainty is precisely the collection of distinctions present in support but closed by that observation; the underlying transition law remains deterministic.

An event is a held support selection. For a nonempty event, probability quantity is the exact positive count of held states relative to the exact positive count of the complete support. The empty event remains empty One and never passes through numerical-zero arithmetic. Conditioning restricts both event and whole to a nonempty held condition before forming the exact part. Independence is not lack of a fitted correlation; it is the structural fact that joint support is the complete pair-cell product and the event pairing factorizes.

Statistical summaries are allowed because they are transparent functions of complete finite records. Ties are retained. A natural-data claim would require a separate empirical registration, frozen source and target custody. The target must remain closed until after the derivation seal; all rows, including failures, must be preserved. No natural dataset was needed or used to admit the formal probability theorem in this paper.

33. Formal proof versus empirical validation

The twenty-two foundational claims and five calculator records are formal structural theorems, exhaustive finite mathematical censuses or executable translations. Their empirical content in this paper is computational execution and direct interface testing: the candidate products are actually generated, decisions actually run, controls are deliberately perturbed, separate validator processes execute and receipts are independently replayed. This execution evidence demonstrates that the declared finite grammars and certificates have the reported machine behavior. It is not observational evidence about an external natural system, and the paper does not mislabel it as such.

When a later branch asserts a relation to natural data, the Foundation measurement boundary applies. Formal law selection and target evaluation are different phases with different custody. A blind opaque predictor cannot replace a derivation chain because a score alone does not identify premises, alternatives, eliminations, source identity, failure conditions or whether the target influenced the law. An empirical SFT claim must expose all of those objects and halt if custody or capability isolation fails.

34. Machine implementation and complete reproduction

The definitive cross-platform logical command is `sft verify-all`, invoked with the host's standard Python launcher:

```
python3 -m sft verify-all   # macOS and most Linux systems
py -m sft verify-all        # standard Windows Python launcher
```

No Docker image, virtual machine, network service or third-party Python package is required for the scientific verification path. The command validates repository structure, runs all unit, unfavorable-control and end-to-end tests under line coverage, requires 100 percent executable-line coverage of the 15-module admission engine, loads the ordered execution manifest, rebuilds each source manifest, reruns every candidate and control, launches every independent validator and compares each new receipt byte-for-structure with the admitted stored receipt.

The proportionate local report for this successor paper is:

```
MATHEMATICS PRIOR-OBLIGATION LEDGER: 71/71 CLOSED
MATHEMATICS REGISTERED CLAIM RECORDS: 27/27
MATHEMATICS CURRENT NON-SUPERSEDED CLAIMS: 26
MATHEMATICS GENERATED CANDIDATES PRESERVED: 21,504
CALCULATOR CLAIMS 004 THROUGH 007: CURRENT
CALCULATOR CLAIM 003: PRESERVED SUPERSEDED ADVERSE EVIDENCE
BRANCH PUBLICATION GATE: run once after final render
```

The coverage statement concerns the engine implementation; mathematical closure comes from the claim-specific candidate grammar, decision vector, unique survivor, minimality, named-shape uniqueness and induction certificate. Neither substitutes for the other.

35. Independent criticism and invalidation routes

A reviewer need not accept the vocabulary or conclusions to reproduce the result. The strongest direct criticism is to identify a candidate inside a declared boundary that the registered product omits. That attacks completeness. A second route is to show that a rejected coordinate actually preserves every requirement, defeating minimality, or that another product member survives, defeating uniqueness. A third is to invalidate the base or successor certificate. A fourth is to provide an input inside the declared domain that violates an operational law.

The repository also supports mechanical attacks: alter a source after registration, change a candidate identity, remove a decision, add a survivor, fail a control, tamper with a certificate, reorder the execution manifest or change a receipt. The engine halts at the exact violated gate. Hashes do not establish truth; they establish which exact sources and artifacts were checked so an altered object cannot silently inherit a prior status.

36. Consequences for the remaining knowledge tree

The current-evidence-complete branch supplies the exact resources downstream branches may cite: symbols can be canonical distinguished forms; encodings can be complete maps; information quantity can be an exact relation between support distinctions and observation; channels can be relations; entropy and uncertainty must respect deterministic support and exact parts; coding can use combinatorial word families, graph paths and algebraic operations. None of those downstream laws is admitted by this paper merely because its mathematical vocabulary now exists.

Formal computation may build states from canonical forms, transitions from relations, machines from dynamical systems, resource orders from exact positive traces, semantics from proof-carrying composition and quantum support from complete Fold word families. Physics and later natural sciences may use finite geometry, dynamics, exact probability and optimization, but must separately derive their laws and pass blind empirical validation where observation is required. Applications remain frontier translations until the relevant science branches close.

37. Scope, limitations and lawful extension

Current-evidence completion is exact at the versioned generated-finite Mathematics boundary. It does not assert a completed universe of all sets, a real or complex continuum, an infinite-dimensional space or every named theorem of conventional mathematics. It derives the general structural kernels needed to generate and test finite exact instances. Named special structures must still expose their carrier and property evidence. Continuum expressions can enter only through separately registered finite approximation or measurement claims.

No official natural empirical claim is made in this branch. The calculator is an executable mathematical translation and validation surface, not a natural-data experiment. Information Science and the computation branches have their own later papers and receipts; after this Mathematics 1.2 publication the active clean-room programme returns to Physics. None of those downstream results can retroactively select a Mathematics law.

38. Open-science status, rights and participation

Mathematical validity must not depend on whether an institution funded the question, whether a journal selected it, whether a reviewer recognizes the author's credential, or whether a prevailing programme finds the conclusion comfortable. A proof either exposes a lawful chain at its declared boundary or it does not. Institutional review can supply valuable criticism, but it cannot substitute for generated completeness, unique survival, exact witnesses and a reproducible certificate. The same rule binds Maria Smith, Ernos Labs, universities, publishers and anonymous critics.

This positioning is not rhetoric detached from evidence. Sponsor-associated outcome differences and sponsor influence on research agendas are documented; grant-review studies report limited agreement, panel sensitivity and unequal outcomes; a Cochrane review found insufficient evidence that grant peer review improves funded-research quality; and null or negative results remain underrepresented in visible literatures. UNESCO's Recommendation on Open Science identifies paywalls and high publication charges as sources of inequality and calls for methods, software, source code and outputs to be available for scrutiny. These findings do not accuse every institution or reviewer of misconduct. They establish that prestige, funding and publication are selection systems with incentives and failure modes, not neutral certificates of mathematical truth.

The exclusion cost is intellectual as well as economic. Scarce grants, credential filters, prestige markets, subscription access and author charges shape who can devote time to a problem, which problems appear respectable, what enters the searchable record and who can inspect it. Maria Smith developed SFT outside conventional academic education and funding. That fact is not evidence that any theorem is correct. It is an indictment of a gate that would have treated the speaker's status as a proxy for whether the work deserved to be heard. The unknown number of minds and questions lost to that proxy is precisely why admission here attaches to public evidence rather than biography.

Opaque computational prediction reproduces the same authority problem in machine form. A hidden model can be a useful instrument and may establish bounded predictive reliability in a genuinely blind trial. A score alone does not reveal the premises, candidate space, exclusions, leakage path, failure boundary or derivation of a law. NIST treats validity, reliability, transparency, accountability, explainability and interpretability as related features of trustworthy AI. For formal mathematics, an oracle answer is therefore never a proof; for empirical science, blindness and transparency must coexist. The target stays closed until the law seals, and the full source-to-result route remains inspectable afterward.

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39. Conclusion

The Mathematics version 1.2 evidence surface preserves twenty-seven dependency-ordered records through 21,504 generated alternatives. Twenty-two foundational claims and calculator claims 004 through 007 are current; calculator claim 003 remains preserved as superseded adverse evidence after its operational parity defect was found. It reconstructs exact arithmetic, finite discrete and combinatorial structure, relations and graphs, witnessed algebra and order, computational geometry and topology, deterministic-support probability, optimization, dynamics, proof and composition without importing conventional answer-producing models.

The unification is methodological as well as mathematical: every object has a generated boundary, every elimination has a reason and proof identity, every closure has an induction certificate, every adverse control is preserved and every admitted result can be independently rerun. The branch is accepted within its exact boundary because that complete chain closes - not because a prior authority, application score or familiar notation selects it.

Appendix A. Authoritative receipt identities

Claim	Engine receipt
SFT-MATH-EXACT-ARITHMETIC-001	sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbae71609cb05c4f40196418
SFT-MATH-DISCRETE-001	sha256:632de46c995329ed86f8397e8cf2cc493d5074027941a50069df66b4ac9e29bb
SFT-MATH-COMBINATORICS-001	sha256:52e1db94bb7865270bb3387c47c609c438de3338b542cc31d910b981e4250968
SFT-MATH-GRAPH-NETWORK-001	sha256:b4072f93147c8cd9b7233fc96cfeadeb791210239ab4e4ff7dec0ed7462e6d2a
SFT-MATH-ALGEBRA-001	sha256:1f08bfaf9f602a3825dc75979070e05f9d51a7d01cdf3c34bab31c06849855bd
SFT-MATH-ORDER-LATTICE-001	sha256:d0e20f40782f51cc85c1f572b624570e65b51bbb80135a5e8a6b3cf69f71a5f6
SFT-MATH-GEOMETRY-TOPOLOGY-001	sha256:a04f7de0ede6e2348896cf16e59981eebb5391198e498a1ec546b68a4a8d3a6e
SFT-MATH-PROBABILITY-STATISTICS-001	sha256:2a86d644881cdacb757327d5aaca01de41c3458c043980c66a16f5084cdfb45b
SFT-MATH-OPTIMIZATION-001	sha256:3385414ceea0fd4212111e42bd6ce7d6d35556b7c97d66563813d5b8d1531849
SFT-MATH-DYNAMICAL-SYSTEMS-001	sha256:3e0e44b65fbe660fe2a29b9927b1870a40da25c4b0c786e2adaa3df9a2f7cabe
SFT-MATH-LOGIC-PROOF-001	sha256:3d366d57b26d002cb27a16e74b022a0856e133c616b7da6a83233c84bdd50813
SFT-MATH-CATEGORY-TYPE-COMPOSITION-001	sha256:47873a83a90e29fa22d065ff13aa0152edb6212414e35cc0435e8eff2c9ea5a8
SFT-MATH-EXACT-RELATIONS-002	sha256:8141546a120b73aa7cac0cd843728bdf3122d1bf3c0048c132760c82fd103238
SFT-MATH-ORBIT-NUMBER-THEORY-002	sha256:8725bce36fe77923288b99b1a978ef3bf84cac9ffe1b172fd67bd221527d19ff
SFT-MATH-LIMIT-CONTINUUM-002	sha256:a32c6d6ee891f97acadd53cbec922bbcad5a6ad8a928264bb654b4caaf997098
SFT-MATH-ALGEBRAIC-BALANCE-002	sha256:ba975fdfe0c8946cb713eabcd9f48e77284941ec08fc257b9d3bb614b701540d
SFT-MATH-BOUNDED-N-BODY-002	sha256:e355bc5aa649df8fe2daf21a3e0ae56d5582680edd7f867a751bb954fca224a4
SFT-MATH-FLOORED-FLUID-REGULARITY-002	sha256:6fb3f40f5631367f23bad6cda58c87b335d33fd212a7c9d37068d8b920ad65e0
SFT-MATH-PRIME-PAIR-CENSUS-002	sha256:52fcc090c336360dcdfd4b95919ffa5926d9d8fcd430ff6bb53d7d18cf5db1d6
SFT-MATH-RIEMANN-MIRROR-002	sha256:27847629dbd2fb1d1c2f14d05017f8fb9395482d6e6ca001a24622e55364dfc4
SFT-MATH-COLLATZ-FINITE-CENSUS-002	sha256:3bb3a9fe4790150369defc5b2dc8568fed100b688dbb6a9c1d5cbd40c4f9a045
SFT-MATH-SELF-SIMILAR-CONVERGENCE-002	sha256:0abf8d557108f37f4d9cfe735db3fd6e0d79a3d9405129f0a40ae86fb606d5b1
SFT-MATH-SCIENTIFIC-CALCULATOR-003	sha256:f95ac235ac96429b36b10e8331478f110ed074e98a338c6f015d3f8874dc787c
SFT-MATH-SCIENTIFIC-CALCULATOR-004	sha256:2d243dbd08260c483e79009067051239f1304ad3998006c1ff82c301062ba28b
SFT-MATH-SCIENTIFIC-CALCULATOR-005	sha256:75b0b5df0397c0aeaba035272561fd0e62949b0ef75580a153c8a109e84a06bb
SFT-MATH-SCIENTIFIC-CALCULATOR-006	sha256:addeb561438c9913313824ba43eb8fceccb53c574f5a9853b3c9c1fac4e4e4b7
SFT-MATH-SCIENTIFIC-CALCULATOR-007	sha256:feeb1879d97191edc8827be6da5421a2c78aba56ba2d3e0fe5dbf0882aac9867

Appendix B. Complete evidence route

For every claim identifier `<CLAIM>`, the complete review route is:

```
claims/<CLAIM>/registration.json
claims/<CLAIM>/WHY_DERIVATION_CHECK.md
claims/<CLAIM>/candidate_census.json
claims/<CLAIM>/elimination_receipt.json
claims/<CLAIM>/controls.json
claims/<CLAIM>/certificate.json
claims/<CLAIM>/execution.py
```

```
generated/<BRANCH>/<VERSIONED-INDEPENDENT-VALIDATOR>.py
receipts/engine/model_admitted/<CLAIM>-<receipt-prefix>.json
```

Executable law sources are under `sft/mathematics/`. `sft/mathematics/catalog.py` fixes the foundational and calculator-kernel order; the versioned claim registrations and execution manifest retain the 007 presentation extension. `census/claims.json` is the model-admitted projection; `census/execution_manifest.json` is the complete replay order; `publications/inventories/mathematics.json` freezes scope. The separate paper evidence map binds every derivation section to exact source, claim-package and receipt hashes.

Appendix C. Reproducibility interpretation

A successful rerun proves that the checked sources deterministically reproduce the registered censuses, decisions, closures, controls, independent certificates and receipts on the reviewing host. It does not compel agreement with the grammar boundary. Scientific review therefore has two complementary tasks: reproduce the artifact and scrutinize whether the declared grammar exhausts the stated boundary. The repository makes both tasks explicit.

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